

Lab question 1 -

Module 2 - Activity 2 - Create a Website on S3

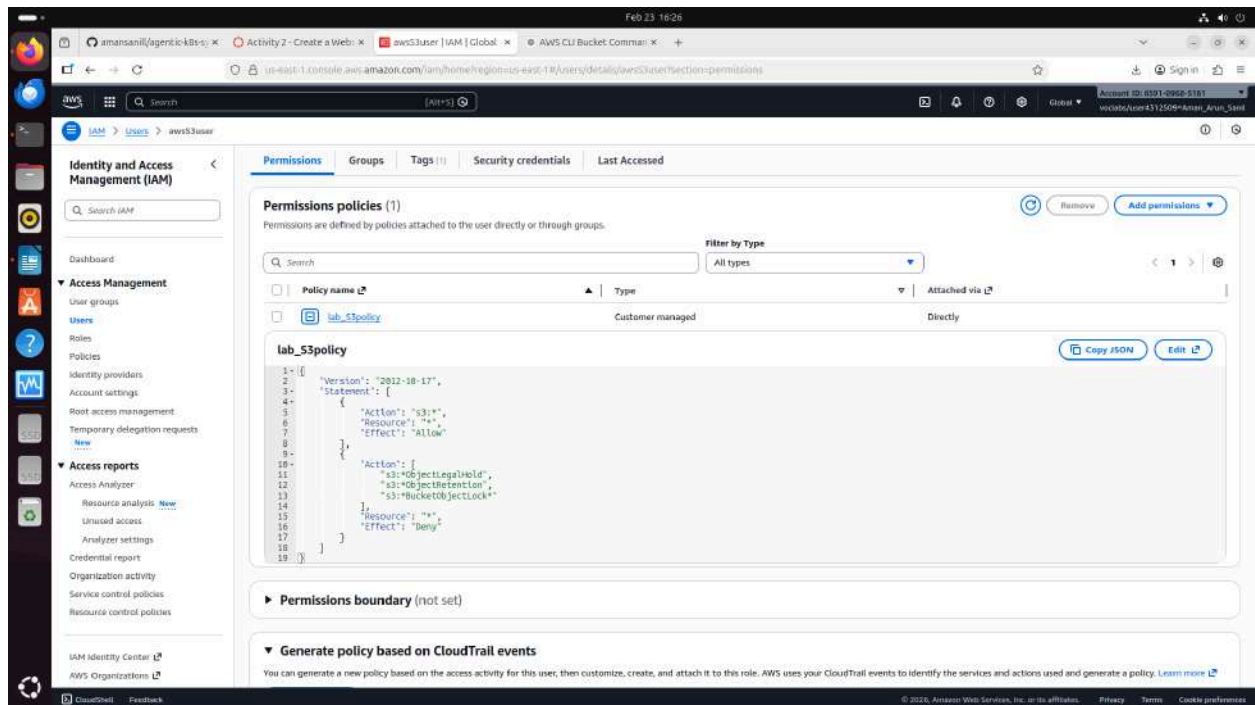
Step 1: Download the pem key and change the permissions, next ssh into the instance and complete the aws configuration. Verify it using aws s3 ls, it shouldn't return anything i.e. indicating that there is no bucket created yet.

```
ec2-user@ip-10-200-0-126:~  
msis@k8s-master-node:~/Downloads$ ls | grep labsuser.pem  
labsuser.pem  
msis@k8s-master-node:~/Downloads$ sudo chmod 600 labsuser.pem  
msis@k8s-master-node:~/Downloads$ sudo ssh -i labsuser.pem ec2-user@107.21.181.153  
The authenticity of host '107.21.181.153 (107.21.181.153)' can't be established.  
ED25519 key fingerprint is SHA256:dSWjzxkYunF6L0q06pEyUnPL2X9C6jzNJcbcCUX7BrE.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '107.21.181.153' (ED25519) to the list of known hosts.  
#_ _  
~\ _ ##### Amazon Linux 2  
~~ \_#####\  
~~ \###| AL2 End of Life is 2026-06-30.  
-- \#/ _ _ _  
~~ V~' ' _->  
~~~ /  
~~~ _ _ _  
~~~ _/_/_/_/ Amazon Linux 2023, GA and supported until 2028-03-15.  
_/_/_/_/ https://aws.amazon.com/linux/amazon-linux-2023/  
[ec2-user@ip-10-200-0-126 ~]$ aws configure  
AWS Access Key ID [None]: AKIA4QBXNO6UUDXW3VKF  
AWS Secret Access Key [None]: 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  
Default region name [None]:  
Default output format [None]: json  
[ec2-user@ip-10-200-0-126 ~]$ aws s3 ls  
[ec2-user@ip-10-200-0-126 ~]$
```

Step 2: Use aws cli commands in order to create a bucket specifying the region.

```
[ec2-user@ip-10-200-0-126 ~]$ aws s3api create-bucket --bucket msis-aman-169 --region eu-west-2 --create-bucket-configuration LocationConstraint=eu-west-2  
{  
  "Location": "http://msis-aman-169.s3.amazonaws.com/",  
  "BucketArn": "arn:aws:s3:::msis-aman-169"  
}  
[ec2-user@ip-10-200-0-126 ~]$
```

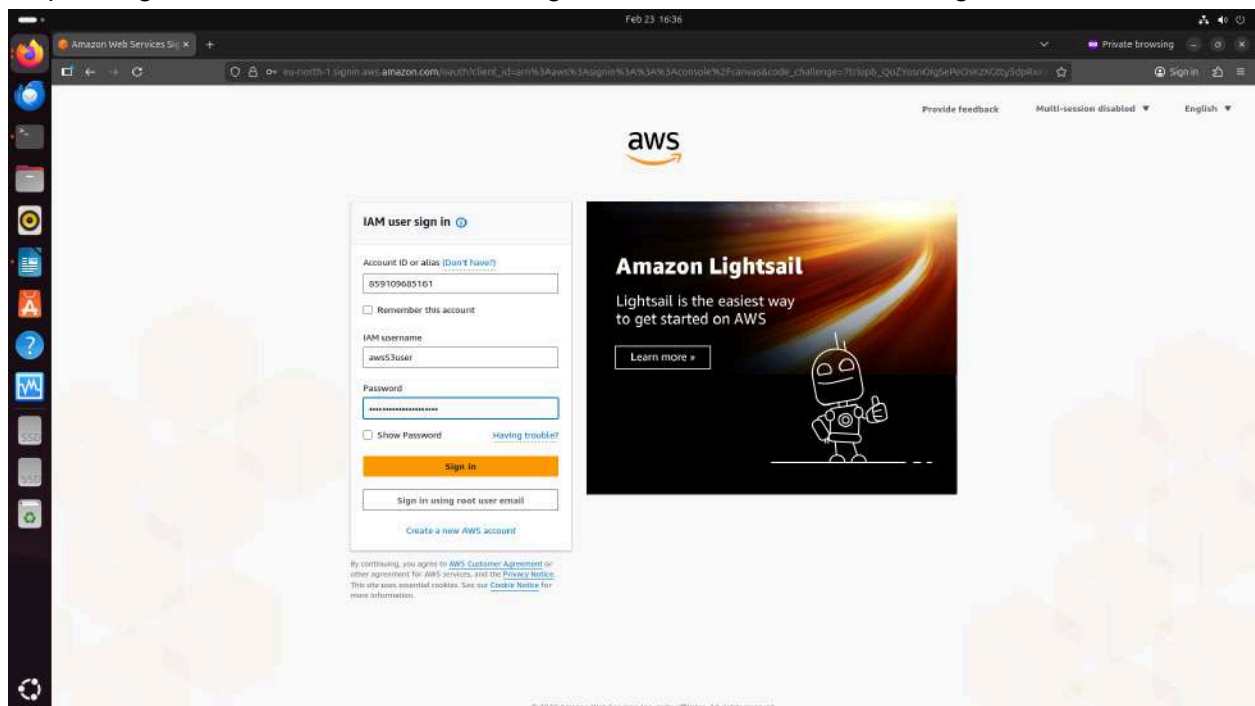
Step 3: Analyze details of an IAM user that has full access to Amazon S3



The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, Access reports, and IAM Identity Center. The main content area is titled 'Permissions policies (1)' and shows a list of policies attached to the user 'aws3user'. The policy 'lab_S3policy' is selected, and its details are displayed. The policy is a customer-managed policy attached directly to the user. The policy document is shown in a code editor, and the permissions boundary is set to 'None'. The policy document is as follows:

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Action": "s3:*",
6-       "Resource": "*",
7-       "Effect": "Allow"
8-     },
9-     {
10-      "Action": [
11-        "s3:objectLegalHold",
12-        "s3:objectRetention",
13-        "s3:bucketObjectLock"
14-      ],
15-      "Resource": "*",
16-      "Effect": "Deny"
17-    }
18-   ]
19- }
```

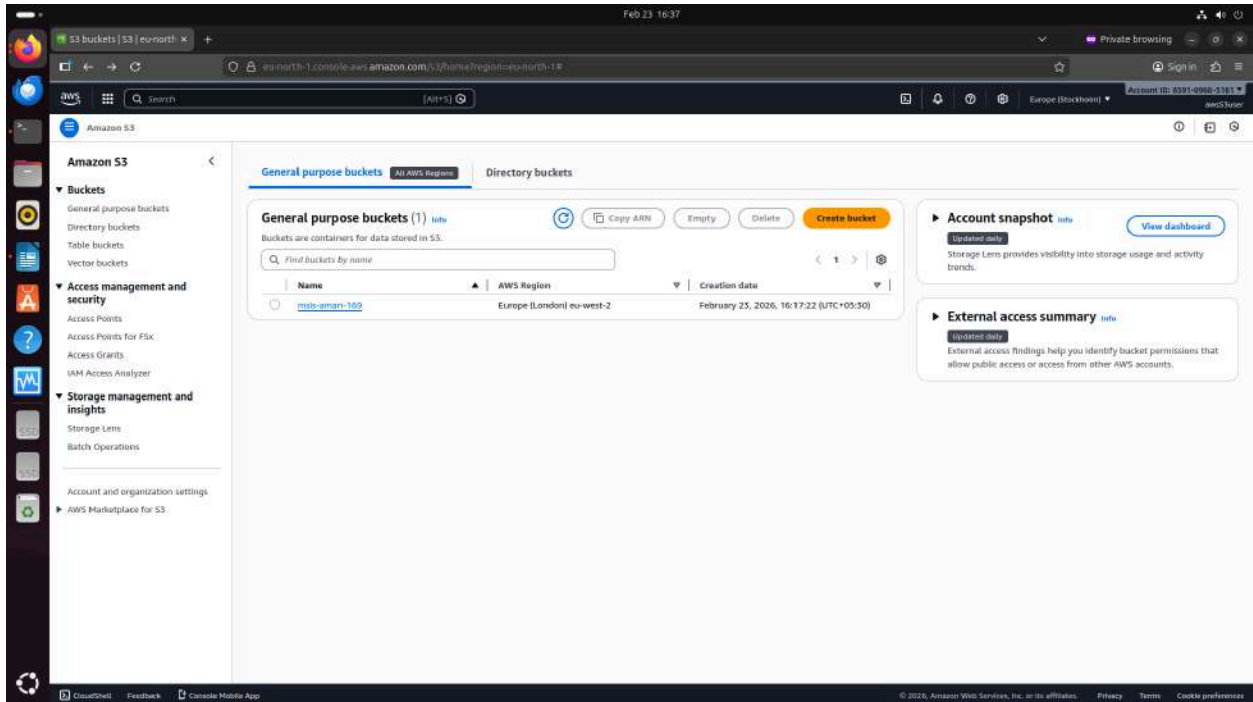
Step 4: Login as that IAM user in order to gain full access to the S3 management console



The screenshot shows the AWS IAM user sign-in page. The page has a dark background with the AWS logo at the top. On the left, there is a sign-in form with the following fields:

- Account ID or alias (Don't have?): 859109685161
- Remember this account: ☐
- IAM username: aws3user
- Password: [Redacted]
- Show Password: ☐ Having trouble?

Below the form are buttons for 'Sign in', 'Sign in using root user email', and 'Create a new AWS account'. On the right, there is a promotional banner for Amazon Lightsail with the text 'Amazon Lightsail' and 'Lightsail is the easiest way to get started on AWS'. The banner also includes a 'Learn more' button and a cartoon robot character.



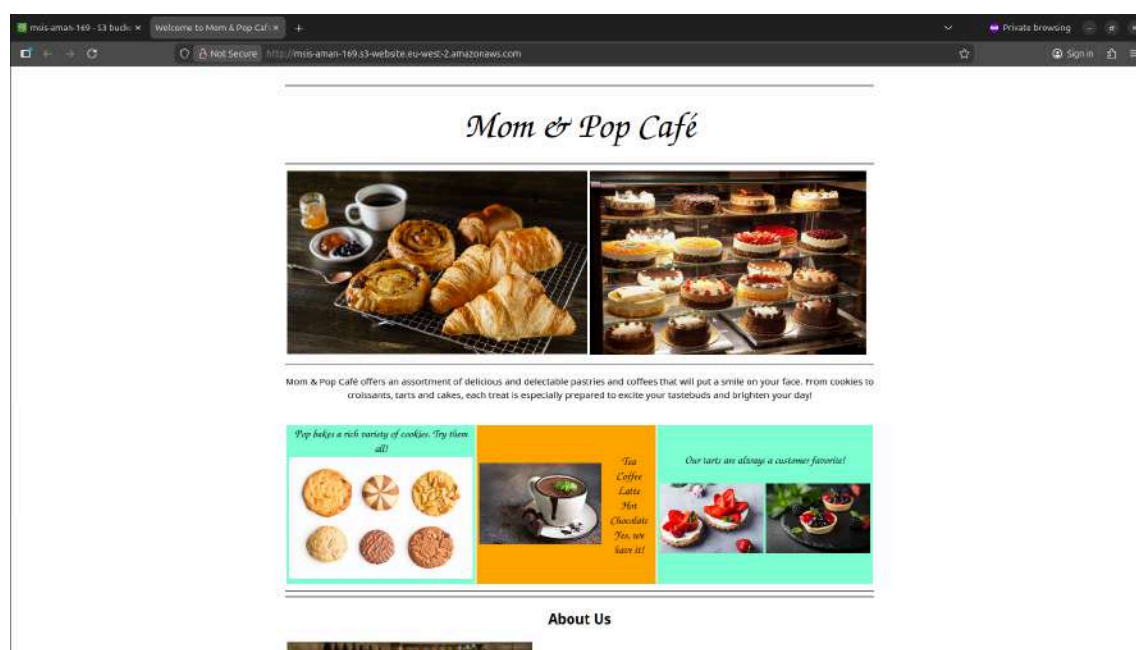
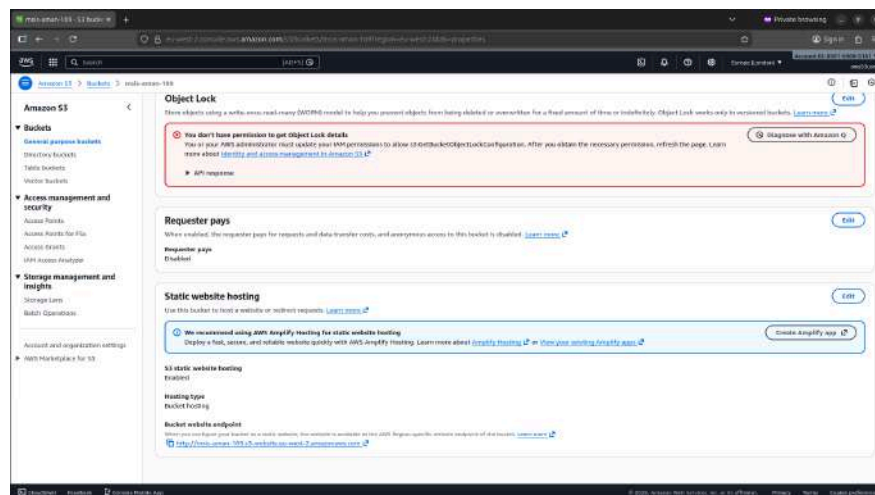
Step 5: cd into sysops-activity-files and extract the gunzip files and remove the gunzip folder once completed.

```
[ec2-user@ip-10-200-0-126 ~]$ cd ~/sysops-activity-files
[ec2-user@ip-10-200-0-126 sysops-activity-files]$
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ tar xvzf static-website.tar.gz
css/
css/styles.css
images/
images/Cake-Vitrine.png
images/Coffee-and-Pastries.png
images/Cookies.png
images/Cup-of-Hot-Chocolate.png
images/Mom-&-Pop-Coffee-Shop.png
images/Mom-&-Pop.png
images/Strawberry-&-Blueberry-Tarts.png
images/Strawberry-Tarts.png
index.html
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ ls
css  images  index.html  static-website.tar.gz
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ sudo rm -rf static-website.tar.gz
[ec2-user@ip-10-200-0-126 sysops-activity-files]$
```

Step 6: Run the first 2 commands to edit the ownership controls and ACL configuration for the S3 bucket. Then run the 3rd command to configure the bucket that you created earlier for static website hosting. This command identifies the index.html file as the index document. Finally run the 4th command to upload the files to the bucket.

```
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ aws s3api put-bucket-ownership-controls --bucket msis-aman-169 --ownership-controls "Rules=[{ObjectOwnership=BucketOwnerPreferred}]"
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ aws s3api put-public-access-block --bucket msis-aman-169 --public-access-block-configuration "BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets=false"
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ aws s3 website s3://msis-aman-169/ --index-document index.html
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ aws s3 cp . s3://msis-aman-169/ --recursive --acl public-read
upload: css/styles.css to s3://msis-aman-169/css/styles.css
upload: ./index.html to s3://msis-aman-169/index.html
upload: images/Mom-&Pop-Coffee-Shop.png to s3://msis-aman-169/images/Mom-&Pop-Coffee-Shop.png
upload: images/Cup-of-Hot-Chocolate.png to s3://msis-aman-169/images/Cup-of-Hot-Chocolate.png
upload: images/Coffee-and-Pastries.png to s3://msis-aman-169/images/Coffee-and-Pastries.png
upload: images/Mom-&Pop.png to s3://msis-aman-169/images/Mom-&Pop.png
upload: images/Strawberry-&Blueberry-Tarts.png to s3://msis-aman-169/images/Strawberry-&Blueberry-Tarts.png
upload: images/Cookies.png to s3://msis-aman-169/images/Cookies.png
upload: images/Cake-Vitrine.png to s3://msis-aman-169/images/Cake-Vitrine.png
upload: images/Strawberry-Tarts.png to s3://msis-aman-169/images/Strawberry-Tarts.png
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ aws s3 ls msis-aman-169
      PRE css/
      PRE images/
2025-02-23 11:11:46      3628 index.html
[ec2-user@ip-10-200-0-126 sysops-activity-files]$
```

Step 7: Open the console and check for the S3 bucket website url endpoint



Step 8: Create a batch file to make updating the website easily repeatable

```
ec2-user@ip-10-200-0-126:~  
GNU nano 2.9.8 update-website.sh  
#!/bin/bash  
  
aws s3 cp -/sysops-activity-files/ s3://msis-aman-169/ --recursive --acl public-read  
  
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ history  
1 aws configure  
2 aws s3 ls  
3 aws s3api create-bucket --bucket your-unique-bucket-name --region eu-west-2 --create-bucket-configuration LocationConstraint=eu-west-2  
4 aws s3api create-bucket --bucket your-unique-bucket-name --region eu-west-2 --create-bucket-configuration LocationConstraint=us-east-1  
5 aws s3api create-bucket --bucket msis-aman-169 --region eu-west-2 --create-bucket-configuration LocationConstraint=eu-west-2  
6 cd -/sysops-activity-files  
7 tar xvzf static-website.tar.gz  
8 ls  
9 sudo rm -rf static-website.tar.gz  
10 aws s3api put-bucket-ownership-controls  
11 --bucket <my-bucket>  
12 aws s3api put-bucket-ownership-controls  
13 --bucket <my-bucket>  
14 aws s3api put-bucket-ownership-controls --bucket msis-aman-169 --ownership-controls "Rules=[{ObjectOwnership=BucketOwnerPreferred}]"  
15 aws s3api put-public-access-block --bucket msis-aman-169 --public-access-block-configuration "BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets=false"  
16 aws s3 website s3://msis-aman-169/ --index-document index.html  
17 aws s3 cp . s3://msis-aman-169/ --recursive --acl public-read  
18 aws s3 ls msis-aman-169  
19 history  
[ec2-user@ip-10-200-0-126 sysops-activity-files]$ cd -  
[ec2-user@ip-10-200-0-126 ~]$ touch update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$ ls  
sysops-activity-files update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$ sudo nano update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$ #!/bin/bash  
[ec2-user@ip-10-200-0-126 ~]$  
[ec2-user@ip-10-200-0-126 ~]$ chmod +x update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$ sudo nano update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$
```

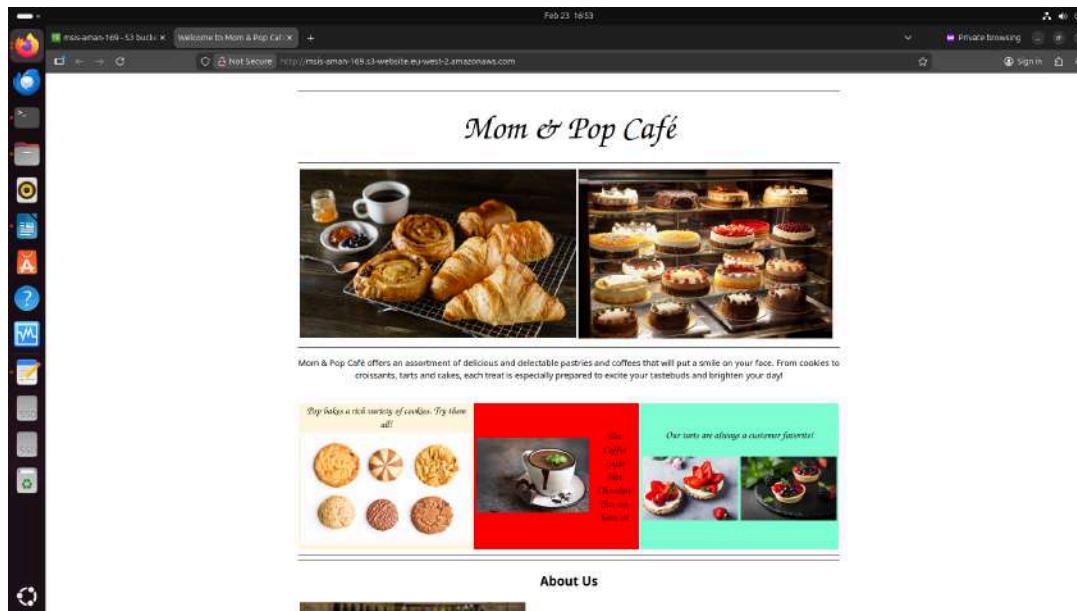
Step 9: Update the index.html file of the cafe website

```
<td bgcolor="cornsilk">  
  <div class="cursiveText">Pop bakes a rich variety of cookies. Try them all!</div>  
  <table>  
    <tr>  
      <td></td>  
    </tr>  
  </table>  
</td>  
<td bgcolor="red">  
  <table>
```

Step 10: Run the batch file to see if the updates take place in the website

```
[ec2-user@ip-10-200-0-126 ~]$ sudo nano sysops-activity-files/index.html  
[ec2-user@ip-10-200-0-126 ~]$ ls  
sysops-activity-files update-website.sh  
[ec2-user@ip-10-200-0-126 ~]$ ./update-website.sh  
upload: sysops-activity-files/css/styles.css to s3://msis-aman-169/css/styles.css  
upload: sysops-activity-files/index.html to s3://msis-aman-169/index.html  
upload: sysops-activity-files/images/Mom-&Pop-Coffee-Shop.png to s3://msis-aman-169/images/Mom-&Pop-Coffee-Shop.png  
upload: sysops-activity-files/images/Strawberry-Tarts.png to s3://msis-aman-169/images/Strawberry-Tarts.png  
upload: sysops-activity-files/images/Strawberry-&Blueberry-Tarts.png to s3://msis-aman-169/images/Strawberry-&Blueberry-Tarts.png  
upload: sysops-activity-files/images/Cup-of-Hot-Chocolate.png to s3://msis-aman-169/images/Cup-of-Hot-Chocolate.png  
upload: sysops-activity-files/images/Mom-&Pop.png to s3://msis-aman-169/images/Mom-&Pop.png  
upload: sysops-activity-files/images/Cookies.png to s3://msis-aman-169/images/Cookies.png  
upload: sysops-activity-files/images/Coffee-and-Pastries.png to s3://msis-aman-169/images/Coffee-and-Pastries.png  
upload: sysops-activity-files/images/Cake-Vitrine.png to s3://msis-aman-169/images/Cake-Vitrine.png  
[ec2-user@ip-10-200-0-126 ~]$
```

Website contents have changed accordingly.

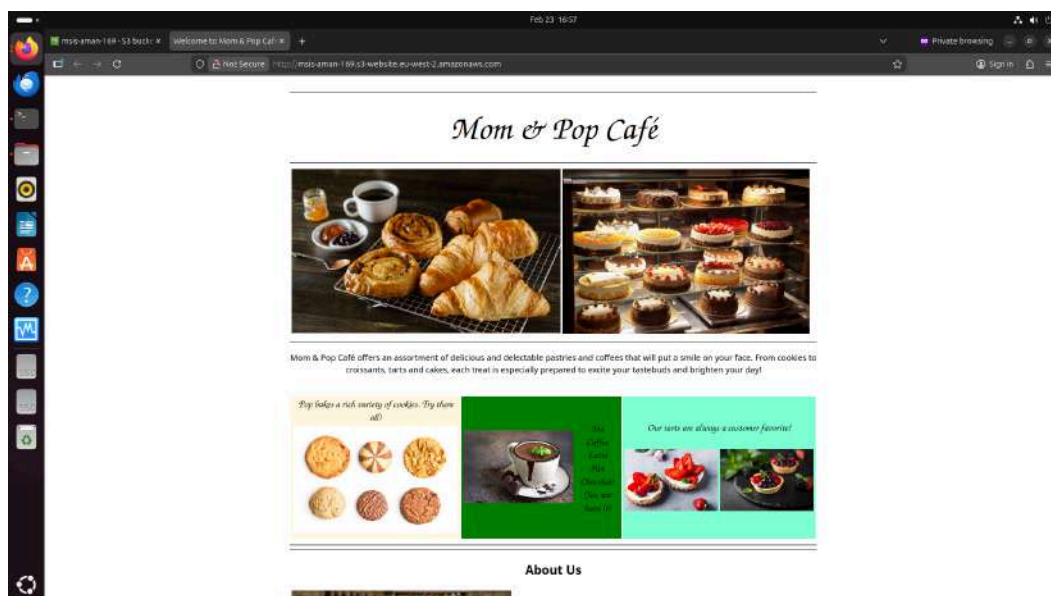


Step 11:

Use the **sync** command in the batch file as it is more efficient than the **cp** command.

```
ec2-user@ip-10-200-0-126:~$ nano update-website.sh
GNU nano 2.9.8
#!/bin/bash
aws s3 sync -/sysops-activity-files/ s3://msis-aman-169/ --acl public-read
```

```
[ec2-user@ip-10-200-0-126 ~]$ sudo nano update-website.sh
[ec2-user@ip-10-200-0-126 ~]$ ./update-website.sh
upload: sysops-activity-files/index.html to s3://msis-aman-169/index.html
```



Creating a dynamic website for cafe

Module 3 : Activity 3 : Troubleshoot create instance

Step 1: Download labusers.pem file and change its permissions

```
msis@k8s-master-node:~$ cd Downloads/
msis@k8s-master-node:~/Downloads$ ls
bird.txt
Cloud_Operations_CDC_5231_Assignments.docx
config.php
dashboard.php
db.php
'DFCL Lab Assignment 1.pdf'
Donuts.jpeg
ec2-user@100.53.144.111
ec2-user@98.93.97.81
Image2.jpg
index.php
init_db.sql
labsuser.pem
login.php
logout.php
mediacouser_accessKeys.csv
profile.php
pymysql-0.9.3.zip
salesAnalysisReportDataExtractor.zip
seed_users.php
task1.yaml
task2.yaml
task3.yaml
'template1(1).yaml'
template1.yaml
'Tools actually doing steganography.docx'
xampp-linux-x64-8.2.12-0-installer.run
msis@k8s-master-node:~/Downloads$ sudo chmod 600 labsuser.pem
[sudo] password for msis:
```

Step 2 : Connect to the CLI host using the pem file and the ssh command, once you have connected to the CLI host successfully configure your CLI host using the aws configure command and paste the Access Key , Secret Access key , Default Region and the Output file format

Step 3: Once you have successfully configured your CLI host check if any files or folders are present in your CLI host . You will find a sysops-activity-files folder inside which you will find a starters subfolder where you will find a create-lamp-instance shell file. Explore the contents of the shell file using the cat command.

```
[ec2-user@cli-host starters]$ cat create-lamp-instance.sh
#!/bin/bash
DATE=`date +%Y-%m-%d %H:%M:%S`
echo
echo "Running create-instance.sh on "$DATE
echo

# Hard coded values
region="eu-west-2"
echo "Region: "$region
instanceType="t2.small"
echo "Instance Type: "$instanceType
profile="default"
echo "Profile: "$profile

echo
echo "Looking up account values..."

# get vpcId
vpc=$(aws ec2 describe-vpcs \
--filters "Name=tag:Name,Values='MomPopCafe VPC'" \
--region $region \
--profile $profile | grep VpcId | cut -d '"' -f4 | sed -n 1p)
echo "VPC: "$vpc

# get subnetId
subnetId=$(aws ec2 describe-subnets \
--filters "Name=tag:Name,Values='MomPopCafe Public Subnet 1'" \
--region $region \
--profile $profile \
--query "Subnets[?contains(AvailabilityZone, 'eu-west-2a')]" | grep SubnetId | cut -d '"' -f4 | sed -n 1p)
echo "Subnet Id: "$subnetId

# Get keypair name
key=$(aws ec2 describe-key-pairs \
--profile $profile | grep KeyName | cut -d '"' -f4 )
echo "Key: "$key

# Get AMI ID
imageId=$(aws ssm get-parameters \
--names '/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2' \
--profile $profile \
--region $region | grep ami- | cut -d '"' -f4 | sed -n 2p)
```

Step 4: Once you have explored the shellscript run it and go through the output. You will observe that there is an error message follow the hints given in the lab description and perform necessary actions to fix the error.

Fix 1. Change the default region from eu-west-1 to us-east-1

fix 2: Change the default region in the instance create command from eu-west-1 to us-east-1

Fix3: Change the AZ from eu-west-1a to us-east-1a in the create subnet command

After making the above change run the shell script again

```

[ec2-user@cli-host starters]$ nano create-lamp-instance.sh
[ec2-user@cli-host starters]$ ./create-lamp-instance.sh

Running create-instance.sh on 2026-02-26 05:20:35

Region: us-east-1
Instance Type: t2.small
Profile: default

Looking up account values...
VPC: vpc-035d8eb2151b5400f
Subnet Id:
Key: vockey
AMI ID: ami-0199fa5fada510433

Creating a new security group...
Security Group: sg-0585396f75d165bea

Opening port 22 in the new security group
Opening port 80 in the new security group

Creating an EC2 instance in us-east-1
Note: AWS CLI version 2, the latest major version of the AWS CLI, is now stable and recommended for general use.
ns at: https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2.html

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

    aws help
    aws <command> help
    aws <command> <subcommand> help
aws: error: argument --subnet-id: expected one argument
[ec2-user@cli-host starters]$ nano create-lamp-instance.sh
[ec2-user@cli-host starters]$ ./create-lamp-instance.sh

Running create-instance.sh on 2026-02-26 05:22:19

Region: us-east-1
Instance Type: t2.small
Profile: default

Looking up account values...
VPC: vpc-035d8eb2151b5400f

```

When prompted if you would like to delete the already existing security group press yes

```
Looking up account values...
VPC: vpc-035d8eb2151b5400f
Subnet Id: subnet-0e5cfa54777532f9e
Key: vockey
AMI ID: ami-0199fa5fada510433

WARNING: Found existing security group with name sg-0585396f75d165bea.
This script will not succeed if it already exists.
Would you like to delete it? [Y/N]
>>
y

Deleting the existing security group...

Creating a new security group...
Security Group: sg-059c93d7d92eb800c

Opening port 22 in the new security group
Opening port 80 in the new security group

Creating an EC2 instance in us-east-1

Instance Details....
{
  "Groups": [],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "CapacityReservationSpecification": {
        "CapacityReservationPreference": "open"
      },
      "ClientToken": "eaa72eaf-ca14-4fd5-8110-9e8ec4da9dbe",
      "CpuOptions": {
        "CoreCount": 1,
        "ThreadsPerCore": 1
      },
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "ImageId": "ami-0199fa5fada510433",
```

Your shell script will run successfully now

```

    "GroupName": "mompopcafeSG"
  }
},
"SourceDestCheck": true,
"State": {
  "Code": 0,
  "Name": "pending"
},
"StateReason": {
  "Code": "pending",
  "Message": "pending"
},
"StateTransitionReason": "",
"SubnetId": "subnet-0e5cfa54777532f9e",
"Tags": [
  {
    "Key": "Name",
    "Value": "mompopcafeserver"
  }
],
"VirtualizationType": "hvm",
"VpcId": "vpc-035d8eb2151b5400f"
}
],
"OwnerId": "i32783143281",
"ReservationId": "r-026d8ea91ca7a0179"
}
instanceId=i-0c6872e9993db1e69

Waiting for a public IP for the new instance...

The public IP of your LAMP instance is: 3.89.81.50

Download the Key Pair from the Qwiklabs page.

Then connect using this command (with .pem or .ppk added to the end of the keypair name):
ssh -i path-to/vockey ec2-user@3.89.81.50

The website should also become available at
http://3.89.81.50/mompopcafe/

Done running create-instance.sh at 2026-02-26 05:22:56

```

Step 6: Check if your webser is loading
if now use the nmap command and check the list of ports that are enabled


```
[ec2-user@cli-host starters]$ sudo yum install -y nmap
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core                               | 3.6 kB    00:00
Resolving Dependencies
--> Running transaction check
--> Package nmap.x86_64 2:6.40-19.amzn2.0.1 will be installed
--> Processing Dependency: nmap-ncat = 2:6.40-19.amzn2.0.1 for package: 2:nmap-6.40-19.amzn2.0.1.x86_64
--> Running transaction check
--> Package nmap-ncat.x86_64 2:6.40-19.amzn2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package                Arch          Version              Repository           Size
=====
Installing:
nmap                   x86_64        2:6.40-19.amzn2.0.1  amzn2-core           4.0 M
Installing for dependencies:
nmap-ncat              x86_64        2:6.40-19.amzn2.0.1  amzn2-core           204 k

Transaction Summary
=====
Install 1 Package (+1 Dependent package)

Total download size: 4.2 M
Installed size: 16 M
Downloading packages:
(1/2): nmap-ncat-6.40-19.amzn2.0.1.x86_64.rpm | 204 kB  00:00
(2/2): nmap-6.40-19.amzn2.0.1.x86_64.rpm      | 4.0 MB  00:00
-----
Total                                          24 MB/s | 4.2 MB  00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 2:nmap-ncat-6.40-19.amzn2.0.1.x86_64 1/2
  Installing : 2:nmap-6.40-19.amzn2.0.1.x86_64      2/2
  Verifying   : 2:nmap-6.40-19.amzn2.0.1.x86_64     1/2
  Verifying   : 2:nmap-ncat-6.40-19.amzn2.0.1.x86_64 2/2

Installed:
nmap.x86_64 2:6.40-19.amzn2.0.1
```

You will observe that port 80 is not enabled go to your webserver's security group and make the necessary changes to enable port 80

```
[ec2-user@cli-host starters]$ nmap -Pn 3.89.81.50

Starting Nmap 6.40 ( http://nmap.org ) at 2026-02-26 05:24 UTC
Nmap scan report for ec2-3-89-81-50.compute-1.amazonaws.com (3.89.81.50)
Host is up (0.0018s latency).
Not shown: 998 filtered ports
PORT      STATE SERVICE
22/tcp    open  ssh
8080/tcp   closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 4.84 seconds
```

Inbound rules (3)										
Search										
☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description		
☐	-	sgr-084278cd8ed6e9da	IPv4	Custom TCP	TCP	8080	0.0.0.0/0	-		
☐	-	sgr-09f6838640b580cbc	IPv4	SSH	TCP	22	0.0.0.0/0	-		
☐	-	sgr-0cfc4c610b34025da	IPv4	HTTP	TCP	80	0.0.0.0/0	-		

Now try to access your webserver

Module 7 - Activity 7 - Troubleshoot VPC

```

amansanill@Amans-MacBook-Air Downloads % sudo chmod 600 labsuser.pem
amansanill@Amans-MacBook-Air Downloads % sudo ssh -i labsuser.pem ec2-user@54.147.185.246
The authenticity of host '54.147.185.246 (54.147.185.246)' can't be established.
ED25519 key fingerprint is SHA256:DmQbsDYA1ng6uT6ZsC1HvZNgFfqoBKZud7VyhxH8cVY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.147.185.246' (ED25519) to the list of known hosts.

  #_
~\_ #####_      Amazon Linux 2
~~ \_#####\
~~   \###|      AL2 End of Life is 2026-06-30.
~~     \#/
~~      V~' ' ~->
      ~~~
      ~~~_._~_/_
      _/_/_/_/_
      _/m/'_

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

```

```
[ec2-user@cli-host ~]$ aws configure
AWS Access Key ID [None]: AKIA4GVUMWQ3URYLLZUZ
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@cli-host ~]$ aws s3 ls
[ec2-user@cli-host ~]$ aws s3api create-bucket --bucket flowlog7272 --region us-east-1
```

```
[ec2-user@cli-host ~]$ aws s3api create-bucket --bucket aman-msis-7272 --region us-east-1
{
  "Location": "/aman-msis-7272"
}
```

Step 4: Run the first command to get the VPC ID for VPC1, which you must have to enable VPC Flow Logs. Run the 2nd command enable VPC Flow Logs on VPC1. Run the 3rd command to confirm the flow log was created.

```
[ec2-user@cli-host ~]$ aws ec2 describe-vpcs --query 'Vpcs[*].[VpcId,Tags[?Key==`Name`].Value,CidrBlock]' --filters "Name=tag:Name,Values='VPC1'"
[
  [
    "vpc-055a348bd6d673bec",
    [
      "VPC1"
    ],
    "10.0.0.0/16"
  ]
]
[ec2-user@cli-host ~]$ aws ec2 create-flow-logs --resource-type VPC --resource-ids vpc-055a348bd6d673bec --traffic-type ALL --log-destination-type s3 --log-destination arn:aws:s3:::aman-msis-7272
{
  "Unsuccessful": [],
  "FlowLogIds": [
    "fl-05fe72969961433f9"
  ],
  "ClientToken": "84yKg9TySqmwGtj3m42SRNg4V9reNB4LJbgnNej9Y5k="
}
[ec2-user@cli-host ~]$ aws ec2 describe-flow-logs
{
  "FlowLogs": [
    {
      "LogDestinationType": "s3",
      "Tags": [],
      "ResourceId": "vpc-055a348bd6d673bec",
      "CreationTime": "2026-02-26T13:19:09.735Z",
      "TrafficType": "ALL",
      "FlowLogStatus": "ACTIVE",
      "LogFormat": "${version} ${account-id} ${interface-id} ${srcaddr} ${dstaddr} ${srcport} ${dstport} ${protocol} ${packets} ${bytes} ${start} ${end} ${action} ${log-status}",
      "FlowLogId": "fl-05fe72969961433f9",
      "MaxAggregationInterval": 600,
      "LogDestination": "arn:aws:s3:::aman-msis-7272",
      "DeliverLogsStatus": "SUCCESS"
    }
  ]
}
```

Step 5: In the terminal where you are connected to the CLI Host, run the following command to find details about the web server instance. Filter the results on the client side—by using the query parameter—to return only relevant details.

The following command will return only the state of the instance, the private IP address, the instance ID, the security groups that are applied to it, the subnet in which it runs, and the key pair name that is associated with it.

```
[ec2-user@cli-host ~]$ aws ec2 describe-instances --filter "Name=ip-address,Values='98.91.1.60'" --query 'Reservations[*].Instances[*].[State,PrivateIpAddress,InstanceId,SecurityGroups,SubnetId,KeyName]'
[
  [
    {
      "Code": 16,
      "Name": "running"
    },
    "10.0.1.188",
    "i-0d4beff61ace7a47e",
    [
      {
        "GroupName": "c193507a4973254113510677t1w838970684471-WebSecurityGroup-yqy17vXFw3wY",
        "GroupId": "sg-0e3b5576f84332183"
      }
    ],
    "subnet-0f8153504b575e802",
    "vockey"
  ]
]
```

Step 6: Try to establish an SSH connection to the web server instance, it gives a timeout message. Now try trouble shooting by

- 1) Use the nmap utility to check what ports are open on the web server EC2 instance.
- 2) Check the security group details by using the **aws ec2 describe-security-groups** command.
- 3) Check the route table settings for the route table that is associated with the subnet where the web server is running.

```
amansanill@Amans-MacBook-Air Downloads % sudo ssh -i labsuser.pem ec2-user@98.91.1.60 -o ConnectTimeout=10
Password:
ssh: connect to host 98.91.1.60 port 22: Operation timed out
```

You will find that there is no route to the internet gateway and that is why the website can not be accessed via the internet.


```

    },
    ],
    "Routes": [
      {
        "GatewayId": "local",
        "DestinationCidrBlock": "10.0.0.0/16",
        "State": "active",
        "Origin": "CreateRouteTable"
      }
    ],
    "OwnerId": "838970684471"
  }
]

```

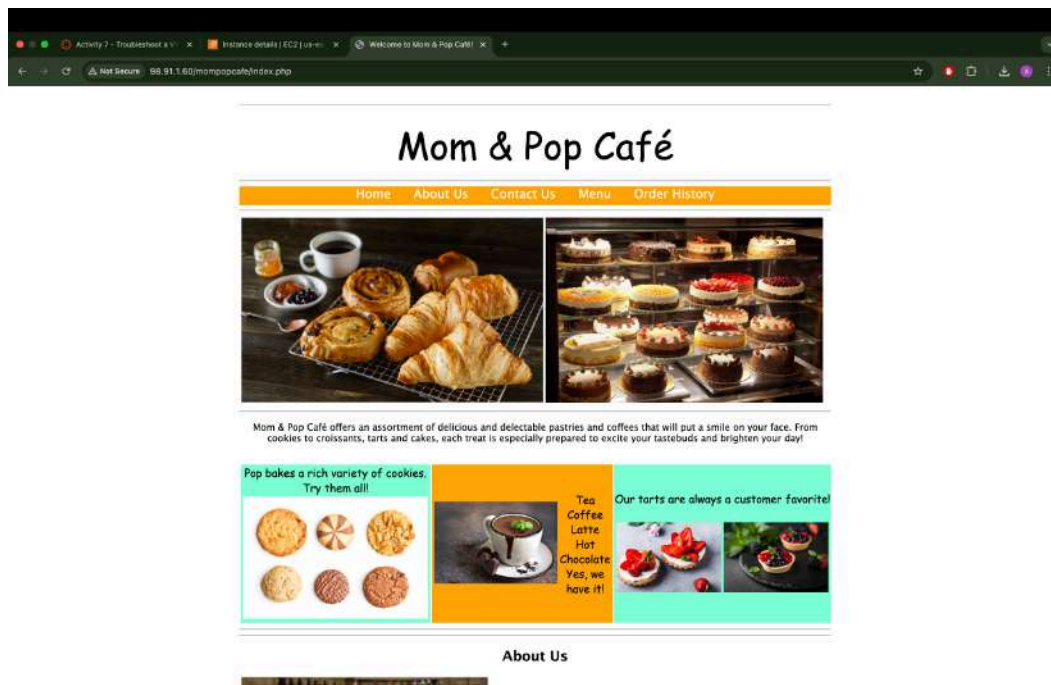
Step 7: **aws ec2 describe-route-tables** , gets the route table id
aws ec2 describe-internet-gateways , gets the internet gateway id. Finally create a route in the route table to direct all internet bound traffic to the internet gateway.

```

[ec2-user@cli-host ~]$ aws ec2 create-route \
> --route-table-id rtb-0768cc961d0f45793 \
> --destination-cidr-block 0.0.0.0/0 \
[> --gateway-id igw-09061520e0ba82330
{
  "Return": true
}

```

Now the website is accessible over http.



Step 8: Try to ssh into the webserver instance now, it still wont work, but we've already checked the ports, security group and route tables. So now try checking the Network ACL, see that ssh access on port 22 is denied.

```
[ec2-user@cli-host ~]$ aws ec2 describe-network-acls --filter "Name=association.subnet-id,Values='subnet-0f8153504b575e802'" --query 'NetworkAcls[*].[NetworkAclId,Entries]'
[
  [
    "acl-08ff844c564088783",
    [
      {
        "RuleNumber": 100,
        "Protocol": "-1",
        "Egress": true,
        "CidrBlock": "0.0.0.0/0",
        "RuleAction": "allow"
      },
      {
        "RuleNumber": 32767,
        "Protocol": "-1",
        "Egress": true,
        "CidrBlock": "0.0.0.0/0",
        "RuleAction": "deny"
      },
      {
        "RuleNumber": 40,
        "Protocol": "6",
        "PortRange": {
          "To": 22,
          "From": 22
        },
        "Egress": false,
        "RuleAction": "deny",
        "CidrBlock": "0.0.0.0/0"
      }
    ]
  ]
]
```

Step 9: Delete this rule.

```
[ec2-user@cli-host ~]$ aws ec2 delete-network-acl-entry \
> --network-acl-id acl-08ff844c564088783 \
> --rule-number 40 \
[> --ingress
```

Now we are able to ssh into our webserver without any problem.

```
amansanill@Amans-MacBook-Air Downloads % sudo ssh -i labsuser.pem ec2-user@98.91.1.60 -o ConnectTimeout=10
Password:
The authenticity of host '98.91.1.60 (98.91.1.60)' can't be established.
ED25519 key fingerprint is SHA256:9yUqtriHHJrR/ne2qZ8iKI0pmpnt92xnmxAkQaqZ5Wk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.91.1.60' (ED25519) to the list of known hosts.

#_
~\_ #####_ Amazon Linux 2
~~ \_#####\
~~ \###| AL2 End of Life is 2026-06-30.
~~ \#/
~~ V~' -->
~~~
~~~. _/ _/ A newer version of Amazon Linux is available!
_/_/_/
_/m/' https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@web-server ~]$
```

Step 10: Now ssh back into the cli host. Now we have created some interesting entries in the flow logs that we created when we enabled VPC Flow Logs at the beginning of this activity. In this final task, we will query the flow logs to observe the activities that they capture. Create a local directory on the CLI Host where you can download the flow log files:

mkdir flowlogs

Change the directory to the new directory:

cd flowlogs

List the S3 buckets to recall the bucket name:

aws s3 ls

Download the flow logs by running the following command (replace <flowlog####> with the actual bucket name).

aws s3 cp s3://<flowlog####>/ . --recursive

If the command is successful, you should see that many files are downloaded.

```
[ec2-user@cli-host ~]$ mkdir flowlogs
[ec2-user@cli-host ~]$
[ec2-user@cli-host ~]$
[ec2-user@cli-host ~]$ cd flowlogs/
[ec2-user@cli-host flowlogs]$ aws s3 ls
2026-02-26 13:17:16 aman-msis-7272
[ec2-user@cli-host flowlogs]$
[ec2-user@cli-host flowlogs]$
[ec2-user@cli-host flowlogs]$ aws s3 cp s3://aman-msis-7272/ . --recursive
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_e4e8def1.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_e4e8def1.log.gz
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_e5ec338b.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_e5ec338b.log.gz
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_11e93a1e.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_11e93a1e.log.gz
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log.gz
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa0dbe3.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa0dbe3.log.gz
download: s3://aman-msis-7272/AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_72071c1b.log.gz to AWSLogs/838970684471/vpcflowlogs/us-east-1/2026/02/26/838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_72071c1b.log.gz
```

Use the **cd** command and **ls** commands repeatedly (or **cd** followed by pressing **TAB** multiple times) as need. The logs will be located in an `AWSLogs/<account-num>/vpcflowlogs/<region>/yyyy/mm/dd` subdirectory.

Notice that the file names all end in `log.gz`, which indicates that they are compressed as GNU zip files.

```
[ec2-user@cli-host flowlogs]$ ls
AWSLogs
[ec2-user@cli-host flowlogs]$ cd AWSLogs/
[ec2-user@cli-host AWSLogs]$ ls
838970684471
[ec2-user@cli-host AWSLogs]$ cd 838970684471/
[ec2-user@cli-host 838970684471]$ ls
vpcflowlogs
[ec2-user@cli-host 838970684471]$ cd vpcflowlogs/
[ec2-user@cli-host vpcflowlogs]$ ls
us-east-1
[ec2-user@cli-host vpcflowlogs]$ cd us-east-1/
[ec2-user@cli-host us-east-1]$ ls
2026
[ec2-user@cli-host us-east-1]$ cd 2026/
[ec2-user@cli-host 2026]$ ls
02
[ec2-user@cli-host 2026]$ cd 02
[ec2-user@cli-host 02]$ ls
26
[ec2-user@cli-host 02]$ cd 26
[ec2-user@cli-host 26]$ ls
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_999963b3.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_e4e8def1.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_11e93a1e.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_bc3a8962.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1330Z_52e28b3a.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1330Z_d97c969f.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_98b0dc2b.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_e5ec338b.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_72071c1b.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_7fcbe5e5.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa00dbe3.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_ffea471.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_a04e7a92.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_e47de3d9.log.gz
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1355Z_03176e8f.log.gz
[ec2-user@cli-host 26]$
```

Run this command to extract the logs:

gunzip *.gz

Run **ls** again.

```
[ec2-user@cli-host 26]$ gunzip *.gz
[ec2-user@cli-host 26]$
[ec2-user@cli-host 26]$ ls
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_999963b3.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1320Z_e4e8def1.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_11e93a1e.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_bc3a8962.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1330Z_52e28b3a.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1330Z_d97c969f.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_98b0dc2b.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1335Z_e5ec338b.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_72071c1b.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_7fcbe5e5.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa00dbe3.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_ffea471.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_a04e7a92.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_e47de3d9.log
838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1355Z_03176e8f.log
```


Enter **head** in the terminal window, followed by a space, and then paste the copied file name. Run the command.

Notice that the header row indicates the kind of data that each log entry contains. Each entry contains information, such as the IP address of the source of the event (in the fourth column), the destination port (seventh column), start and end timestamps (in Unix timestamp format), and the action that resulted (one of ACCEPT or REJECT).

```
[ec2-user@cli-host 26]$ head 838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log
version account-id interface-id srcaddr dstaddr srcport dstport protocol packets bytes start end action log-status
2 838970684471 eni-0e0b800df6ee9e74e 92.63.197.66 10.0.1.188 8080 7226 6 1 40 1772111958 1772111989 REJECT OK
2 838970684471 eni-0e0b800df6ee9e74e 206.168.35.158 10.0.1.188 17375 22622 6 1 60 1772111958 1772111989 REJECT OK
2 838970684471 eni-0e0b800df6ee9e74e 107.172.7.140 10.0.1.188 49372 22 6 1 40 1772111958 1772111989 REJECT OK
```

Run a **grep** command that looks in each log file in the current directory, and returns lines that contain the word **REJECT**:

grep -rn REJECT

This command should return a large dataset because it includes every event where the VPC settings rejected the request.

Find out how many records were returned:

grep -rn REJECT . | wc -l

The result shows the number of lines in your result set.

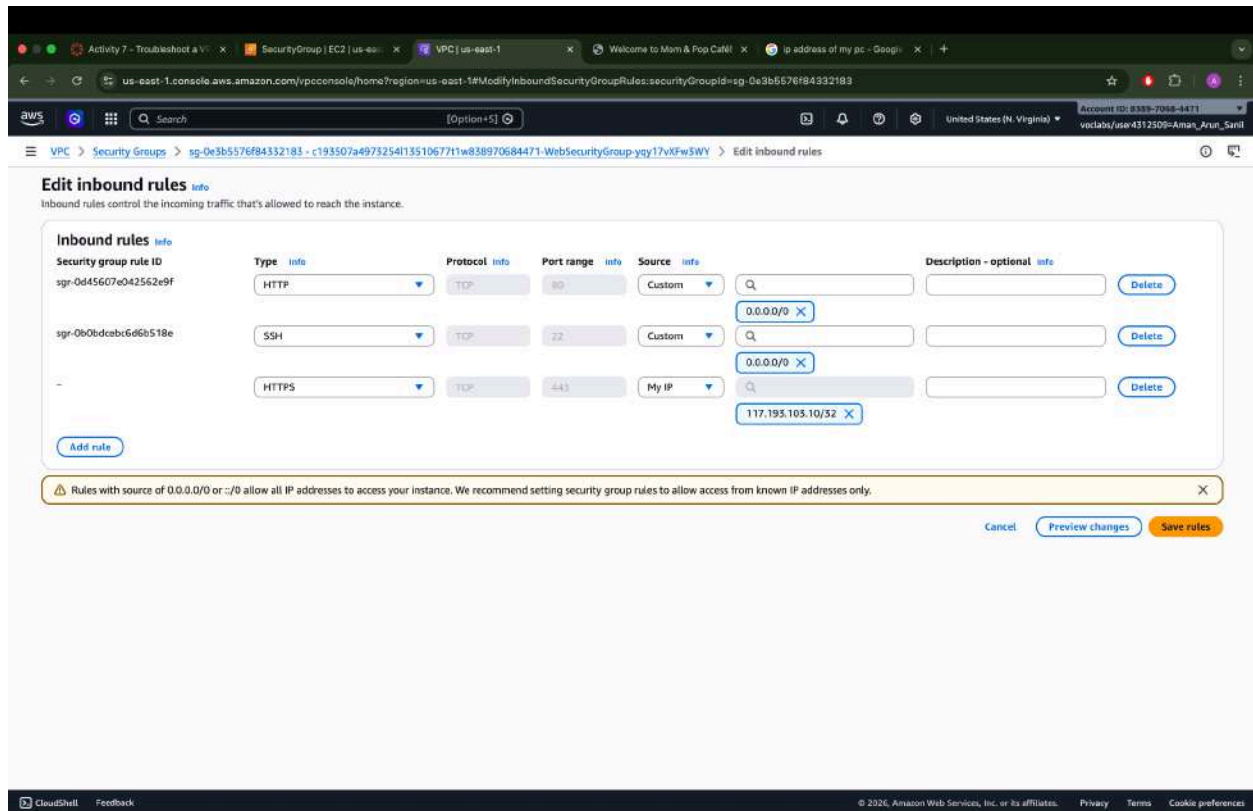
Refine your search by only looking for lines that contain 22 (which is the port number where you attempted to connect to the web server when access was blocked):

grep -rn '22' . | grep REJECT

This command should return a smaller number of results.

```
[ec2-user@cli-host 26]$ grep -rn REJECT . | wc -l
688
[ec2-user@cli-host 26]$ grep -rn '22' . | grep REJECT
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_11e93a1e.log:24:2 838970684471 eni-005c946744571e54d 34.47.117.84 10.0.1.146 2633
8 22 6 1 48 1772112277 1772112316 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1336Z_52e29b3a.log:18:2 838970684471 eni-0e0b800df6ee9e74e 117.193.103.10 10.0.1.188 50
105 22 6 7 448 1772112685 1772112640 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1340Z_7fcbefe5.log:74:2 838970684471 eni-0e0b800df6ee9e74e 45.148.10.192 10.0.1.188 345
35 22 6 1 40 1772113449 1772113455 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_ffea471.log:6:2 838970684471 eni-005c946744571e54d 45.148.10.240 10.0.1.146 5906
2 22 6 1 40 1772113555 1772113605 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_ffea471.log:66:2 838970684471 eni-0e0b800df6ee9e74e 117.193.103.10 10.0.1.188 50
218 22 6 7 448 1772113683 1772113717 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa00dbe3.log:57:2 838970684471 eni-0e0b800df6ee9e74e 194.180.48.63 10.0.1.188 407
87 22 6 1 40 1772113539 1772113573 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_aa00dbe3.log:66:2 838970684471 eni-005c946744571e54d 45.148.10.240 10.0.1.146 515
41 22 6 1 40 1772113536 1772113574 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_e47de3d9.log:37:2 838970684471 eni-0e0b800df6ee9e74e 64.227.7.165 10.0.1.188 4895
1 22 6 1 48 1772113962 1772113993 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1350Z_e47de3d9.log:78:2 838970684471 eni-005c946744571e54d 64.227.7.165 10.0.1.146 4895
1 22 6 1 48 1772113957 1772113998 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1315Z_0ee6208a.log:4:2 838970684471 eni-0e0b800df6ee9e74e 107.172.7.140 10.0.1.188 4937
2 22 6 1 40 1772111958 1772111989 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1325Z_bc3a8962.log:69:2 838970684471 eni-005c946744571e54d 46.62.238.59 10.0.1.146 5027
22 6 1 48 1772112480 1772112513 REJECT OK
[ec2-user@cli-host 26]$
```

Step 11: Create a fake inbound rule for the webserver security group, eg:- https access on my ip and note down the ip. Later delete the rule.



Back in the CLI Host SSH terminal session, run a more refined query on the flow logs (replace <ip-address> with the IP address that you copied):

grep -rn '22' . | grep REJECT | grep <ip-address>

The number of lines in the result set should now match the number of times you tried and failed to use SSH to connect the web server instance.

Notice that the elastic network interface ID is in each of the log entries that were returned by your query. Run the following command, the result set should confirm that the network interface ID that is recorded in the flow log matches the network interface that is assigned to the web server instance (as part of the network interface).

```
[ec2-user@cli-host 26]$ grep -rn '22' . | grep REJECT | grep 117.193.103.10
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1330Z_52e28b3a.log:18:2 838970684471 eni-0e0b800df6ee9e74e 117.193.103.10 10.0.1.188 50
105 22 6 7 448 1772112605 1772112640 REJECT OK
./838970684471_vpcflowlogs_us-east-1_fl-05fe72969961433f9_20260226T1345Z_ffea3471.log:66:2 838970684471 eni-0e0b800df6ee9e74e 117.193.103.10 10.0.1.188 50
218 22 6 7 448 1772113683 1772113717 REJECT OK
[ec2-user@cli-host 26]$ aws ec2 describe-network-interfaces --filters "Name=association.public-ip,Values='98.91.1.60'" --query 'NetworkInterfaces[*].[NetworkInterfaceId, Association.PublicIp]'
[
  [
    "eni-0e0b800df6ee9e74e",
    "98.91.1.60"
  ]
]
```

Run the **date** command to compare the result to the current time.

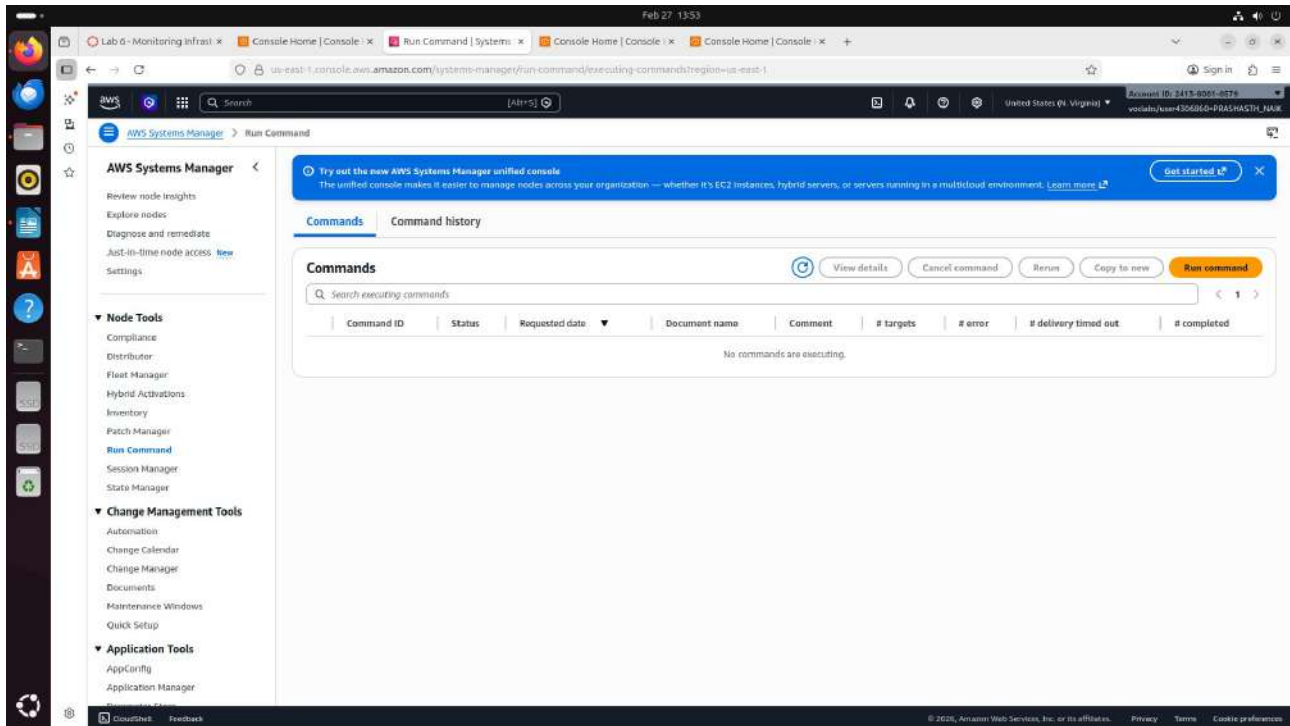
```
[ec2-user@cli-host 26]$ date -d @1772113683  
Thu Feb 26 13:48:03 UTC 2026
```

Question 4

Module 9 – Lab 6 (Monitoring Infrastructure)

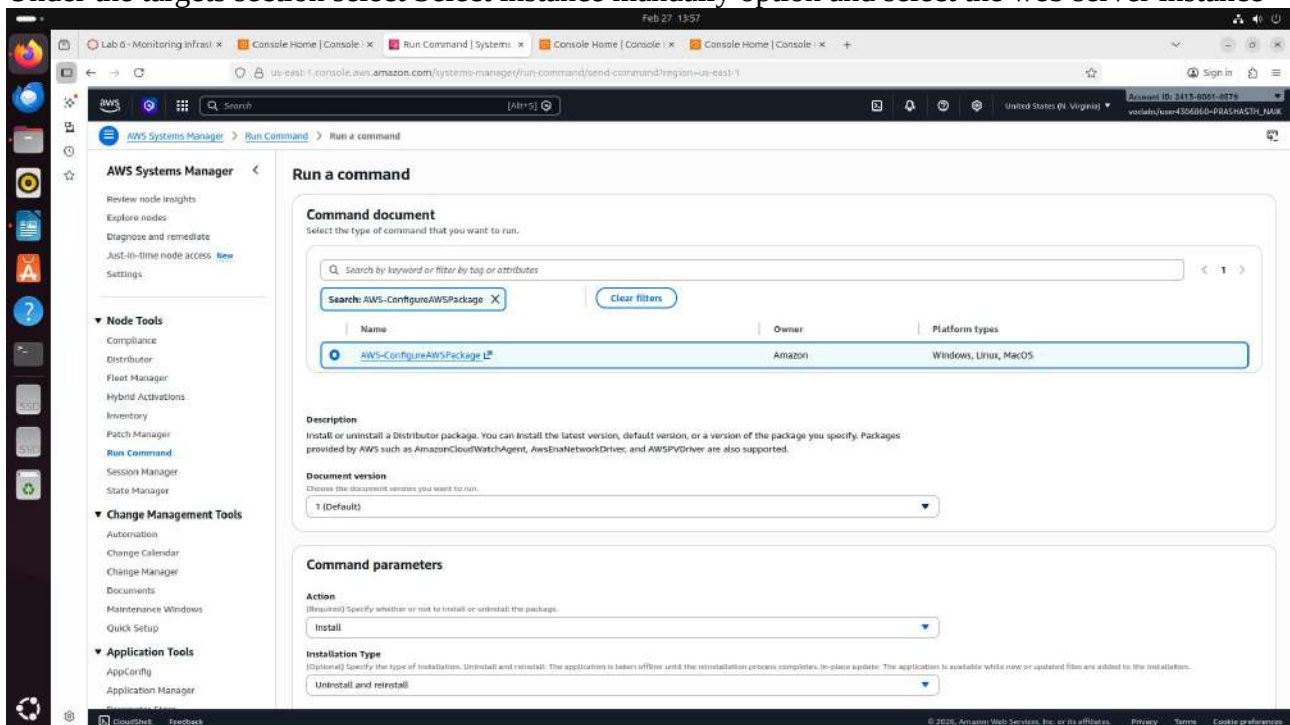
Step 1:

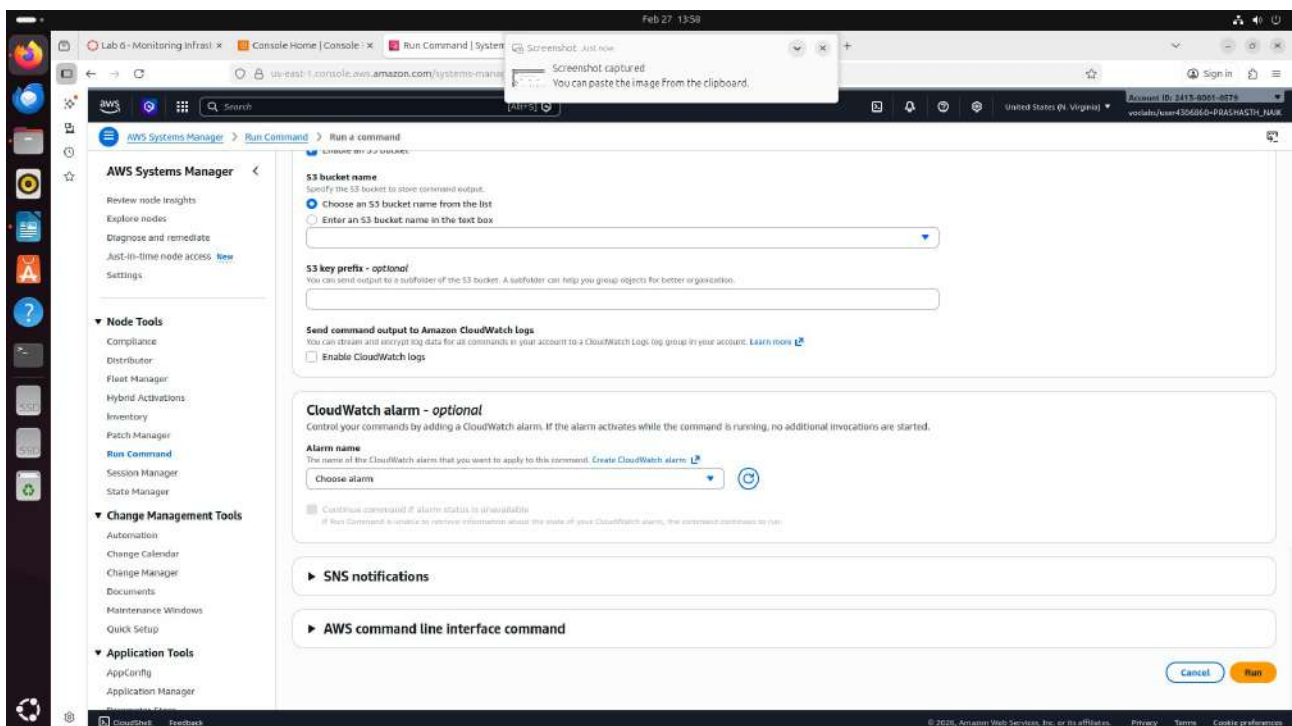
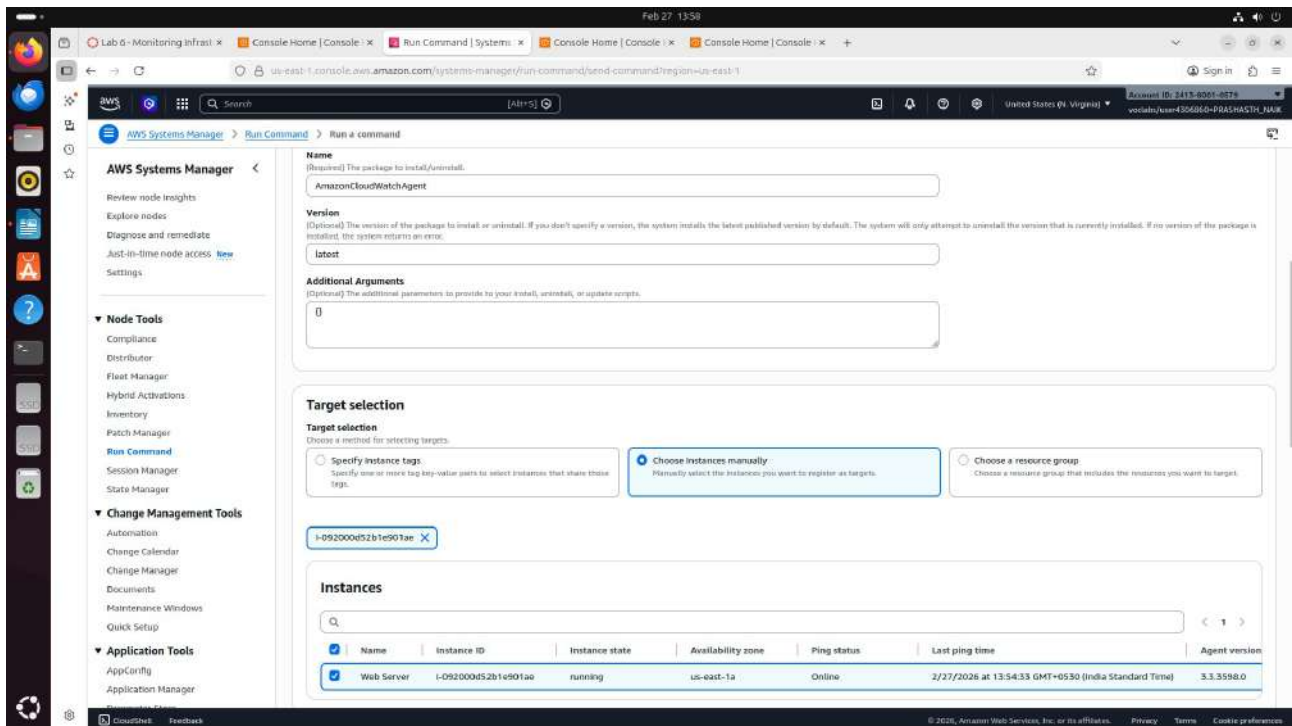
In the AWS services list select Systems Manager
Select the Run Command option on the left panel



Step 2:

Select the AWS-ConfigureAWSPackage Radio button and fill the actions, name and version fields.
Under the targets section select Select instance manually option and select the web server instance





Click the run button

Step 3:
Under the targets and output section select the instanceID

Lab 6 - Monitoring Infrast

Console Home | Console

Run Command | Systems

Console Home | Console

Console Home | Console

us-east-1.console.aws.amazon.com/systems-manager/run-command/3f8cfb3e-deab-47b7-a5a8-8f2a5a27752c?region=us-east-1

Sign in

Account ID: 2415-8301-0276
vocalis/iam43668d-0-P0A5HAG5tr_HAWK

AWS Systems Manager

Review node insights

Explore nodes

Diagnose and remediate

Just-in-time node access

Settings

Node Tools

Compliance

Distributor

Fleet Manager

Hybrid Activations

Inventory

Patch Manager

Run Command

Session Manager

State Manager

Change Management Tools

Automation

Change Calendar

Change Manager

Documents

Maintenance Windows

Quick Setup

Application Tools

AppConfig

Application Manager

Command ID: 3f8cfb3e-deab-47b7-a5a8-8f2a5a27752c was successfully sent!

Command ID: 3f8cfb3e-deab-47b7-a5a8-8f2a5a27752c

Cancel command

Run

Copy to new

Command status

Overall status: Success

Detailed status: Success

targets: 1

completed: 1

error: 0

delivery timed out: 0

Targets and outputs

Search command invocations

View output

Instance ID	Instance name	Status	Detailed Status	Start time	Finish time
i-092000e52b1e901ae	ip-10-0-0-214.ec2.internal	Success	Success	Fri, 27 Feb 2026 08:28:25 GMT	Fri, 27 Feb 2026 08:28:32 GMT

CloudWatch alarm

Command description

Command parameters

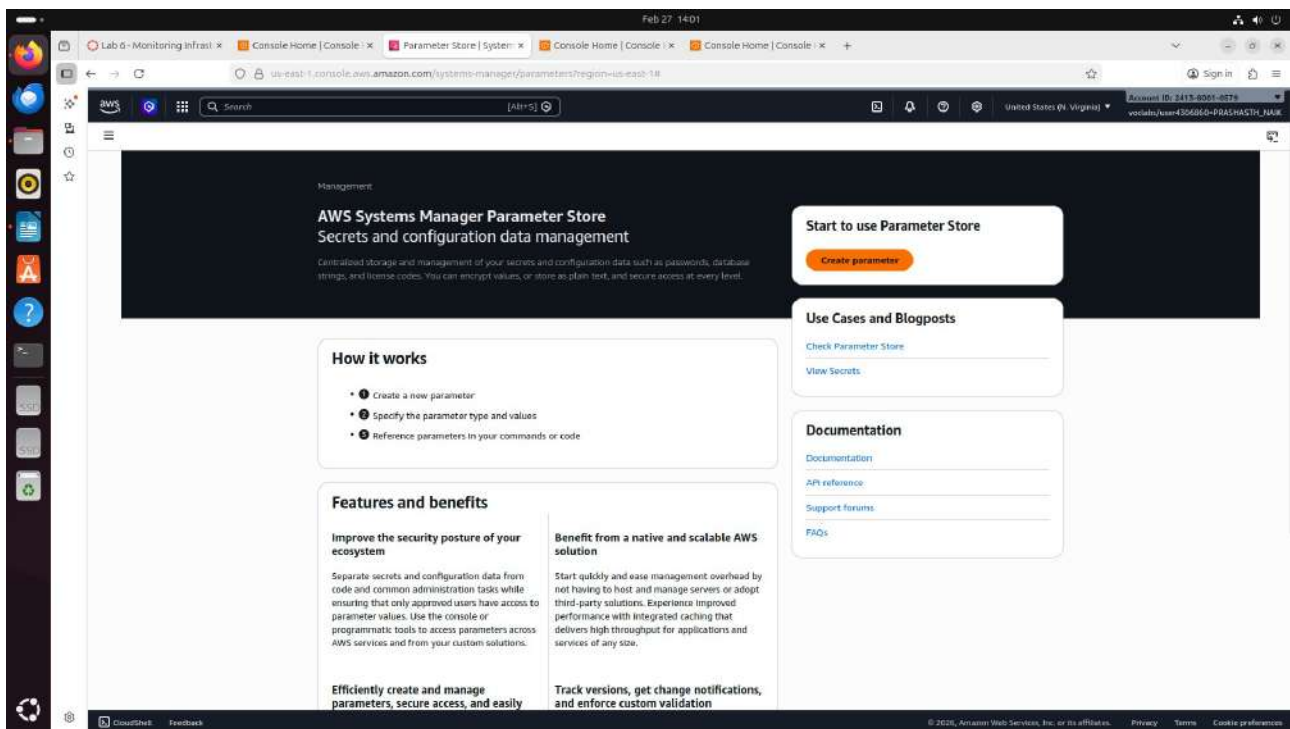
CloudWatch

Feedback

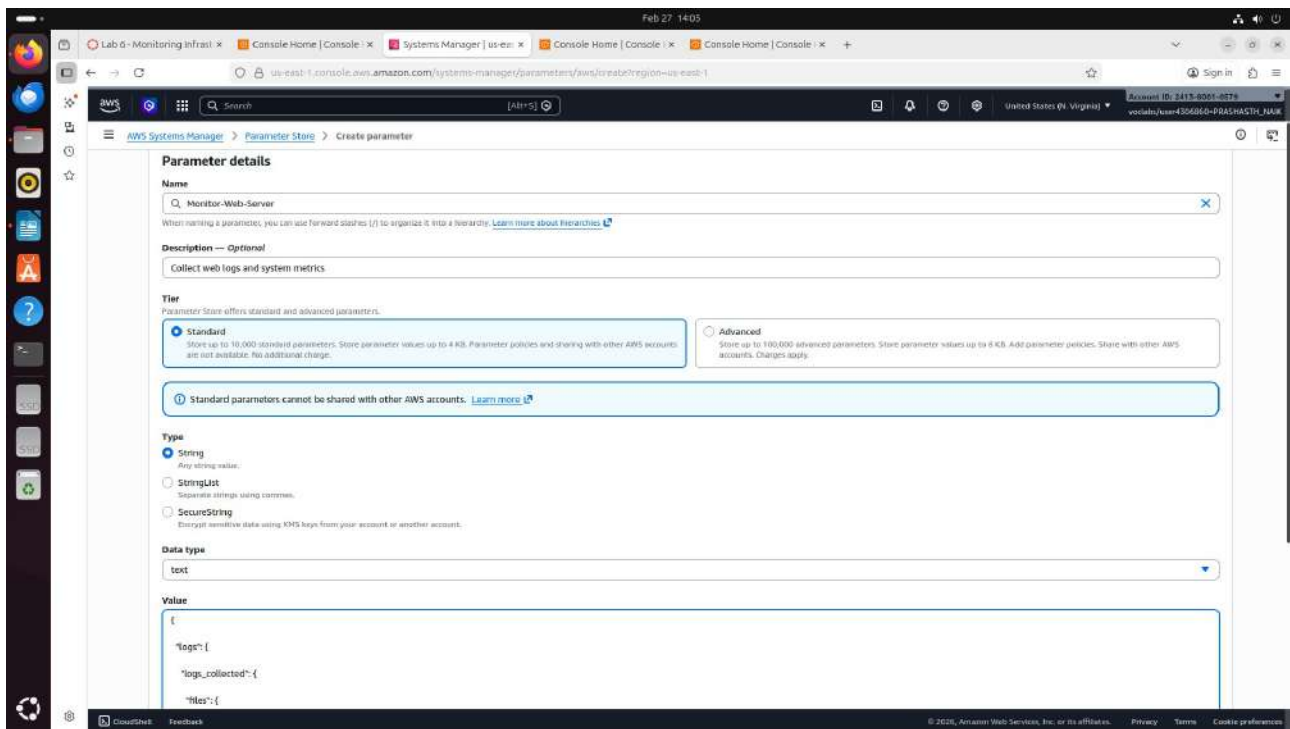
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Step 2 - Command description and status					
Status	Detailed status	Response code	Step name	Start time	Finish time
Success	Success	0	configurePackage	Fri, 27 Feb 2026 08:28:25 GMT	Fri, 27 Feb 2026 08:28:32 GMT
► Output					
► Error					

Step 4: On the Left panel select the parameter store option



Step 5:
Click on the create parameter option
And enter the Name , description and value fields and select the create parameter option



Value

```
{
  "logs": {
    "logs_collected": {
      "files": {
        "log_group_name": "httpAccessLog",
        "file_path": "/var/log/httpd/access_log",
        "log_stream_name": "instance_id",
        "timestamp_format": "%b %d %H:%M:%S"
      }
    }
  }
}
```

Maximum length: 4096 characters.

Tags — Optional
You can use tags to organize and restrict access to your parameter.

[Add tag](#)

[Cancel](#) [Create parameter](#)

Step 6:

On the left panel select the run command option

click the run command option and enter the following details in the search bar

Document name prefix : Equals : AmazonCloudWatch-ManagedAgent
and press enter

In the command parameter section enter the following details

Action: *configure*

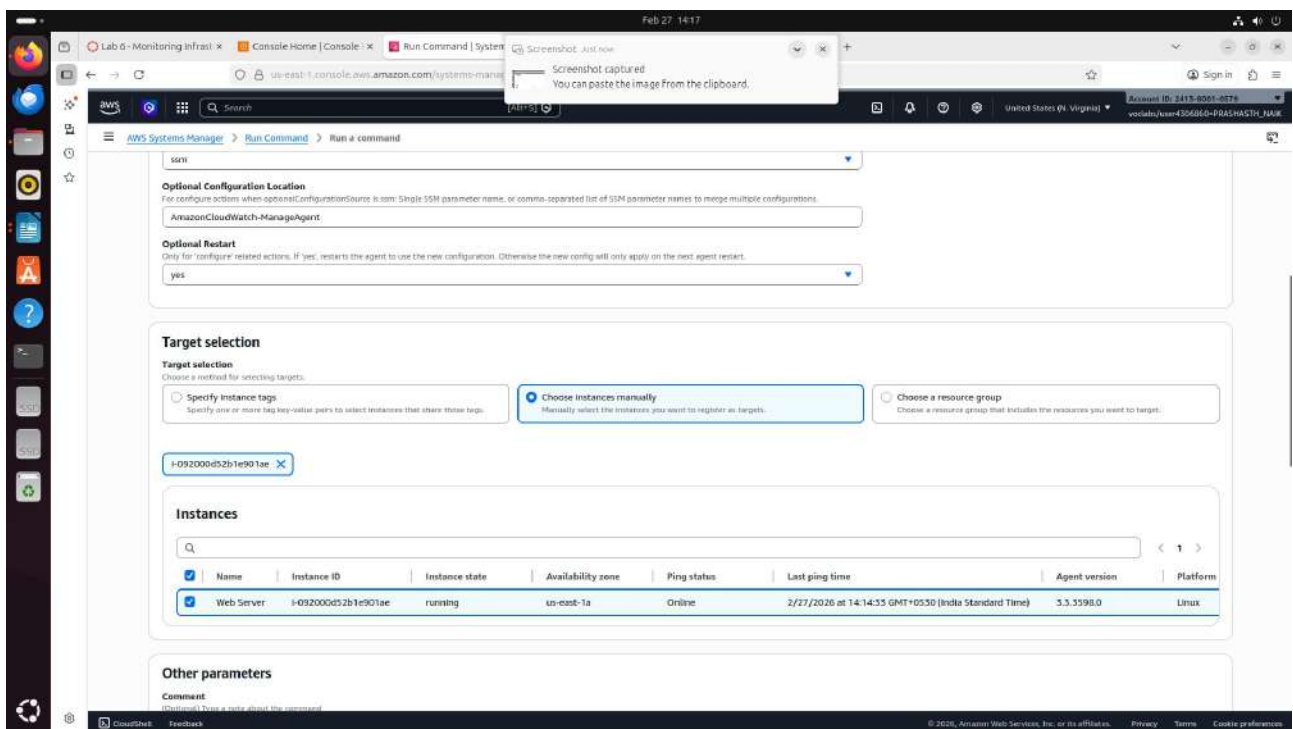
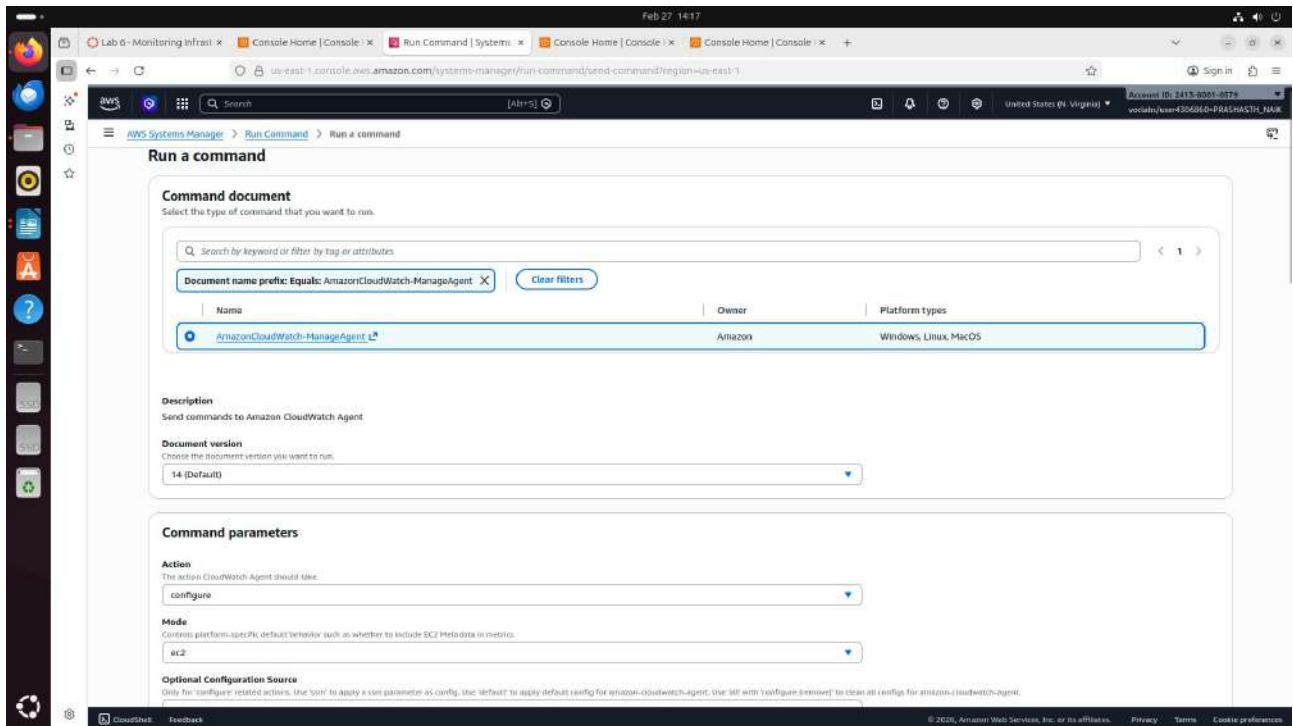
Mode: *ec2*

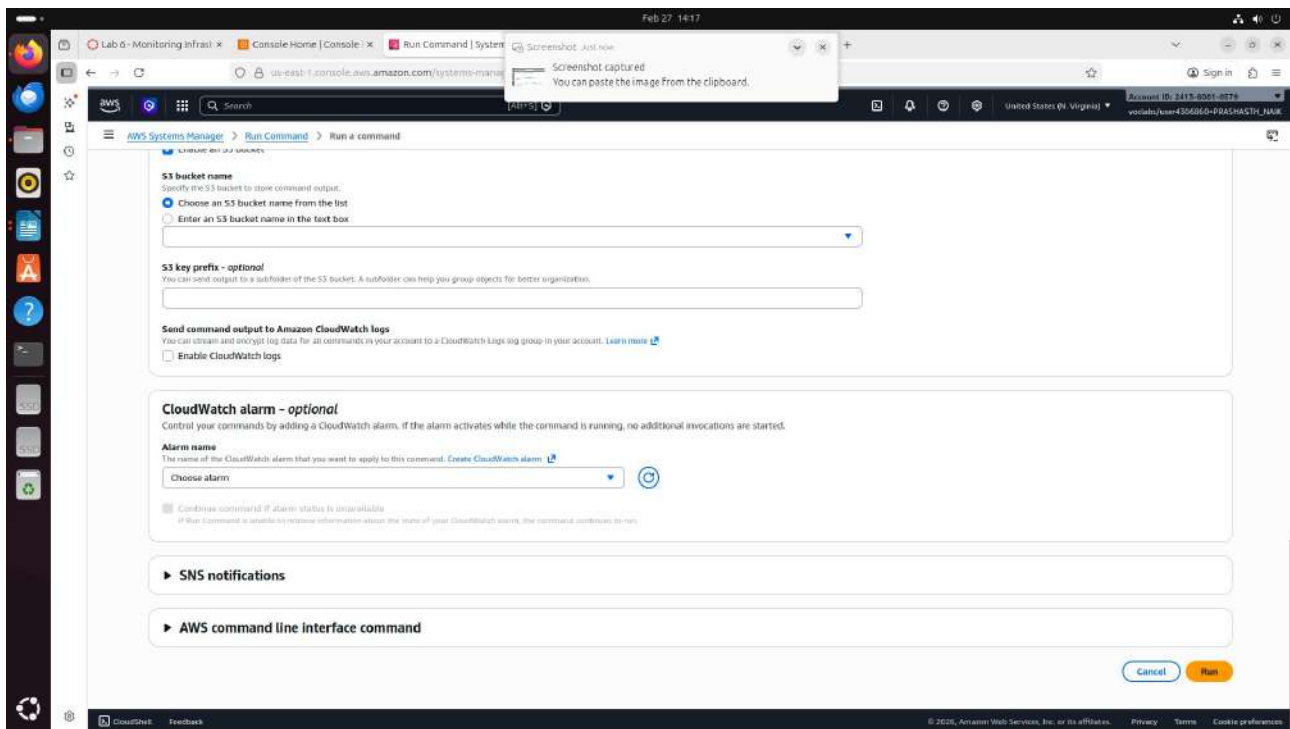
Optional Configuration Source: *ssm*

Optional Configuration Location: *Monitor-Web-Server*

Optional Restart: *yes*

Press the choose instances manually and select the webserver option

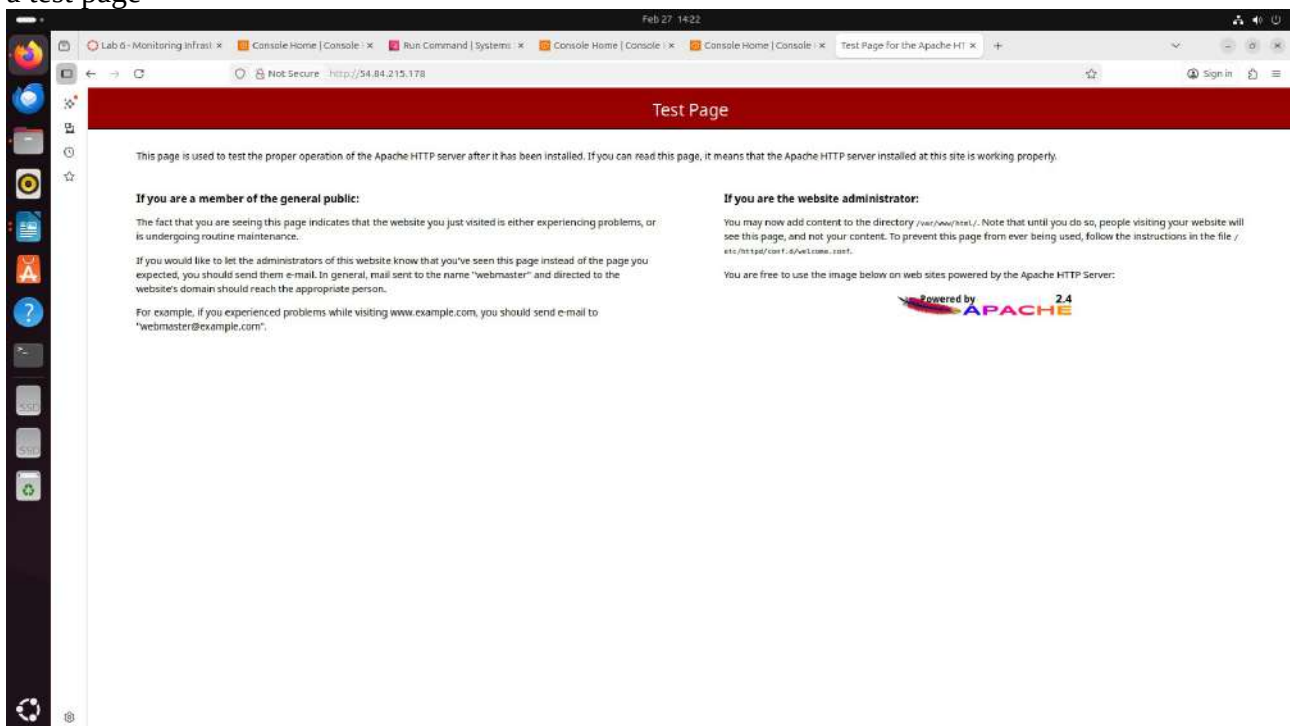




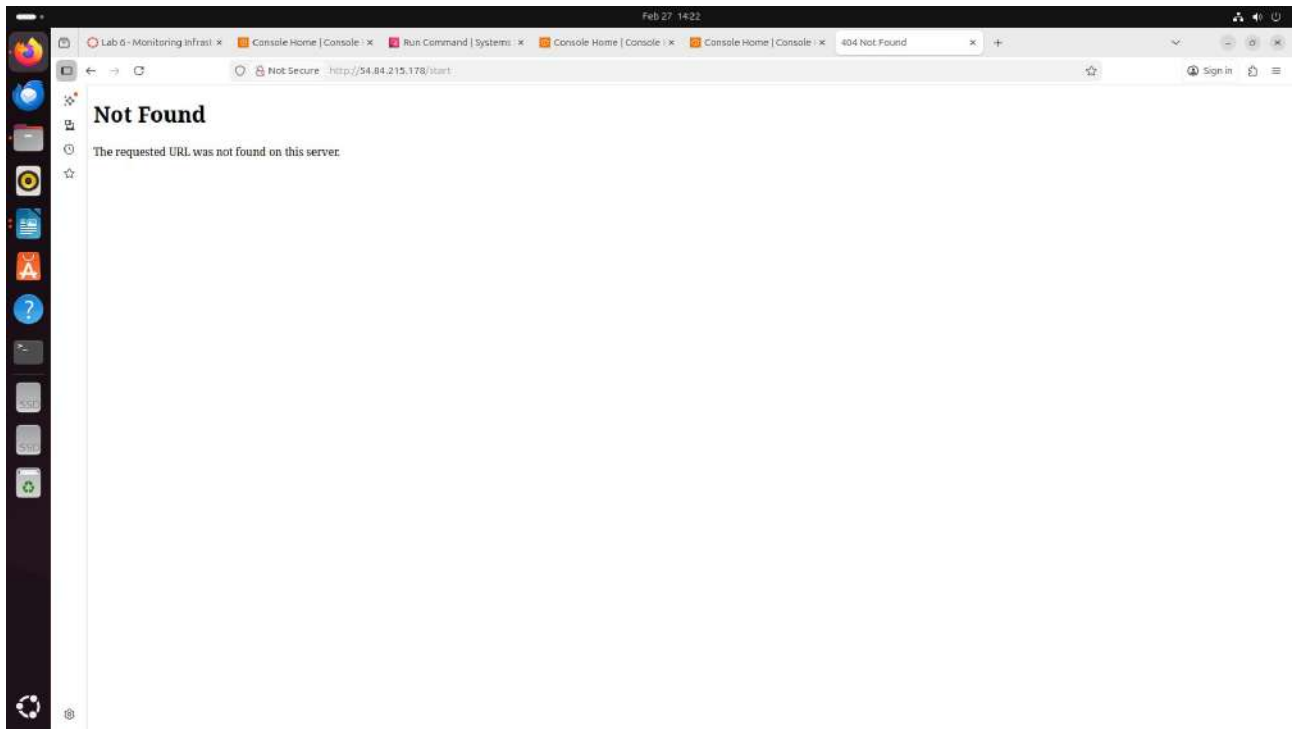
and press the run command

Step 7:

Copy paste the webserver url from the AWS details page and paste in the web browser you will get a test page



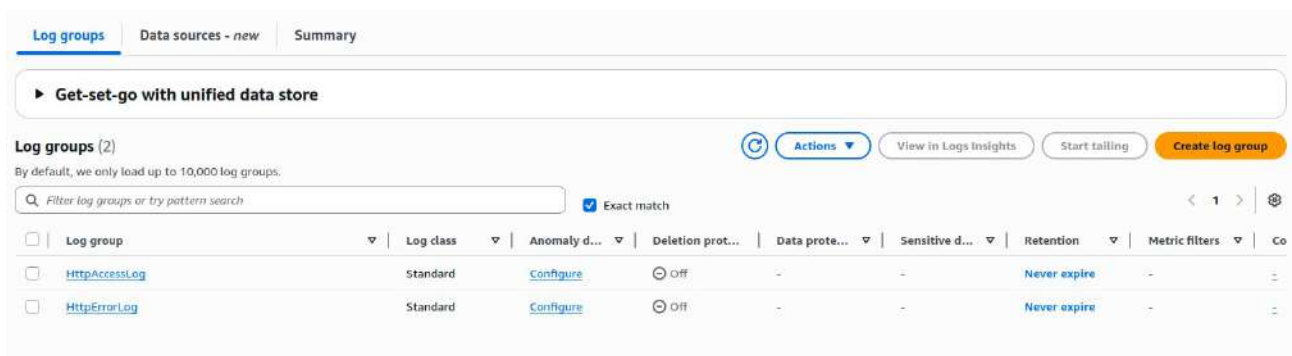
append /start at the end of the webserver url you will receive an error message

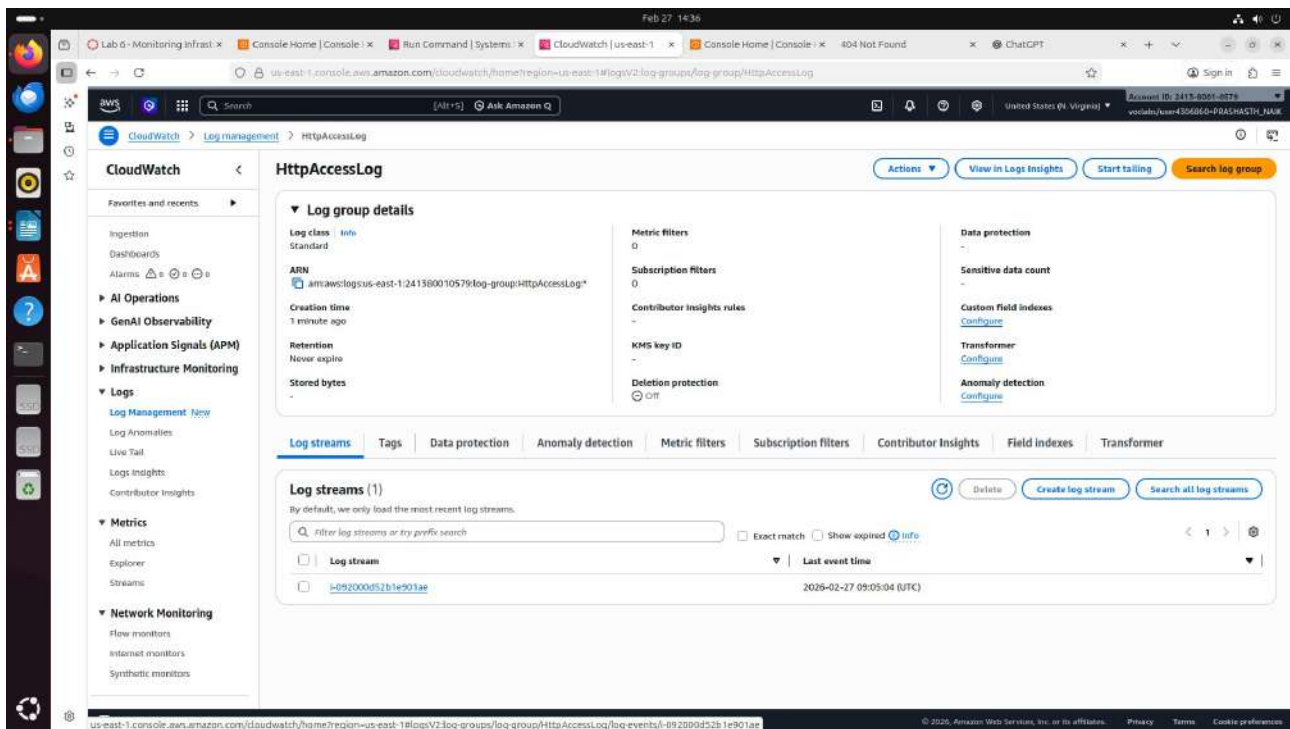


Step 8:

Under the Log group you will see two options `HttpErrorLog` and `HttpAccessLog`.

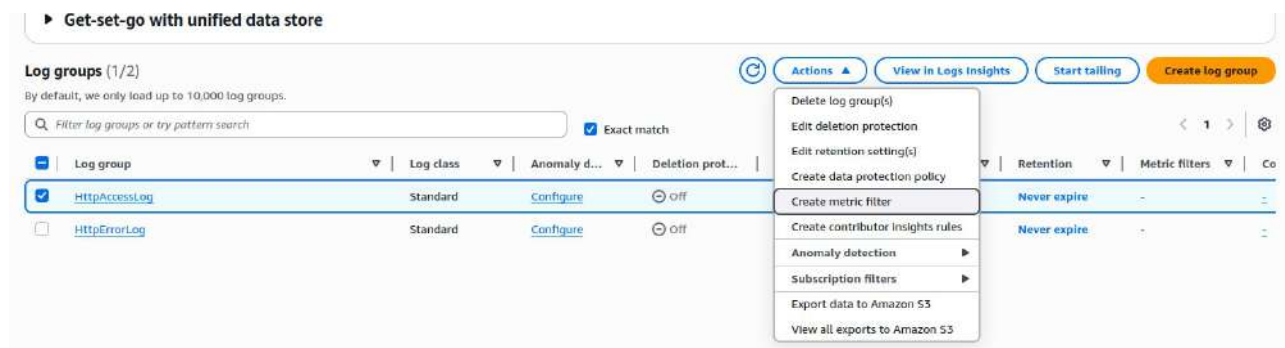
Select `HttpAccessLog` and open the Log Stream you will see the get requests made by your browser





Step 9:

In the Log group select HttpAccessLogs and press actions and select the option Create Metrics Filter



Enter the below details as the filter

[ip, id, user, timestamp, request, status_code=404, size]

Define pattern



Under Test pattern select the instance ID and press the test pattern button

Test pattern

The test does not include the field selection criteria

Select log data to test
All log data fetched will be in original log format.

Log event messages
Type log data to test with your Filter Pattern. Please use line breaks to separate log events.

```
204.76.203.206 - - [27/Feb/2026:08:26:52 +0000] "GET / HTTP/1.1" 403 3630 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/90.0.4430.85 Safari/537.36 Edg/90.0.818.46"
170.106.11.6 - - [27/Feb/2026:08:27:00 +0000] "GET / HTTP/1.1" 403 3630 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 13_2_3 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/13.0.3 Mobile/15E148 Safari/604.1"
218.248.47.10 - - [27/Feb/2026:08:51:59 +0000] "GET / HTTP/1.1" 403 3630 "-" "Mozilla/5.0 (X11; Ubuntu; x86_64; rv:148.0) Gecko/20100101 Firefox/148.0"
```

Results
Found 2 matches out of 7 event(s) in the sample log.

Show test results

Event number	\$id	\$ip	\$request	\$size
5	-	218.248.47.10	GET /favicon.ico HTTP/1.1	236 "http://54.84.215.178/" "Moz
6	-	218.248.47.10	GET /start HTTP/1.1	236 "-" "Mozilla/5.0 (X11; Ubuntu;

press the next button and enter the filter name , filter namespace , Metrics name and metrics values fields

Feb 27, 14:41

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logV2-log-groups/log-group/HttpAccessLog/create-metric-filter

CloudWatch > Log management > HttpAccessLog > Create metric filter

CloudWatch

Step 1: Define pattern
Step 2: **Assign metric**
Step 3: Review and create

Assign metric

Create filter name
Log events that match the pattern you define are recorded to the metric that you specify. You can graph the metric and set alarms to notify you.

Filter name

Filter pattern

☐ Enable metric Filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Metric details

Metric namespace
Namespaces let you group similar metrics. [Learn more](#)

☒ Create new
Namespaces can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric name
Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value
Metric value is the value published to the metric name when a Filter Pattern match occurs.

Valid metric values are: floating point number (1, 69.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. StatusCode for predefined Filter patterns or \$status for 255N-based Filter patterns - dollar (\$) or dollar dot (\$) followed by alphanumeric and/or underscore (_) characters).

Default value - optional

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Press next and press create metrics filter button

Step 10:

In the panel titled **404Errors**, choose the check box in the top right corner.

404Errors

Filter pattern

[ip, id, user, timestamp, request, status_code=404, size]

Field selection criteria

-

Metric

[LogMetrics](#) / [404Errors](#)

Metric value

1

Default value

-

Applied on transformed logs

-

Unit

-

Emit system field dimensions

-

Dimensions

-

Alarms

None.

Choose create alarm option and configure the following details
Period , conditon and press next

Feb 27 14:46

Lab 6 - Monitoring in AWS

Console Home | Com: x

Run Command | Syst: x

CloudWatch | (view all) x

Create alarm | Alarm: x

Console Home | Com: x

404 Not Found

ChatGPT

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2/create?--Page--MetricSelection--AlarmType--MetricAlarm--AlarmData--(Namespace--LogMe...

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 2415-8951-0174

username=4306060-PDASHASTY_MAK

CloudWatch

Alarms

Create alarm

Configure actions

Step 3: Add alarm details

Step 4: Preview and create

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

5.5

5

4.5

06:50

07:00

07:50

08:00

08:50

09:00

404Errors

Namespace

LogMetrics

Metric name

404Errors

Statistic

Sum

Period

1 minute

Conditions

Threshold type

Static

Use a value as a threshold

Anomaly Detection

Use a band as a threshold

Whenever 404Errors is...

Greater

> threshold

Greater/Equal

>= threshold

Lower/Equal

<= threshold

Lower

< threshold

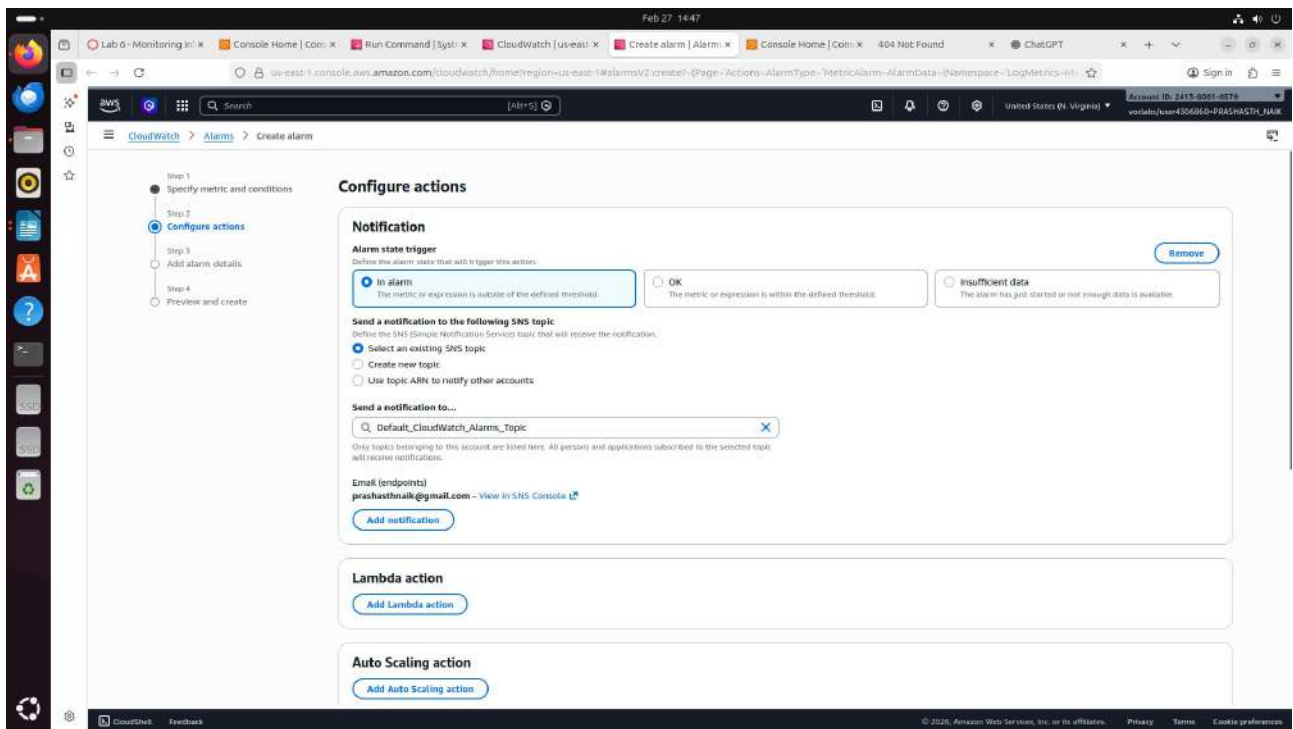
than...

Define the threshold value.

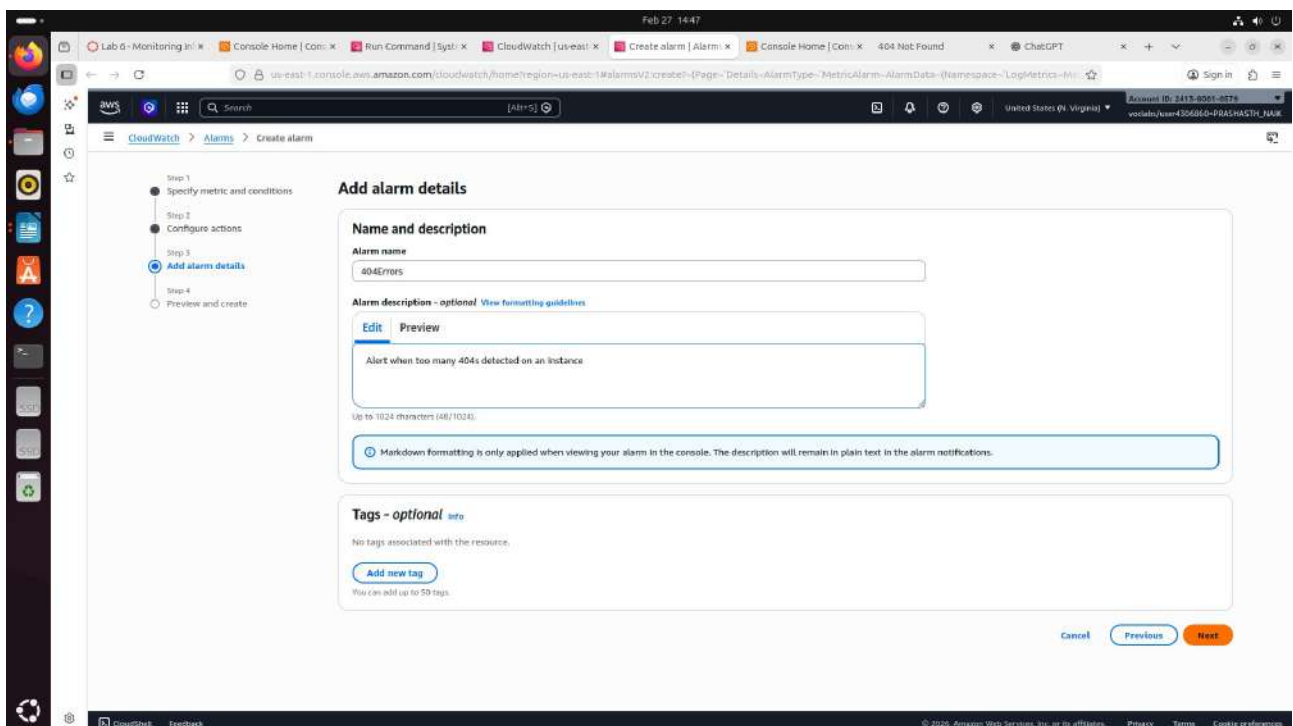
5

Must be a numeric.

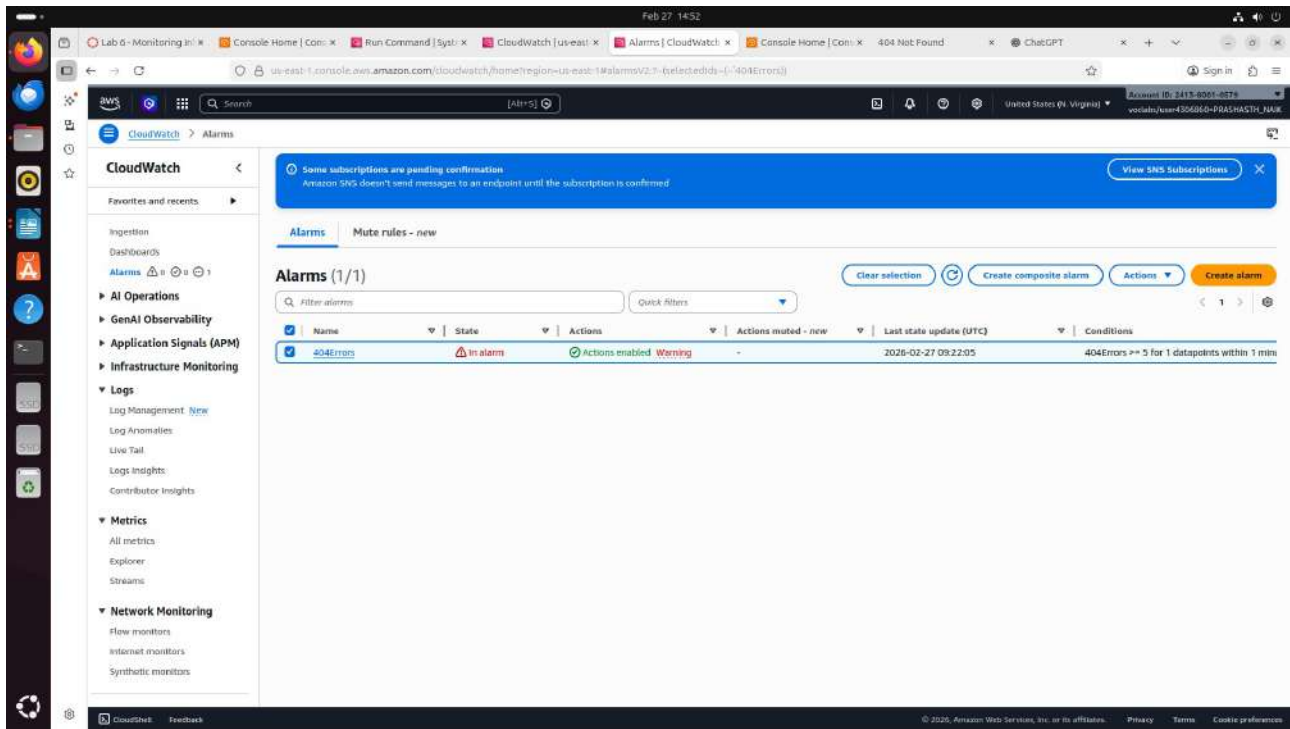
Create a new notification topic



Add the following fields alarm name , alarm description and press next and create alarm



Step 11:
Go to the alarms section you will observe that there is insufficient data
refresh the webpage couple of times you will go to in alarm stage



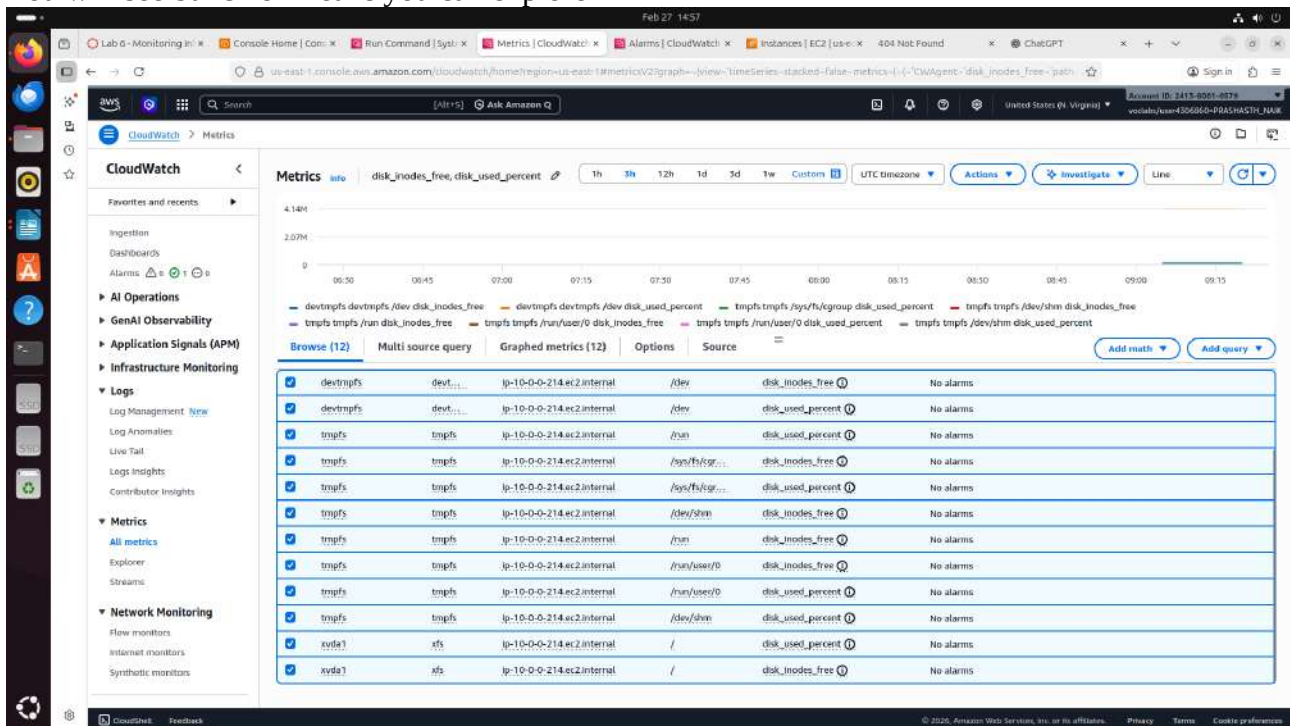
Step 12: Open EC2

select webserver you can monitor its preformance under the monitor section

Open cloudwatch and select all metrics

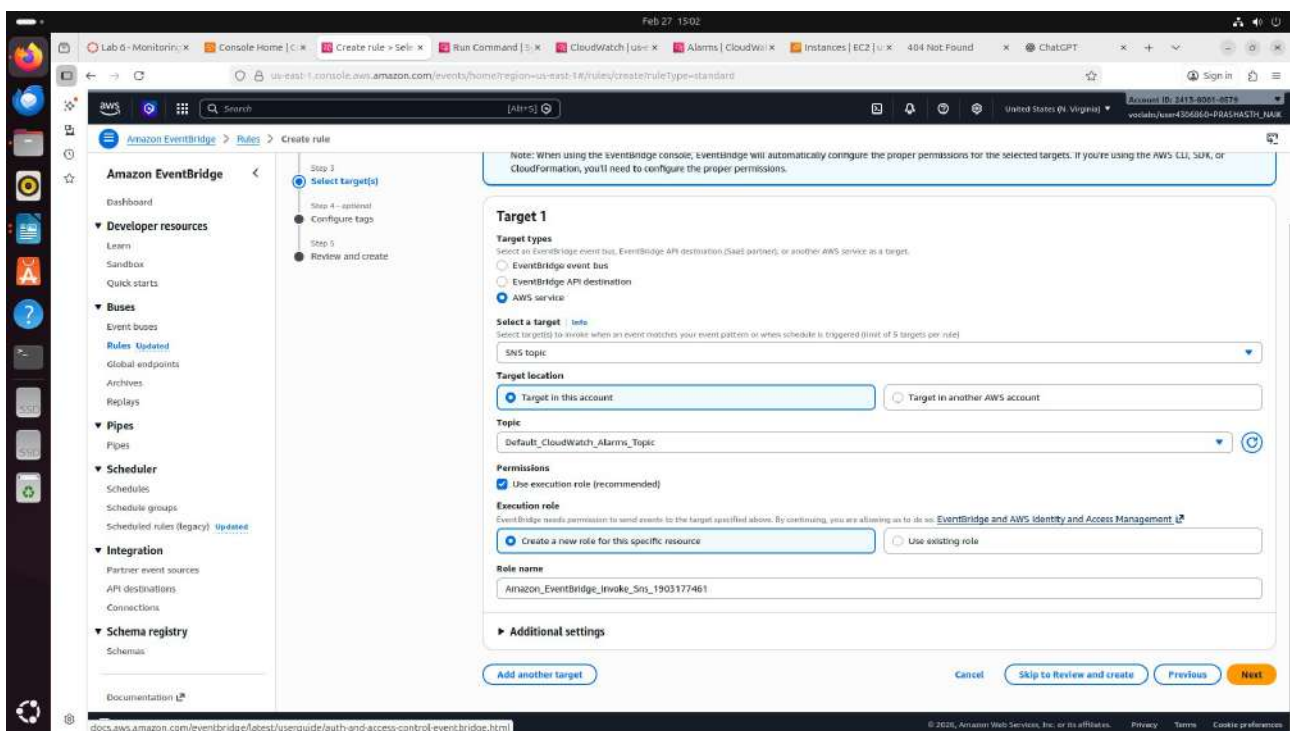
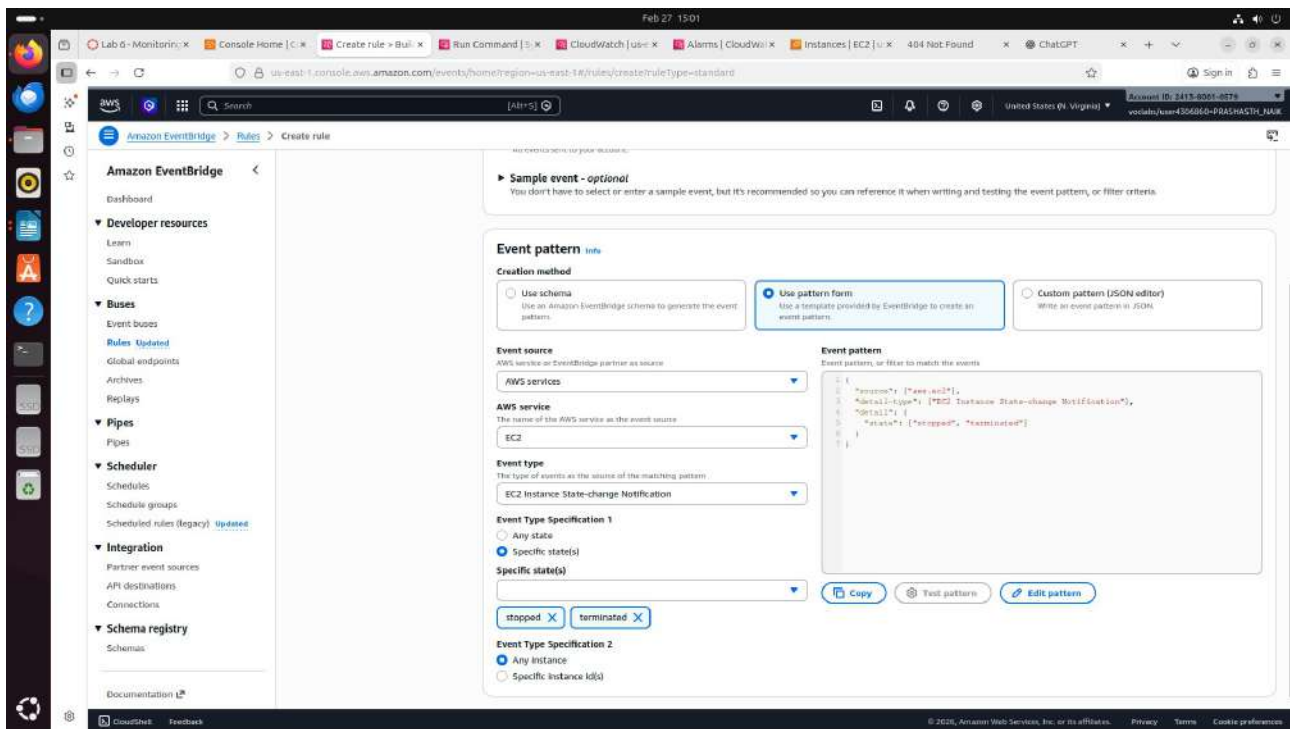
Choose CWAgent > device,fstype,host,path

You will see bunch of metric you can explore



Step 13:

Open Amazon Event Bridge and create a rule



Step 14:
Go to amazon SNS and see the topic created earlier
Go to EC2 and stop your instance

Step 15:
Go to AWS config and create two AWS managed rules
ec2-volume-inuse-check and required tags

Feb 27 15:05

Lab 6 - Monitor x Default_Cloud x Create rule > R x Run Command x CloudWatch x Alarms | Cloud x Instances | EC2 x 404 Not Found x ChatGPT x AWS Config | x

us-east-1.console.aws.amazon.com/config/home?region=us-east-1#/home

Sign in

United States (N. Virginia)

Account ID: 2415-8561-0276

violata/iam430686d-PRASHASTH_HAJK

AWS Config

Record and evaluate configurations of your AWS resources

AWS Config provides a detailed view of the resources associated with your AWS account, including how they are configured, how they are related to one another, and how the configurations and their relationships have changed over time.

Set up AWS Config

A summarized view of AWS and non-AWS resources and the compliance status of the rules and the resources in each AWS Region.

[Get started](#) [1-click setup](#)

Pricing

[AWS Config](#) [Pricing details](#)

[AWS Config rules](#) [Pricing details](#)

[AWS Pricing Calculator](#)

Learn more

[Documentation](#)

[FAQs](#)

[Partners](#)

[What's new](#)

How it works

Configuration change occurs in your AWS resources.

AWS Config records and normalizes the changes into a consistent format.

AWS Config automatically evaluates the recorded configurations against the configurations you specify.

Services that integrate with AWS Config:

- AWS Config APIs & Console
- Amazon SNS
- Amazon CloudWatch
- Amazon S3

Access change history and compliance results using the console or APIs. CloudWatch Events or SNS alert you when changes occur. Deliver change history and snapshot files to your S3 bucket for analysis.

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Feb 27 15:06

Lab 6 - Monitor x Default_Cloud x Create rule > R x Run Command x CloudWatch x Alarms | Cloud x Instances | EC2 x 404 Not Found x ChatGPT x Set up AWS Co x

us-east-1.console.aws.amazon.com/config/home?region=us-east-1#/getstarted/quickSetup=true

Sign in

United States (N. Virginia)

Account ID: 2415-8561-0276

violata/iam430686d-PRASHASTH_HAJK

Rules

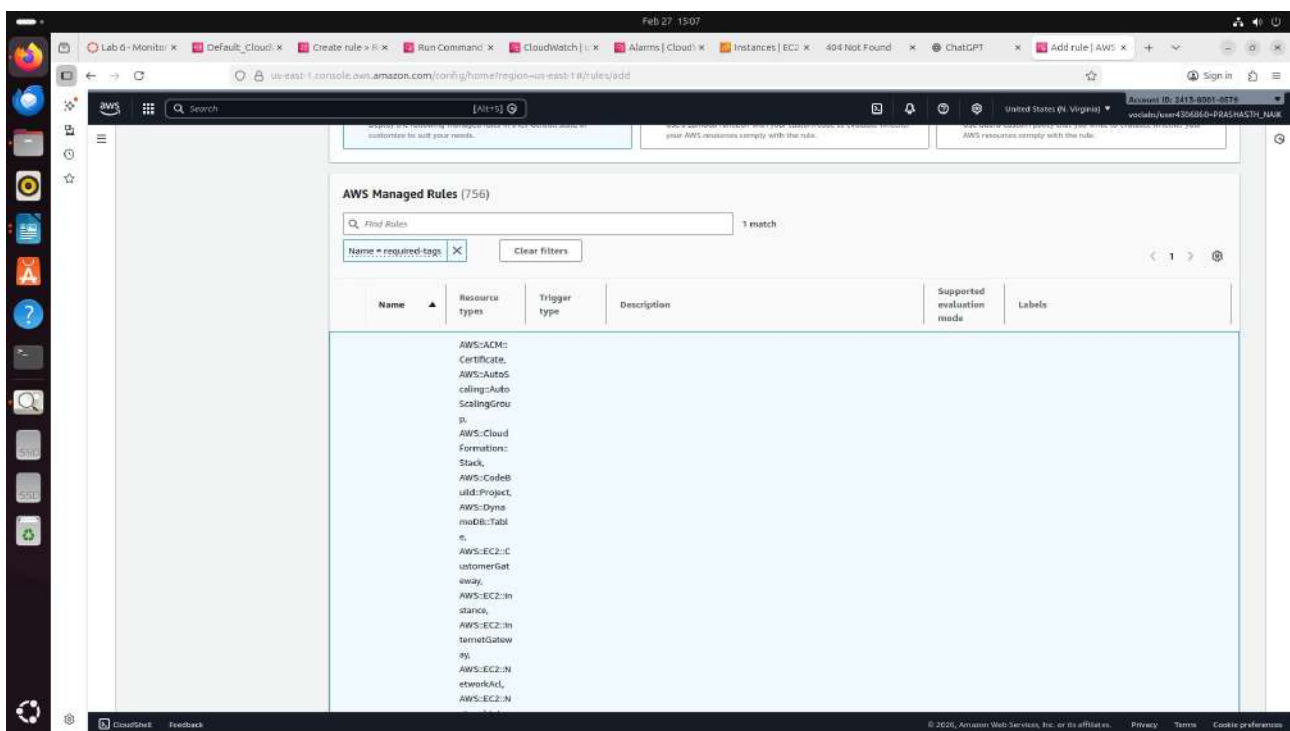
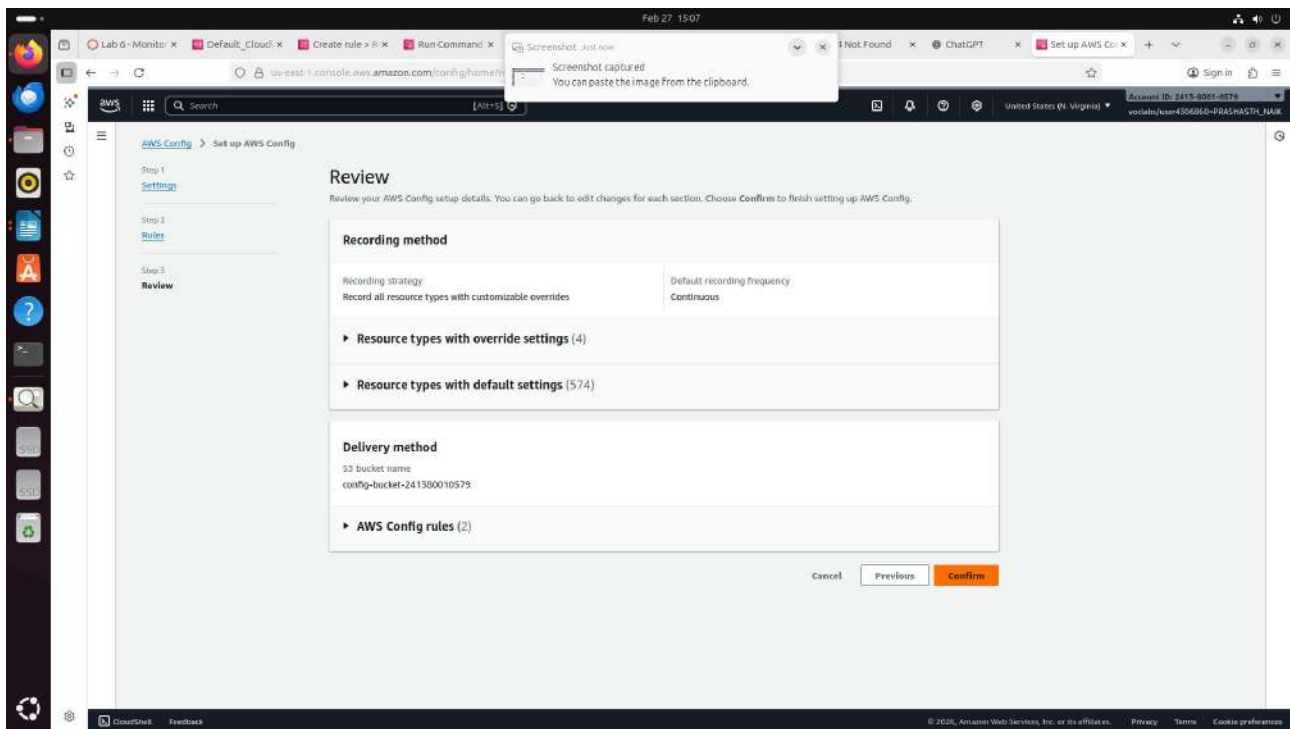
AWS Managed Rules (687)

1 match

[Clear filters](#)

☒	Name	Resource types	Trigger type	Description	Supported evaluation mode
☒	ec2-volume-in-use-check	AWS::EC2::Volume	CHANGE-TRIGGERED	Checks whether EBS volumes are attached to EC2 instances.	DETECTIVE

[Cancel](#) [Previous](#) [Next](#)



Lab 6 - Monitor

Default_Cloud

Create rule > R

Run Command

CloudWatch

Alarms | Cloud

Instances | EC2

404 Not Found

ChatGPT

Rules | AWS Co

us-east-1.console.aws.amazon.com/config/home?region=us-east-1#/rules

Sign in

United States (N. Virginia)

Account ID: 2415-8391-8276

username:43668d-0-P0A5HAG5tr_HJJK

AWS Config

Dashboard

Compliance packs

Rules

Resources

Aggregators

Compliance Dashboard

Compliance packs

Rules

Inventory Dashboard

Resources

Authorizations

Advanced queries

Settings

What's new

Documentation

Partners

FAQs

Pricing

The rule `required-tags` has been added to your account.

Rules

A rule is a compliance check that helps you manage your ideal configuration settings. AWS Config evaluates whether your resource configurations comply with relevant rules and displays the compliance results.

Rules

Filter by compliance status

All

View details

Edit rule

Actions

Add rule

Name	Remediation action	Type	Enabled evaluation mode	Detective
☐ ec2-volume-in-use-check	Not set	AWS managed	DETECTIVE	1 Non
☐ required-tags	Not set	AWS managed	DETECTIVE	-

CloudShell

Feedback

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Lab 6 - Monitor

Default_Cloud

Create rule > R

Run Command

CloudWatch

Alarms | Cloud

Instances | EC2

404 Not Found

ChatGPT

Rule details | A

us-east-1:console.amazonaws.com/config/home?region=us-east-1#/rules/details/configRuleName=required-tags

Sign in

United States (N. Virginia)

Account ID: 2415-8391-0276

username=4366860-P0A5HAG57r_HJUK

AWS Config

Dashboard

Compliance packs

Rules

Resources

Aggregators

Compliance Dashboard

Compliance packs

Rules

Inventory Dashboard

Resources

Authorizations

Advanced queries

Settings

What's new

Documentation

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FAQs

Pricing

tag1Key	String	-	Key of a fourth required tag.
tag1Value	CSV	-	Optional value of the fourth required tag. Separate multiple values with commas.
tag5Key	String	-	Key of a fifth required tag.
tag5Value	CSV	-	Optional value of the fifth required tag. Separate multiple values with commas.
tag6Key	String	-	Key of a sixth required tag.
tag6Value	CSV	-	Optional value of the sixth required tag. Separate multiple values with commas.
tag1Key	String	project	Key of the required tag.

Resources in scope

All

View details

Remediate

ID	Type	Status	Annotation	Compliance
i-092000652b1e901ae	EC2 Instance	-	-	Compliant
c-193507a497326011361093111w...	CloudFormation Stack	-	-	Noncompliant
igw-05323a4cd2ade8615	EC2 InternetGateway	-	-	Noncompliant
igw-095b612a3b8435446	EC2 InternetGateway	-	-	Noncompliant
ad-035017220f1f6c1bb	EC2 NetworkAd	-	-	Noncompliant
ad-0664a53c82b062094	EC2 NetworkAd	-	-	Noncompliant
eni-0897bf709d9500483	EC2 NetworkInterface	-	-	Noncompliant
rtb-044bd90f11745204c8	EC2 RouteTable	-	-	Noncompliant
rtb-07d2d8a4b22a6647f	EC2 RouteTable	-	-	Noncompliant
rtb-Defa3ce3c3db52f5a	EC2 RouteTable	-	-	Noncompliant

CloudWatch

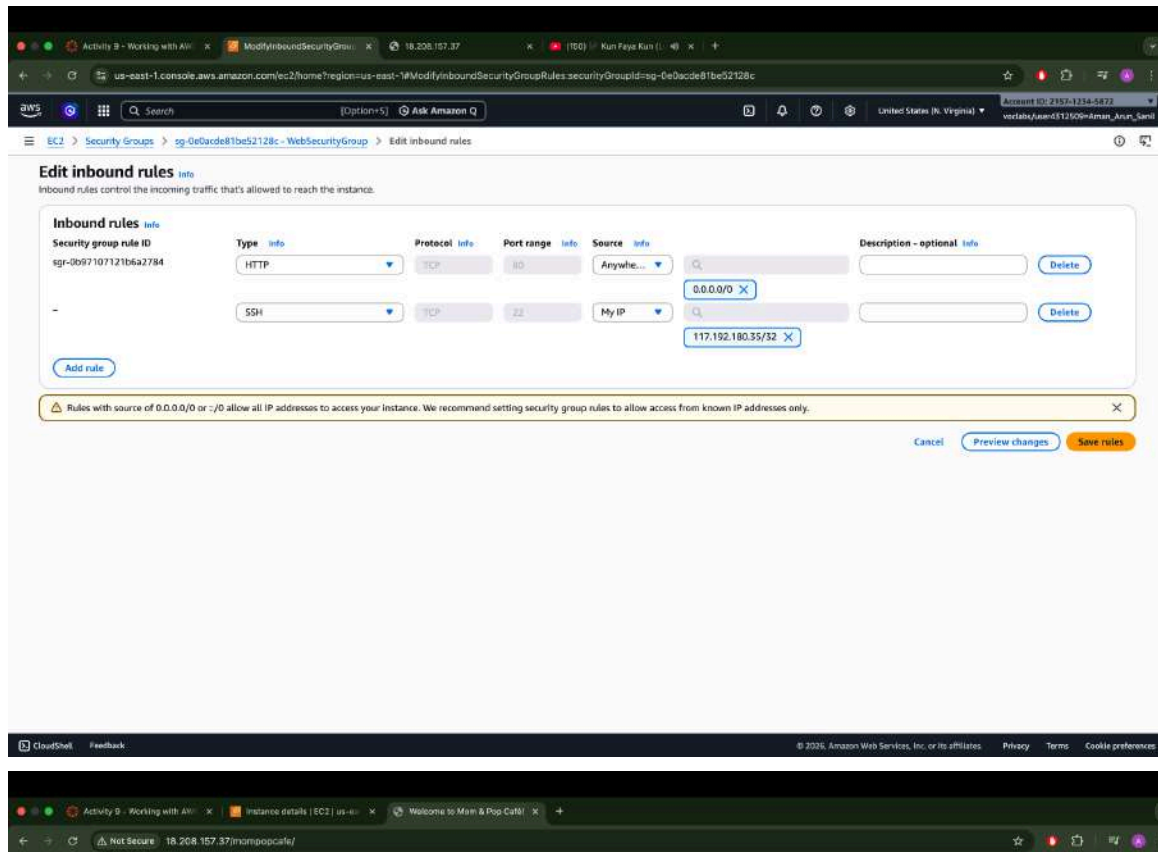
Feedback

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Lab question 5-

Module 9 - Activity 9 - Working with AWS Cloudtrail

Step 1: Add Inbound rule on port 22 for ssh access to the webserver instance from your ip.



Step 2: Create an AWS CloudTrail log

The screenshot displays the AWS CloudTrail console interface for creating a new trail. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/cloudtrailv2/home?region=us-east-1#/create`. The console header includes the AWS logo, a search bar, and navigation links for CloudTrail, Trails, and Create trail. The account ID is 2157-1234-5672, and the user is Aman_Arun_Sunil.

Step 1: Choose trail attributes

Choose trail attributes

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To renew accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☒ Create new S3 bucket
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

Logs will be stored in aman-mals-7272/AmazonLogs/215712345672

Log file SSE-KMS encryption [Info](#)
☐ Enabled

Additional settings

Log file validation [Info](#)
☒ Enabled

SNS notification delivery [Info](#)
☐ Enabled

CloudWatch Logs - optional
Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

Step 2: Choose log events

Choose log events

Events [Info](#)
Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)

Event type
Choose the type of events that you want to log.

☒ **Management events**
Capture management operations performed on your AWS resources.

☐ **Data events**
Log the resource operations performed on or within a resource.

☐ **Insights events**
Identify unusual activity, errors, or user behavior in your account.

☐ **Network activity events**
Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Management events [Info](#)
Management events show information about management operations performed on resources in your AWS account.

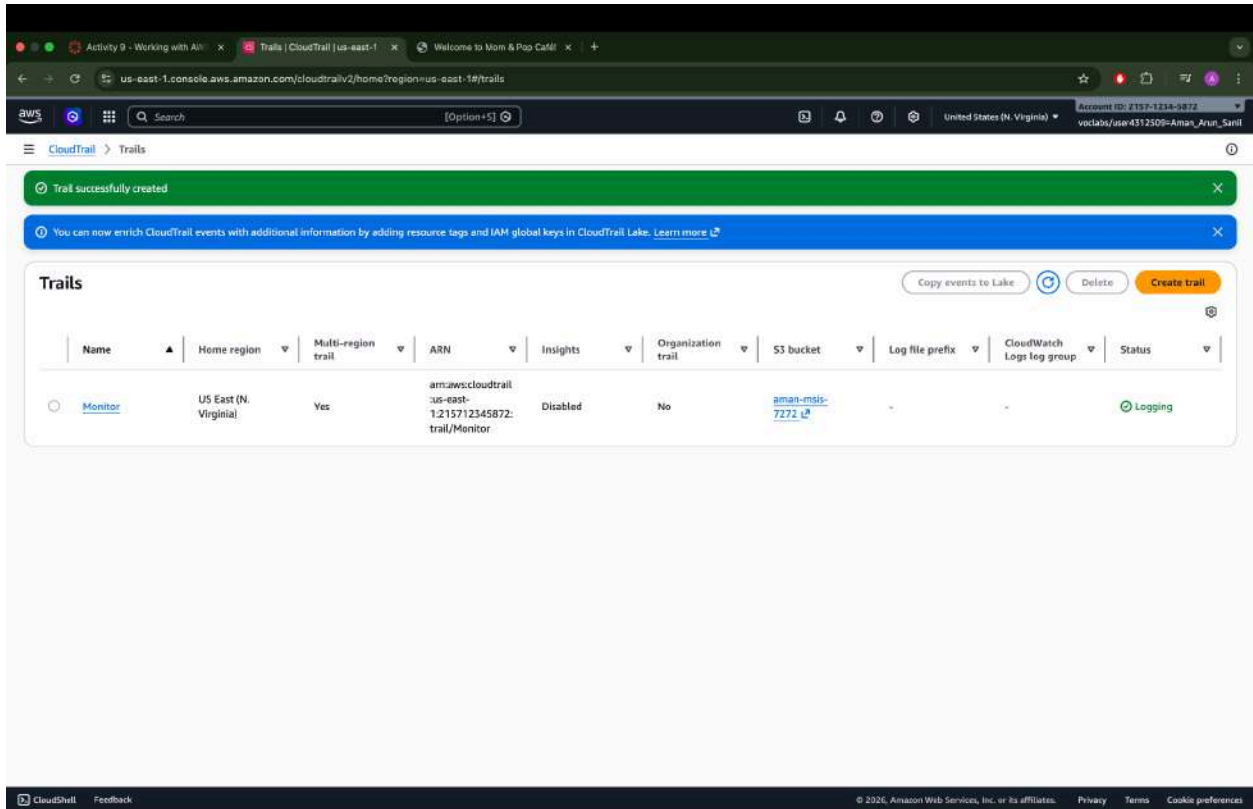
API activity
Choose the activities you want to log.

☒ Read ☒ Write

☐ Exclude AWS KMS events

☐ Exclude Amazon RDS Data API events

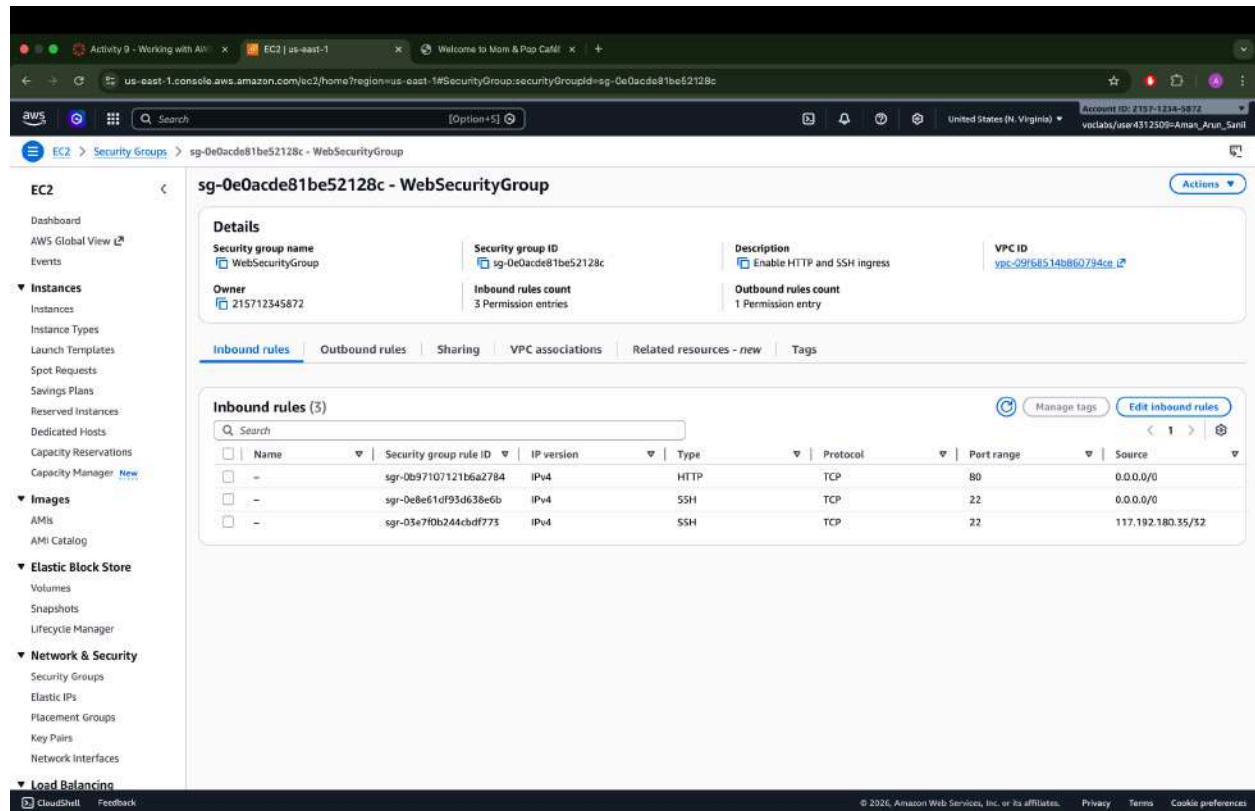
[Cancel](#) [Previous](#) [Next](#)



After creating the AWS CloudTrail, go back to your website and observe that someone has hacked our website and changed the image.



Check the webserver security group and observe that an extra inbound rule has been added which wasn't there before. Someone has enabled ssh access to anywhere.



Step 3: ssh into the webserver instance, Analyze the CloudTrail Logs by using grep.
Create a local directory on the web server to download the CloudTrail Log files to:

mkdir ctraillogs

Change the directory to the new directory:

cd ctraillogs

List the buckets to recall the bucket name:

aws s3 ls

Download the CloudTrail logs by running the following command. Replace <monitoring####> with the actual bucket name that starts with monitoring (the bucket name is part of the output from the ls command that you ran).

aws s3 cp s3://<monitoring####>/. --recursive

If the command is successful, you should see that a few log files are downloaded.

```

amansanill@Amans-MacBook-Air Downloads % chmod 600 labsuser.pem
amansanill@Amans-MacBook-Air Downloads % sudo ssh -i labsuser.pem ec2-user@18.208.157.37
The authenticity of host '18.208.157.37 (18.208.157.37)' can't be established.
ED25519 key fingerprint is SHA256:/9TbSboSYwoyBpppb+4MYgrXKbXQKSEjK3Hg8bMcVfg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.208.157.37' (ED25519) to the list of known hosts.

#
_#_
~\_####_ Amazon Linux 2
~~~\_####\_
~~~\_####\_ AL2 End of Life is 2026-06-30.
~~~\_####\_
~~~\_####\_
~~~\_####\_
~~~\_####\_
~~~\_####\_ A newer version of Amazon Linux is available!
~~~\_####\_
~~~\_####\_ Amazon Linux 2023, GA and supported until 2028-03-15.
~~~\_####\_ https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@web-server ~]$ mkdir ctraillogs
[ec2-user@web-server ~]$ cd ctraillogs
[ec2-user@web-server ctraillogs]$ aws s3 ls
2026-02-27 17:32:41 aman-msis-7272
2026-02-27 17:29:07 aws-cloudtrail-logs-215712345872-43755935
2026-02-27 16:58:35 mompopimagerfiles65694
[ec2-user@web-server ctraillogs]$ aws s3 cp s3://aman-msis-7272/ . --recursive
download: s3://aman-msis-7272/AWSLogs/215712345872/CloudTrail-Digest/ap-northeast-1/2026/02/27/215712345872_CloudTrail-Digest_ap-northeast-1_Monitor_us-east-1_20260227T173243Z.json.gz to AWSLogs/215712345872/CloudTrail-Digest/ap-northeast-1/2026/02/27/215712345872_CloudTrail-Digest_ap-northeast-1_Monitor_us-east-1_20260227T173243Z.json.gz
download: s3://aman-msis-7272/AWSLogs/215712345872/CloudTrail-Digest/ap-south-1/2026/02/27/215712345872_CloudTrail-Digest_ap-south-1_Monitor_us-east-1_20260227T173243Z.json.gz to AWSLogs/215712345872/CloudTrail-Digest/ap-south-1/2026/02/27/215712345872_CloudTrail-Digest_ap-south-1_Monitor_us-east-1_20260227T173243Z.json.gz
download: s3://aman-msis-7272/AWSLogs/215712345872/CloudTrail-Digest/eu-north-1/2026/02/27/215712345872_CloudTrail-Digest_eu-north-1_Monitor_us-east-1_20260227T173243Z.json.gz to AWSLogs/215712345872/CloudTrail-Digest/eu-north-1/2026/02/27/215712345872_CloudTrail-Digest_eu-north-1_Monitor_us-east-1_20260227T173243Z.json.gz
download: s3://aman-msis-7272/AWSLogs/215712345872/CloudTrail-Digest/ap-northeast-2/2026/02/27/215712345872_CloudTrail-Digest_ap-northeast-2_Monitor_us-east-1_20260227T173243Z.json.gz to AWSLogs/215712345872/CloudTrail-Digest/ap-northeast-2/2026/02/27/215712345872_CloudTrail-Digest_ap-northeast-2_Monitor_us-east-1_20260227T173243Z.json.gz
download: s3://aman-msis-7272/AWSLogs/215712345872/CloudTrail-Digest/ca-central-1/2026/02/27/215712345872_CloudTrail-Digest_ca-central-1_Monitor_us-east-1_20260227T173243Z.json.gz to AWSLogs/215712345872/CloudTrail-Digest/ca-central-1/2026/02/27/215712345872_CloudTrail-Digest_ca-central-1_Monitor_us-east-1_20260227T173243Z.json.gz

```

Use the cd command and ls commands repeatedly (or enter cd followed by pressing TAB multiple times) as necessary to change the directory to the subdirectory where the logs were downloaded. When you run ls, all of the downloaded log files should display. They will be located in an AWSLogs/<account-num>/CloudTrail-<Region>/<yyyy>/<mm>/<dd> subdirectory.

Notice that the log files end in json.gz, which indicates that they are compressed as GNU zip files.

Run this command to extract the logs:

gunzip *.gz. Run ls again.

Use this format as it is more readable,

cat <filename.json> | python -m json.tool

```
[ec2-user@web-server 27]$ ls
215712345872_CloudTrail_us-east-1_20260227T1735Z_roOoDiWKhAWP46pF.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_jvZE0lJXLOPOf5HE.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_s5embTtjUM0u6r9h.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_kc7l129QZi3Tygyq.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_WYdiPjyFC2AHlM7G.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_mrM3ithIVvmE8Z2R.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_YQIqcihD29mXFRFg.json 215712345872_CloudTrail_us-east-1_20260227T1750Z_PsLi0CMHwBP1MjbY.json
215712345872_CloudTrail_us-east-1_20260227T1740Z_ANoCbR5VQcjkWIPC.json 215712345872_CloudTrail_us-east-1_20260227T1750Z_R9FpeBLhzRh8o5l6.json
[ec2-user@web-server 27]$ cat 215712345872_CloudTrail_us-east-1_20260227T1735Z_roOoDiWKhAWP46pF.json | python -m json.tool
{
  "Records": [
    {
      "awsRegion": "us-east-1",
      "eventCategory": "Management",
      "eventID": "daea0009-4fa1-4103-bf81-227e5552a019",
      "eventName": "GetParametersByPath",
      "eventSource": "ssm.amazonaws.com",
      "eventTime": "2026-02-27T17:34:01Z",
      "eventType": "AwsApiCall",
      "eventVersion": "1.11",
      "managementEvent": true,
      "readOnly": true,
      "recipientAccountId": "215712345872",
      "requestID": "24d63e63-2f6b-446f-95c8-9ff794293735",
      "requestParameters": {
        "path": "/mompopcafe"
      },
      "resources": [
        {
          "ARN": "arn:aws:ssm:us-east-1:215712345872:parameter/mompopcafe",
          "accountId": "215712345872"
        }
      ],
      "responseElements": null,
      "sourceIPAddress": "18.208.157.37",
      "tlsDetails": {
        "cipherSuite": "ECDHE-RSA-AES128-GCM-SHA256",
        "clientProvidedHostHeader": "ssm.us-east-1.amazonaws.com",
        "tlsVersion": "TLSv1.2"
      },
      "userAgent": "aws-sdk-php/3.87.2 GuzzleHttp/6.3.3 curl/8.3.0 PHP/7.2.34",
      "userIdentity": {
        "accessKeyId": "ASIATEOLVF4IC4XYVKYX",
        "accountId": "215712345872",
        "arn": "arn:aws:sts::215712345872:assumed-role/c193507a4973262113682811t1w2157123458-MomPopIamRole-PIJ15zdFaeqY/i-0c3786fd0ff5de12d",
        "inScopeOf": {
          "credentialsIssuedTo": "arn:aws:ec2:us-east-1:215712345872:instance/i-0c3786fd0ff5de12d",

```

```
[ec2-user@web-server 27]$ ls
215712345872_CloudTrail_us-east-1_20260227T1735Z_roOoDiWKhAWP46pF.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_jvZE0lJXLOPOf5HE.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_s5embTtjUM0u6r9h.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_kc7l129QZi3Tygyq.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_WYdiPjyFC2AHlM7G.json 215712345872_CloudTrail_us-east-1_20260227T1745Z_mrM3ithIVvmE8Z2R.json
215712345872_CloudTrail_us-east-1_20260227T1735Z_YQIqcihD29mXFRFg.json 215712345872_CloudTrail_us-east-1_20260227T1750Z_PsLi0CMHwBP1MjbY.json
215712345872_CloudTrail_us-east-1_20260227T1740Z_ANoCbR5VQcjkWIPC.json 215712345872_CloudTrail_us-east-1_20260227T1750Z_R9FpeBLhzRh8o5l6.json
[ec2-user@web-server 27]$ cat 215712345872_CloudTrail_us-east-1_20260227T1735Z_roOoDiWKhAWP46pF.json | python -m json.tool
{
  "Records": [
    {
      "awsRegion": "us-east-1",
      "eventCategory": "Management",
      "eventID": "daea0009-4fa1-4103-bf81-227e5552a019",
      "eventName": "GetParametersByPath",
      "eventSource": "ssm.amazonaws.com",
      "eventTime": "2026-02-27T17:34:01Z",
      "eventType": "AwsApiCall",
      "eventVersion": "1.11",
      "managementEvent": true,
      "readOnly": true,
      "recipientAccountId": "215712345872",
      "requestID": "24d63e63-2f6b-446f-95c8-9ff794293735",
      "requestParameters": {
        "path": "/mompopcafe"
      },
      "resources": [
        {
          "ARN": "arn:aws:ssm:us-east-1:215712345872:parameter/mompopcafe",
          "accountId": "215712345872"
        }
      ],
      "responseElements": null,
      "sourceIPAddress": "18.208.157.37",
      "tlsDetails": {
        "cipherSuite": "ECDHE-RSA-AES128-GCM-SHA256",
        "clientProvidedHostHeader": "ssm.us-east-1.amazonaws.com",
        "tlsVersion": "TLSv1.2"
      },
      "userAgent": "aws-sdk-php/3.87.2 GuzzleHttp/6.3.3 curl/8.3.0 PHP/7.2.34",
      "userIdentity": {
        "accessKeyId": "ASIATEOLVF4IC4XYVKYX",
        "accountId": "215712345872",
        "arn": "arn:aws:sts::215712345872:assumed-role/c193507a4973262113682811t1w2157123458-MomPopIamRole-PIJ15zdFaeqY/i-0c3786fd0ff5de12d",
        "inScopeOf": {
          "credentialsIssuedTo": "arn:aws:ec2:us-east-1:215712345872:instance/i-0c3786fd0ff5de12d",

```

Step 4: Consider how you want to target the search. You are not interested in everything that is happening in this account. Instead, your interest is in an action that was taken on a particular EC2 instance (that is, the web server that was hacked).

Start by filtering the log results where the sourceIP address matches the IP address of the Cafe Web Server instance.

Run the following command to set the WebServerIP address as a variable that you can use in future commands (replace <WebServerIP> with the actual Public IPv4 address of the Cafe Web Server instance):

ip=<WebServerIP>

Run the following command:

for i in \$(ls); do echo \$i && cat \$i | python -m json.tool | grep sourceIP address ; done

The command you ran does the following:

It creates a for loop that includes the names of the files in the current directory.

During each iteration of the for loop, it echoes the file name and then prints the contents of the file in JSON format.

However, only the lines of JSON that contain the sourceIP address tag are printed.

Note that there are several log entries in the trail where the sourceIP address was the Cafe Web Server instance.

Now run a similarly structured command, but where the command returns the eventName of every captured event:

for i in \$(ls); do echo \$i && cat \$i | python -m json.tool | grep eventName ; done

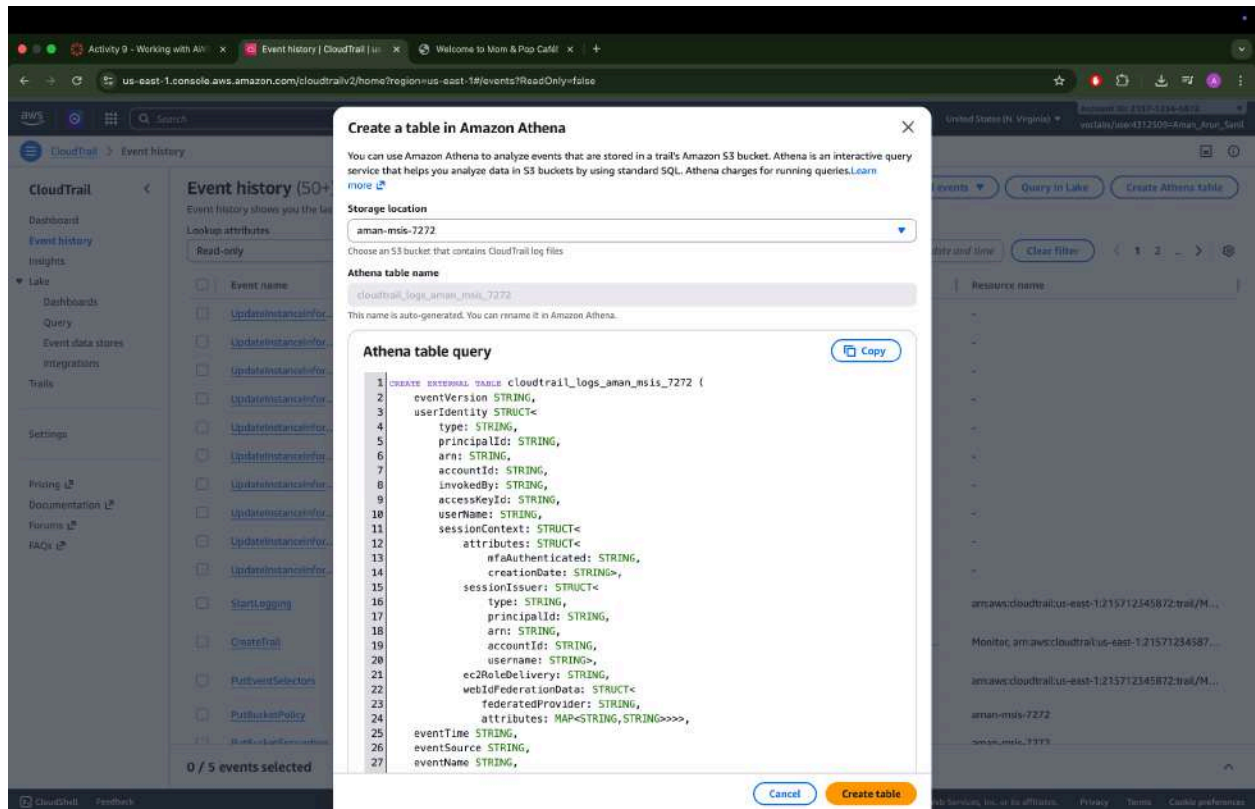
The command you ran follows the same logic as the command you ran before, but this time, it filters log entries for the eventName.

Step 5: Analyze the logs by using AWS CLI cloudtrail commands

```
[ec2-user@web-server 27]$ aws cloudtrail lookup-events --lookup-attributes AttributeKey=EventName,AttributeValue=ConsoleLogin
{
  "Events": [
    {
      "EventId": "b53962da-9603-4fca-bd5c-dd29bfeed465",
      "Username": "user4312509-Aman_Arun_Sanil",
      "EventTime": 177212358.0,
      "CloudTrailEvent": {
        "eventVersion": "1.11",
        "userIdentity": {
          "type": "AssumedRole",
          "principalId": "AROATEOLVF4IWFH00AFB:user4312509-Aman_Arun_Sanil",
          "arn": "arn:aws:sts:215712345872:assumed-role/voclabs/user4312509-Aman_Arun_Sanil",
          "accountId": "215712345872",
          "accessKeyId": "ASIATEOLVF4ILOU4FFHS",
          "sessionContext": {
            "sessionIssuer": {
              "type": "Role",
              "principalId": "AROATEOLVF4IWFH00AFB",
              "arn": "arn:aws:iam:215712345872:role/voclabs",
              "accountId": "215712345872",
              "userName": "voclabs",
              "attributes": {
                "creationDate": "2026-02-27T16:54:22Z",
                "mfaAuthenticated": false
              },
              "eventTime": "2026-02-27T17:12:38Z",
              "eventSource": "signin.amazonaws.com",
              "eventName": "ConsoleLogin",
              "awsRegion": "us-east-1",
              "sourceIPAddress": "117.192.180.35",
              "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36",
              "requestParameters": null,
              "responseElements": {
                "ConsoleLogin": {
                  "Success": true,
                  "additionalEventData": {
                    "MobileVersion": "No",
                    "MFAUsed": "No",
                    "eventID": "b53962da-9603-4fca-bd5c-dd29bfeed465",
                    "readOnly": false,
                    "eventType": "AwsConsoleSignIn",
                    "managementEvent": true,
                    "recipientAccountId": "215712345872",
                    "eventCategory": "Management",
                    "tlsDetails": {
                      "tlsVersion": "TLSv1.3",
                      "cipherSuite": "TLS_AES_128_GCM_SHA256",
                      "clientProvidedHostHeader": "signin.aws.amazon.com"
                    },
                    "accessKeyId": "ASIATEOLVF4ILOU4FFHS",
                    "eventName": "ConsoleLogin",
                    "readOnly": false,
                    "eventSource": "signin.amazonaws.com",
                    "resources": []
                  }
                }
              }
            }
          }
        },
        "EventId": "dfe971fc-b5db-4d53-ba59-1295bb0a4134",
        "Username": "user4312509-Aman_Arun_Sanil",
        "EventTime": 1770194036.0,
        "CloudTrailEvent": {
          "eventVersion": "1.11",
          "userIdentity": {
            "type": "AssumedRole",
            "principalId": "AROATEOLVF4IDQPHYCOI5:user4312509-Aman_Arun_Sanil",
            "arn": "arn:aws:sts:215712345872:assumed-role/voclabs/user4312509-Aman_Arun_Sanil",
            "accountId": "215712345872",
            "accessKeyId": "ASIATEOLVF4IIPGVNSO",
            "sessionContext": {
              "sessionIssuer": {
                "type": "Role",
                "principalId": "AROATEOLVF4IDQPHYCOI5",
                "arn": "arn:aws:iam:215712345872:role/voclabs",
                "accountId": "215712345872",
                "userName": "voclabs",
                "attributes": {
                  "creationDate": "2026-02-04T08:28:51Z",
                  "mfaAuthenticated": false
                },
                "eventTime": "2026-02-04T08:33:56Z",
                "eventSource": "signin.amazonaws.com",
                "eventName": "ConsoleLogin",
                "awsRegion": "us-east-1",
                "sourceIPAddress": "45.112.148.130",
                "userAgent": "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:147.0) Gecko/20100101 Firefox/147.0",
                "requestParameters": null,
                "responseElements": {
                  "ConsoleLogin": {
                    "Success": true,
                    "additionalEventData": {
                      "MobileVersion": "No",
                      "MFAUsed": "No",
                      "eventID": "dfe971fc-b5db-4d53-ba59-1295bb0a4134",
                      "readOnly": false,
                      "eventType": "AwsConsoleSignIn",
                      "managementEvent": true,
                      "recipientAccountId": "215712345872",
                      "eventCategory": "Management",
                      "tlsDetails": {
                        "tlsVersion": "TLSv1.3",
                        "cipherSuite": "TLS_AES_128_GCM_SHA256",
                        "clientProvidedHostHeader": "signin.aws.amazon.com"
                      },
                      "accessKeyId": "ASIATEOLVF4IIPGVNSO",
                      "eventName": "ConsoleLogin",
                      "readOnly": false,
                      "eventSource": "signin.amazonaws.com",
                      "resources": []
                    }
                  }
                }
              }
            }
          }
        }
      }
    ]
  }
}
```

```
[ec2-user@web-server 27]$ sgId=$(aws ec2 describe-instances --filters "Name=tag:Name,Values=Cafe Web Server" --query 'Reservations[*].Instances[*].SecurityGroups[*].[GroupID]' --region $region --output text)
[ec2-user@web-server 27]$ echo $sgId
sg-0e0acde81be52128c
[ec2-user@web-server 27]$ aws cloudtrail lookup-events --lookup-attributes AttributeKey=ResourceType,AttributeValue=AWS::EC2::SecurityGroup --region $region --output text | grep $sgId
EVENTS AKIATEOLVF4IHAWFIMVU {"eventVersion": "1.11", "userIdentity": { "type": "IAMUser", "principalId": "AIDATEOLVF4II4U2DL05S", "arn": "arn:aws:iam:215712345872:user/chaos", "accountId": "215712345872", "accessKeyId": "AKIATEOLVF4IHAWFIMVU", "userName": "chaos", "eventTime": "2026-02-27T17:30:12Z", "eventSource": "ec2.amazonaws.com", "eventName": "AuthorizeSecurityGroupIngress", "awsRegion": "us-east-1", "sourceIPAddress": "18.208.157.37", "userAgent": "aws-cli/1.18.147 Python/2.7.18 Linux/4.14.355-280.714.amzn2.x86_64 botocore/1.18.6", "requestParameters": { "groupId": "sg-0e0acde81be52128c", "ipPermissions": { "items": [ { "ipProtocol": "tcp", "fromPort": 22, "toPort": 22, "groups": { }, "ipRanges": { "items": [ { "cidrIp": "0.0.0.0/0" } ] }, "ipv6Ranges": { }, "prefixListIds": { } } } }, "responseElements": { "requestId": "b6f5be77-78a7-475a-b99c-448a061de701", "return": true, "securityGroupRuleSet": { "items": [ { "groupOwnerId": "215712345872", "groupId": "sg-0e0acde81be52128c", "securityGroupRuleId": "sgr-0e8e61df93dc38e6b", "isEgress": false, "ipProtocol": "tcp", "fromPort": 22, "toPort": 22, "cidrIpv4": "0.0.0.0/0", "securityGroupRule":
```

Step 6: Analyze the CloudTrail logs by using Amazon Athena, create an Athena Table. Storage location: Choose the monitoring#### Amazon S3 bucket where you configured CloudTrail to store log files.



Step 7: Start by setting up a query results location and then running a simple query to get an idea of the data that is available in the logs.

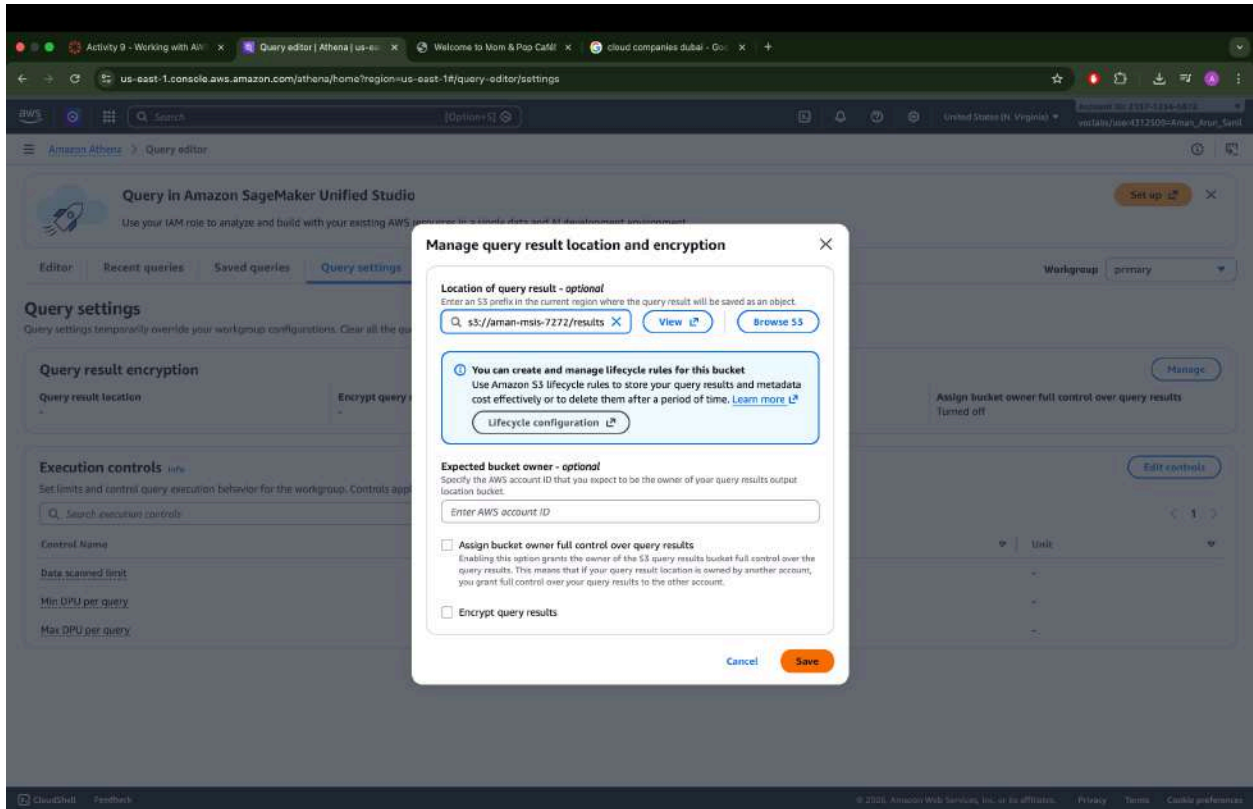
Choose the View settings button that appears above the query panel, then choose Manage.

Choose Browse S3, select your monitoring##### bucket, and select Choose.

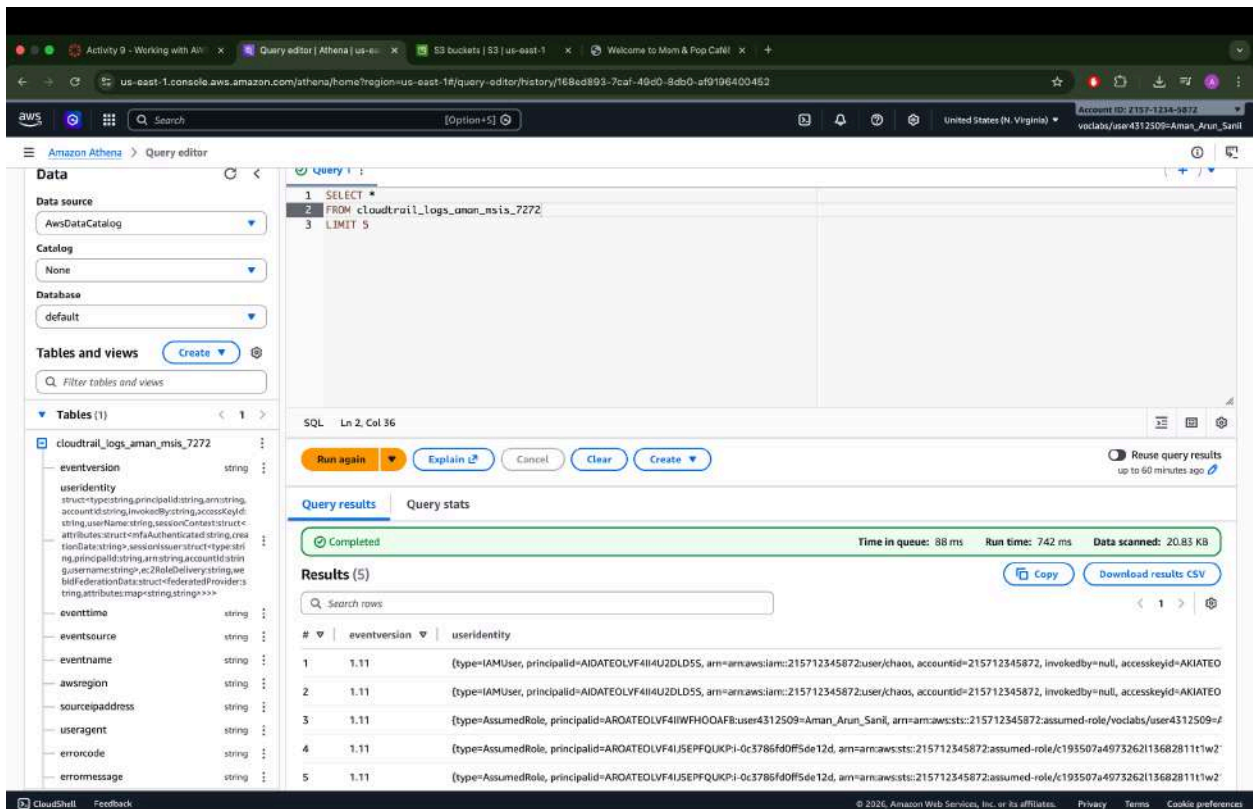
In the Location of query result box, add /results/ to the value, so that it now reads

s3://monitoring#####/results/where monitoring##### is the name of the bucket you created earlier.

Choose Save.



Return to the Editor tab. Paste the following SQL query into the New query 1 panel. Replace ##### with the numbers in your actual table, and choose Run.



Run a new query that selects only those columns that were previously mentioned. This time, limit the results to 30 rows:

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' panel shows the 'cloudtrail_logs_aman_msia_7272' table selected. The 'Query' panel on the right contains the following SQL query:

```
1 SELECT useridentity.userName, eventtime, eventsources, eventname, requestparameters
2 FROM cloudtrail_logs_aman_msia_7272
3 LIMIT 30
```

Below the query, the 'Query results' tab is active, showing a table with 30 rows. The table has the following columns: #, userName, eventtime, eventsources, eventname, and requestparameters. The first five rows are displayed:

#	userName	eventtime	eventsources	eventname	requestparameters
1		2026-02-27T18:06:41Z	logs.amazonaws.com	DescribeMetricFilters	{\"limit\":50}
2		2026-02-27T18:06:41Z	logs.amazonaws.com	DescribeMetricFilters	{\"limit\":50}
3		2026-02-27T18:10:10Z	logs.amazonaws.com	DescribeMetricFilters	{\"limit\":50}
4		2026-02-27T18:27:08Z	ssm.amazonaws.com	UpdateInstanceInformation	{\"instanceId\":\"i-0c3786fd0ff5de12d\",\"agentVersion\":\"3.3.3598.0\",\"platformTyp
5		2026-02-27T18:32:50Z	ssm.amazonaws.com	UpdateInstanceInformation	{\"instanceId\":\"i-0c3786fd0ff5de12d\",\"agentVersion\":\"3.3.3598.0\",\"agentStatus

It returns a list of all users who were active in the account in the past day, and the distinct actions they have taken.

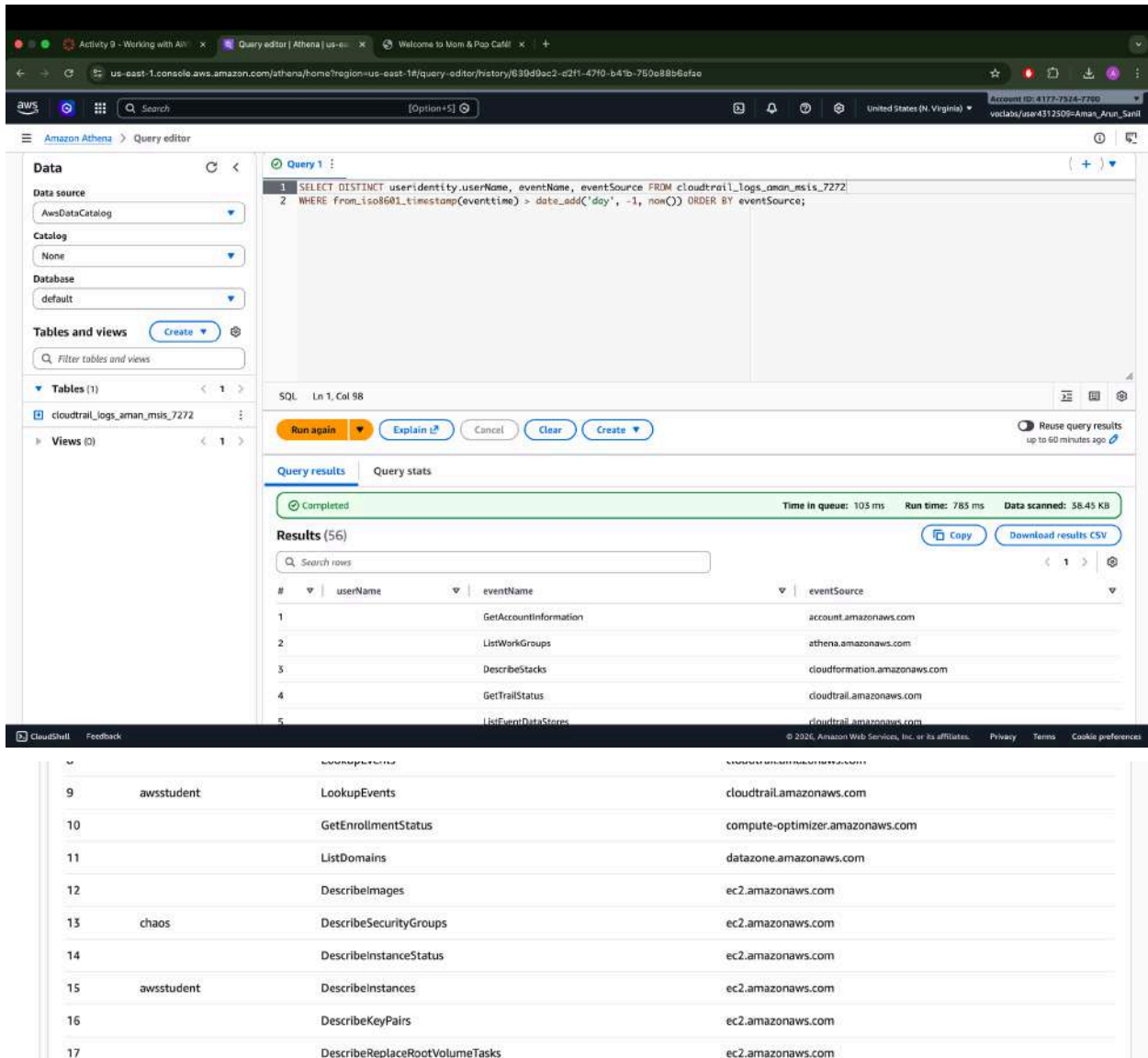
You have successfully completed the challenge if you can identify the following information:

The name of the AWS user who created the security hole in the Cafe Web Server security group

The exact time that they hacked the security group

The IP address from which they hacked it (copy this value to a text file for later reference)

The method they used to perform the hack (console or programmatic access)



The screenshot displays the Amazon Athena console interface. On the left, the 'Data' sidebar shows the 'Data source' as 'AwsDataCatalog', 'Catalog' as 'None', and 'Database' as 'default'. Below this, 'Tables and views' are listed, including 'cloudtrail_logs_aman_msis_7272'. The main area shows 'Query 1' with the following SQL query:

```
1 SELECT DISTINCT useridentity.userName, eventName, eventSource FROM cloudtrail_logs_aman_msis_7272
2 WHERE from_iso8601_timestamp(eventtime) > date_add('day', -1, now()) ORDER BY eventSource;
```

The query status is 'Completed'. The 'Query results' section shows 56 results. The 'Query stats' section indicates: Time in queue: 105 ms, Run time: 785 ms, Data scanned: 38.45 KB. The 'Results (56)' section shows a table with columns: #, userName, eventName, and eventSource. The results are as follows:

#	userName	eventName	eventSource
1		GetAccountInformation	account.amazonaws.com
2		ListWorkGroups	athena.amazonaws.com
3		DescribeStacks	cloudformation.amazonaws.com
4		GetTrailStatus	cloudtrail.amazonaws.com
5		ListEventDataStores	cloudtrail.amazonaws.com
9	awsstudent	LookupEvents	cloudtrail.amazonaws.com
10		GetEnrollmentStatus	compute-optimizer.amazonaws.com
11		ListDomains	datazone.amazonaws.com
12		DescribeImages	ec2.amazonaws.com
13	chaos	DescribeSecurityGroups	ec2.amazonaws.com
14		DescribeInstanceStatus	ec2.amazonaws.com
15	awsstudent	DescribeInstances	ec2.amazonaws.com
16		DescribeKeyPairs	ec2.amazonaws.com
17		DescribeReplaceRootVolumeTasks	ec2.amazonaws.com

Step 8: In the terminal where you have an active SSH session to the web server instance, run the following command to find out who has recently logged into this OS:

sudo aureport --auth

That is interesting! There is evidence that a user other than ec2-user has logged in. Who is that chaos-user?

Run the who command to figure out who is currently logged in:

who

Whoa! The user is still logged in! Let's get them off this instance right away.

Try removing the chaos-user OS user:

sudo userdel -r chaos-user

That didn't work, because they are still logged in. However, it did return the process number they are connected as.

Stop the process that has the active chaos-user login session (replace ProcNum with the process number returned by the last command):

sudo kill -9 ProcNum

Run the who command again to verify they are no longer connected:

who

Now you (the ec2-user) should be the only user connected.

Try again to delete the chaos-user OS user:

sudo userdel -r chaos-user

It should succeed this time.

Verify no other suspicious OS users who can login:

sudo cat /etc/passwd | grep -v nologin

sudo ls -l /etc/ssh/sshd_config

Notice the last modified timestamp for the file. This file was modified today! That is concerning.

Edit the SSH configuration file in the VI editor:

sudo vi /etc/ssh/sshd_config and comment out password authentication

```
[ec2-user@web-server 28]$ sudo aureport --auth

Authentication Report
=====
# date time acct host term exe success event
=====
??>&aturday 28 February 2026
    ??P ec2-user 117.192.176.150 ? /usr/sbin/sshd yes 157
??>&aturday 28 February 2026
    ??P ec2-user 117.192.176.150 ? /usr/sbin/sshd yes 158
??>&aturday 28 February 2026
    ??P ec2-user 117.192.176.150 ssh /usr/sbin/sshd yes 161
[ec2-user@web-server 28]$ who
ec2-user pts/0      2026-02-28 09:28 (117.192.176.150)
[ec2-user@web-server 28]$ sudo userdel -r chaos-user
[ec2-user@web-server 28]$ sudo cat /etc/passwd | grep -v nologin
root:x:0:0:root:/root:/bin/bash
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
[ec2-user@web-server 28]$
```

```
# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no
#PasswordAuthentication no

# Change to no to disable s/key passwords
#ChallengeResponseAuthentication yes
ChallengeResponseAuthentication no
```

Restart the SSH service so that the changes go into effect

```

[ec2-user@web-server 28]$ sudo vi /etc/ssh/sshd_config
[ec2-user@web-server 28]$ sudo service sshd restart
Redirecting to /bin/systemctl restart sshd.service
[ec2-user@web-server 28]$ exit
logout
Connection to 67.202.57.45 closed.
amansanill@Amans-MacBook-Air Downloads % sudo ssh -i labsuser.pem ec2-user@67.202.57.45
Password:
Last login: Sat Feb 28 09:28:56 2026 from 117.192.176.150

      #_
     ~\  #####_      Amazon Linux 2
    ~~ \_#####\
    ~~  \###|      AL2 End of Life is 2026-06-30.
    ~~   \#/
    ~~    V~'  '-->
    ~~~
    ~~~.  .  /
    ~~~ _/  /
    ~~~ _/m/'

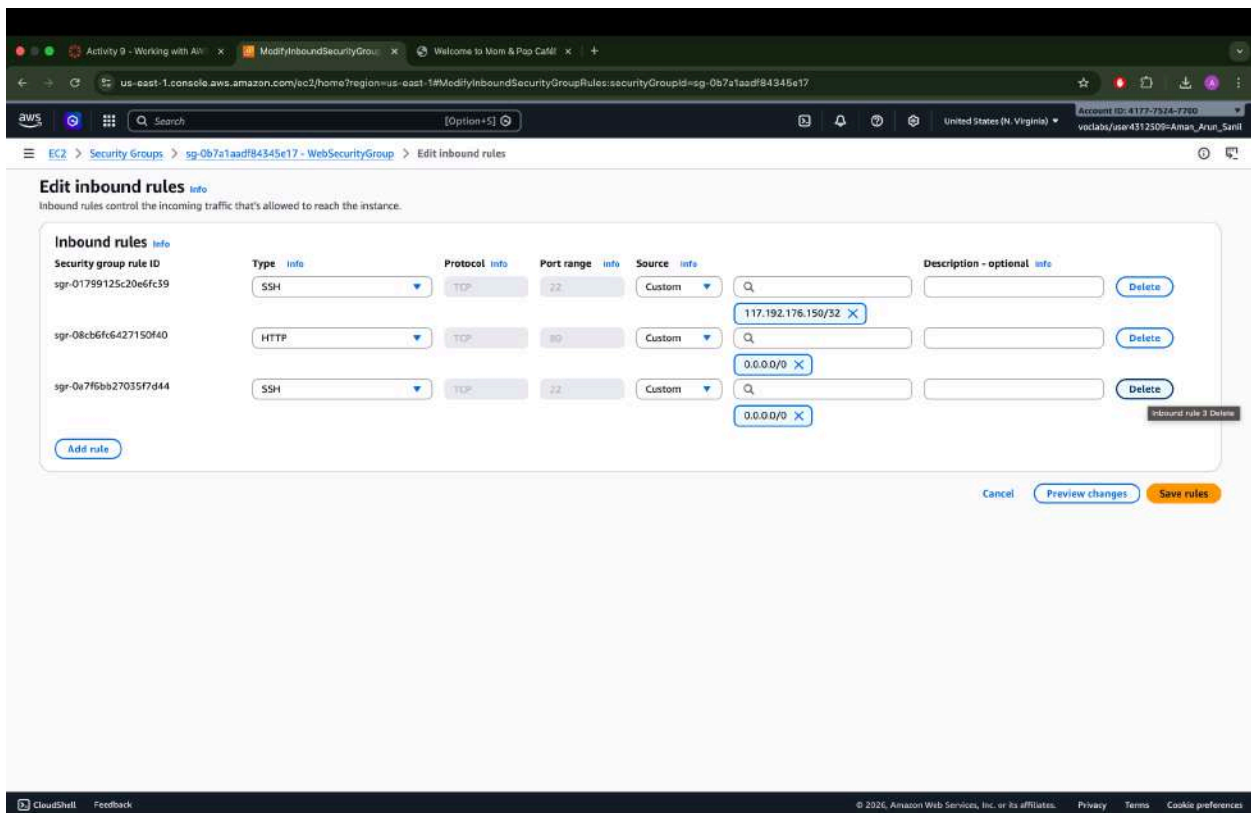
A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@web-server ~]$

```

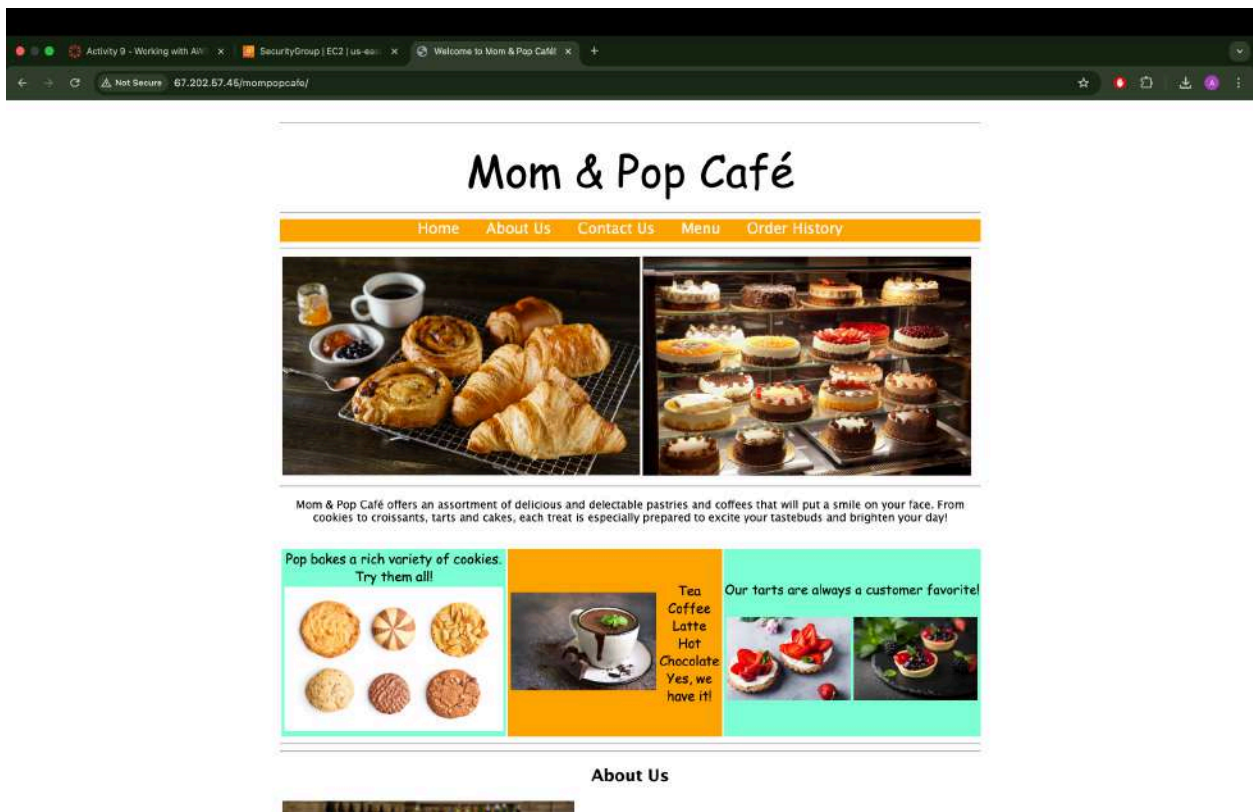
Step 9: Fix the website. Delete the unwanted ssh rule.



Restore the original graphic file.

```
[ec2-user@web-server ~]$ cd /var/www/html/mompopcafe/images/
[ec2-user@web-server images]$ ls -l
total 43864
-rwxrwxrwx 1 root root 3932543 Jan 24 2019 Cake-Vitrine.png
-rwxrwxrwx 1 root root 4029565 Jan 24 2019 Chocolate-Chip-Cookies.png
-rwxrwxrwx 1 1001 root 3275378 Jan 24 2019 Coffee-and-Pastries.backup
-rw-r--r-- 1 1001 root 285715 Feb 28 08:43 Coffee-and-Pastries.png
-rwxrwxrwx 1 root root 4749304 Jan 24 2019 Coffee.png
-rwxrwxrwx 1 root root 1494128 Jan 24 2019 Cookies.png
-rwxrwxrwx 1 root root 2812000 Jan 24 2019 Croissants.png
-rwxrwxrwx 1 root root 3762806 Jan 24 2019 Cup-of-Hot-Chocolate.png
-rwxrwxrwx 1 root root 564440 Jan 26 2019 default-image.png
-rwxrwxrwx 1 root root 3393260 Jan 24 2019 Donuts.png
-rwxrwxrwx 1 root root 3262031 Jan 24 2019 Latte.png
-rwxrwxrwx 1 root root 744262 Jan 24 2019 Mom-&-Pop-Coffee-Shop.png
-rwxrwxrwx 1 root root 2864098 Jan 24 2019 Mom-&-Pop.png
-rwxrwxrwx 1 root root 3012481 Jan 24 2019 Muffins.png
-rwxrwxrwx 1 root root 3090387 Jan 24 2019 Strawberry-&-Blueberry-Tarts.png
-rwxrwxrwx 1 root root 3615146 Jan 24 2019 Strawberry-Tarts.png
[ec2-user@web-server images]$ sudo mv Coffee-and-Pastries.backup Coffee-and-Pastries.png
[ec2-user@web-server images]$
```

Check the website.



Delete the login profile of the hacker and then delete the IAM user.


```
[ec2-user@web-server images]$ aws iam delete-login-profile --user-name=chaos
[ec2-user@web-server images]$
```

The screenshot shows the AWS IAM console interface. The left sidebar contains the 'Identity and Access Management (IAM)' navigation menu with options like Dashboard, Access Management, Policies, and Access reports. The main content area displays the 'Users (1/2)' page. A table lists the users, with 'chaos' selected. The table columns include User name, Path, Groups, Last activity, MFA, Password age, Console last sign-in, and Access key ID. The 'chaos' user is shown with a status of 'Access denied' and an active key ID.

User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
awsstudent	/	1	-	-	Access denied	-	Active - AKIAWCRKYMFGT4 Active - AKIAWCRKYMFGKLEI
☒ chaos	/	0	-	-	Access denied	-	Active - AKIAWCRKYM...

Lab question 6

Module 8 – activity 8 – Working with S3

Step 1: Establish an ssh connection with the CLI host and configure you AWS CLI

```
msis@k8s-master-node:~$ cd Downloads/
msis@k8s-master-node:~/Downloads$ sudo chmod 600 labsuser.pem
[sudo] password for msis:
msis@k8s-master-node:~/Downloads$ ssh -i labsuser.pem ec2-user@3.219.56.242
The authenticity of host '3.219.56.242 (3.219.56.242)' can't be established.
ED25519 key fingerprint is SHA256:92t2s+Wv9Z/pXR4BEHcoKEgsVPd00og7pCQNeuYry1U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.219.56.242' (ED25519) to the list of known hosts.

      #_
     _/  #####      Amazon Linux 2
    ~~~~ \#####\
    ~~~~  \####|
    ~~~~   \##|
    ~~~~    \#/_
    ~~~~     V~'  _.->
    ~~~~      /
    ~~~~     /
    ~~~~    /
    ~~~~   /
    ~~~~  /
    ~~~~ /
    ~~~/

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-200-0-139 ~]$ curl http://169.254.169.254/latest/dynamic/instance-identity/document | grep region
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload    Total     Spent    Left     Speed
100    476   100    476    0     0   84217      0 --:--:-- --:--:-- --:--:--  95200
"region" : "us-east-1",
[ec2-user@ip-10-200-0-139 ~]$ aws configure
AWS Access Key ID [None]: ggg
AWS Secret Access Key [None]: ggg
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-10-200-0-139 ~]$ sudo nano .aws/credentials
```

Step 2: create an S3 bucket in us-east-1 region and load the sample images stored in the CLI host to the /images endpoint of your S3 bucket and list the bucket contents

```
[ec2-user@ip-10-200-0-139 ~]$ aws s3 mb s3://prashnaik0013 --region us-east-1
make_bucket: prashnaik0013
[ec2-user@ip-10-200-0-139 ~]$ ls
initial-images  initial-images.tar.gz  new-images  new-images.tar.gz
[ec2-user@ip-10-200-0-139 ~]$ aws s3 sync ~/initial-images/ s3://prashnaik0013/images
upload: initial-images/Cup-of-Hot-Chocolate.jpg to s3://prashnaik0013/images/Cup-of-Hot-Chocolate.jpg
upload: initial-images/Strawberry-Tarts.jpg to s3://prashnaik0013/images/Strawberry-Tarts.jpg
upload: initial-images/Donuts.jpg to s3://prashnaik0013/images/Donuts.jpg
[ec2-user@ip-10-200-0-139 ~]$ aws s3 ls s3://prashnaik0013/images/ --human-readable --summarize
2026-03-01 09:51:36   308.7 KiB Cup-of-Hot-Chocolate.jpg
2026-03-01 09:51:36   371.8 KiB Donuts.jpg
2026-03-01 09:51:36   468.0 KiB Strawberry-Tarts.jpg

Total Objects: 3
Total Size: 1.1 MiB
[ec2-user@ip-10-200-0-139 ~]$
```

Step 3:

Open the IAM service in the console and review the users and groups created
open the Mediacourse user and explore it go to the security credentials tab and create an access key
for it and download the .csv file and copy the account number.

Activity 8 - Work with Am... Console Home | Console | x medicouser | IAM | Glob... medicouser | IAM | Glob... Instances | EC2 | us-east-1 x +

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1&users/details/medicouser/section=permissions

Search [Alt+S] Ask Amazon Q

Account ID: 7717-8817-7681
console/user/4306860-P0ASHASTH_HAM

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ Access reports

- Access Analyzer
- Resource analysis [New](#)
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

IAM Identity Center [IAM Identity Center](#)

AWS Organizations [AWS Organizations](#)

medicouser info

[Delete](#)

Summary

ARN: [arn:aws:iam::771708277691:user/medicouser](#)

Created: March 01, 2026, 15:15 (UTC+05:30)

Console access: [Enabled without MFA](#)

Last console sign-in: [Never](#)

Access key 1: [Create access key](#)

Permissions

[Groups \(1\)](#) [Tags \(1\)](#) [Security credentials](#) [Last Accessed](#)

Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: [All types](#)

Search:

☐	Policy name ↗	Type	Attached via ↗
☐	IAMUserChangePassword	AWS managed	Group medicaco
☐	medicacoPolicy	Customer inline	Group medicaco

► Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Access denied to access-analyzer:ListPolicyGenerations [Diagnose with Amazon Q](#)

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Activity 8 - Work with Am... Console Home | Console | x medicouser | IAM | Glob... medicouser | IAM | Glob... Instances | EC2 | us-east-1 x +

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1&users/details/medicouser/section=groups

Search [Alt+S] Ask Amazon Q

Account ID: 7717-8817-7681
console/user/4306860-P0ASHASTH_HAM

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- User groups
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IAM Identity Center [IAM Identity Center](#)

AWS Organizations [AWS Organizations](#)

medicouser info

[Delete](#)

Summary

ARN: [arn:aws:iam::771708277691:user/medicouser](#)

Created: March 01, 2026, 15:15 (UTC+05:30)

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Access key 1: [Create access key](#)

Groups

[Permissions](#) [Tags \(1\)](#) [Security credentials](#) [Last Accessed](#)

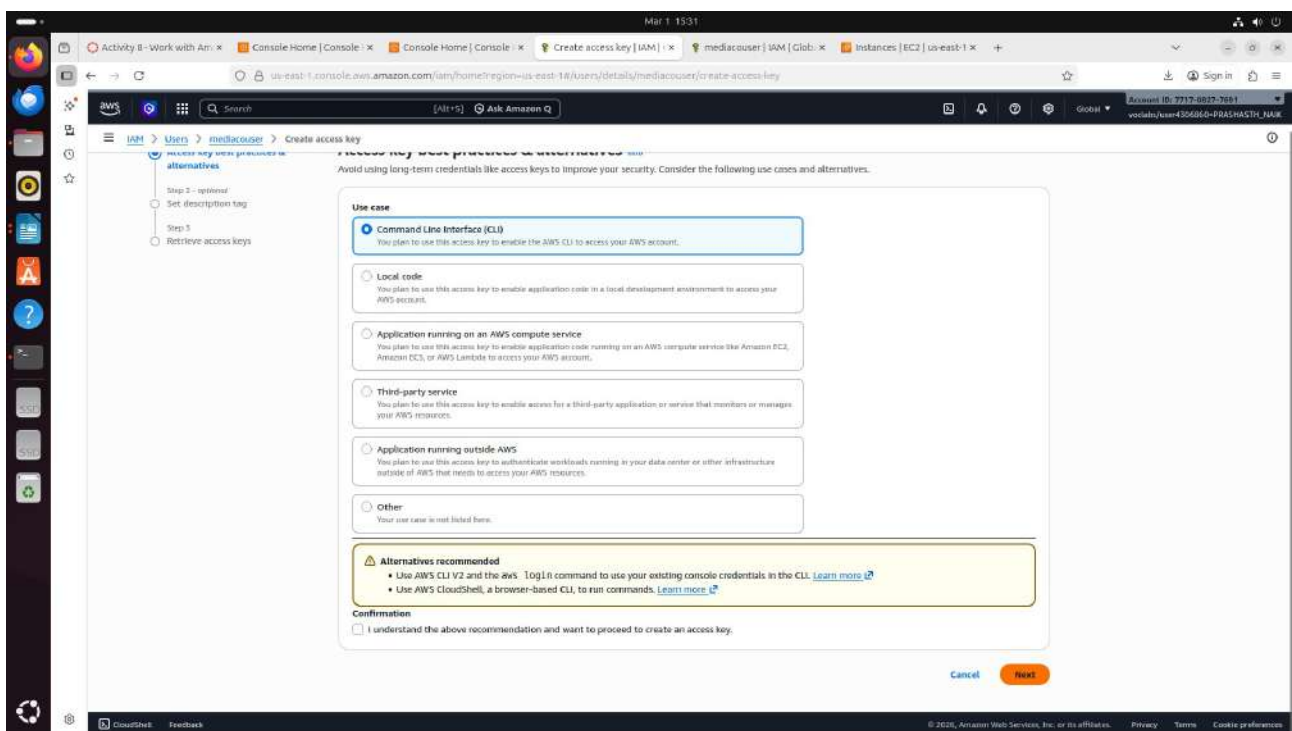
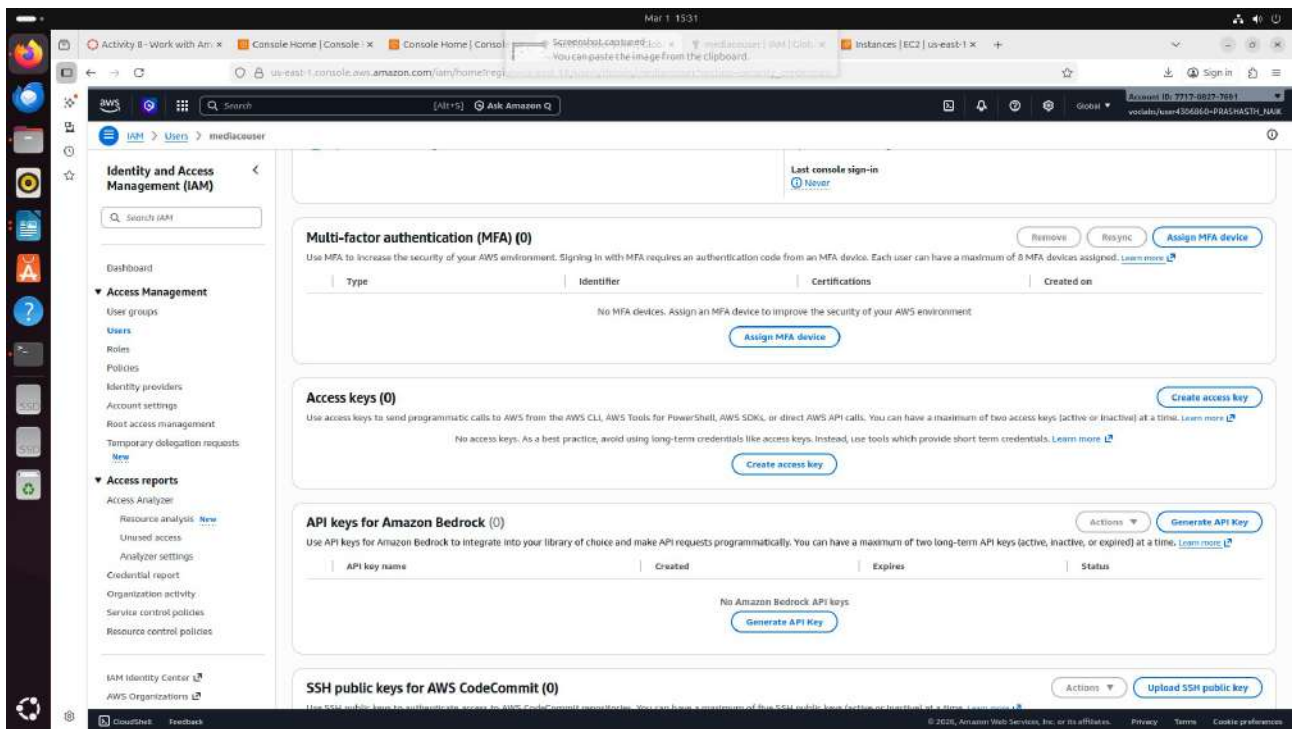
User groups membership

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

[Remove](#) [Add user to groups](#)

☐	Group name	Attached policies ↗
☐	medicaco	IAMUserChangePassword

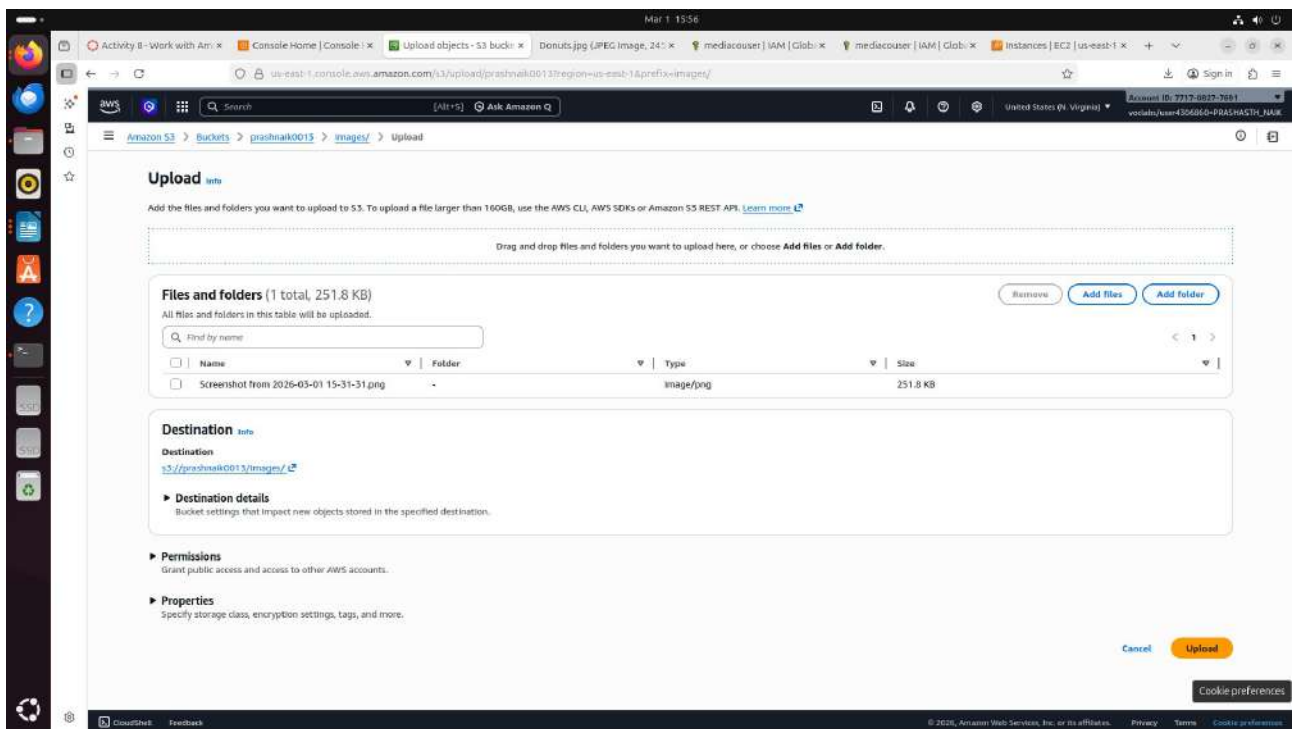
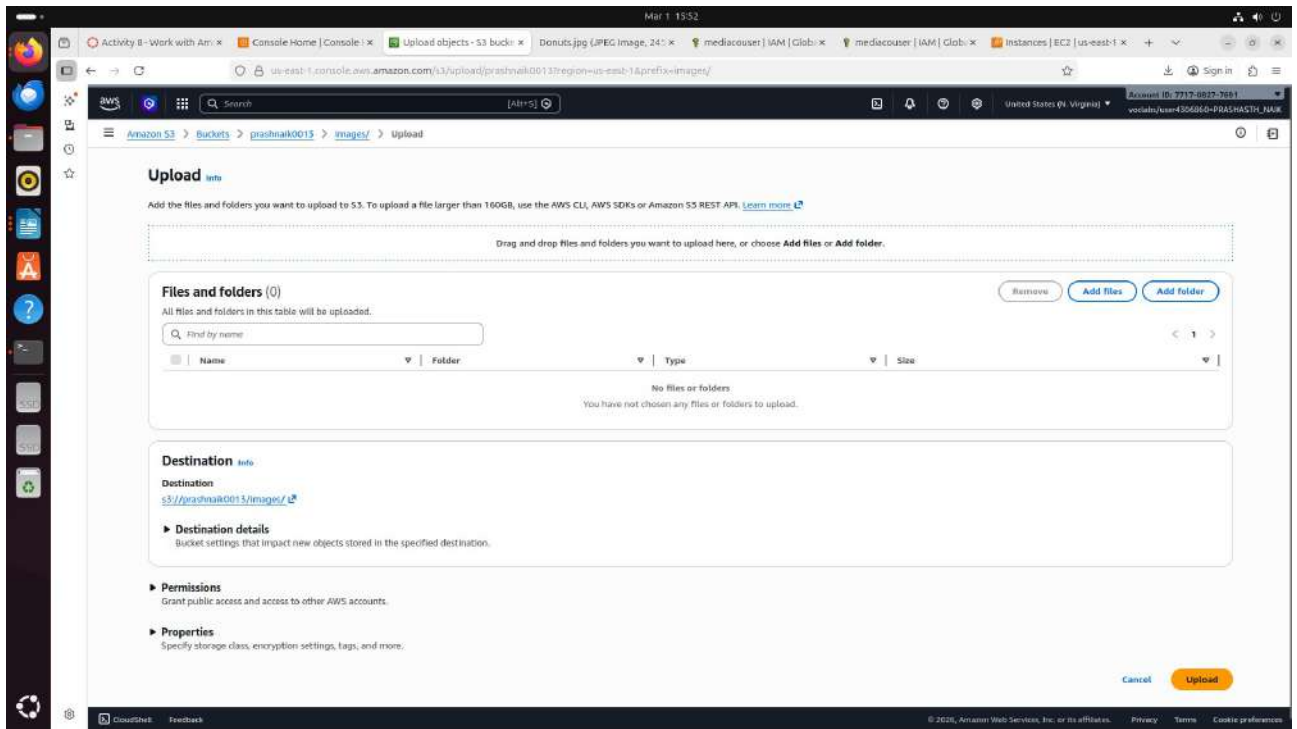
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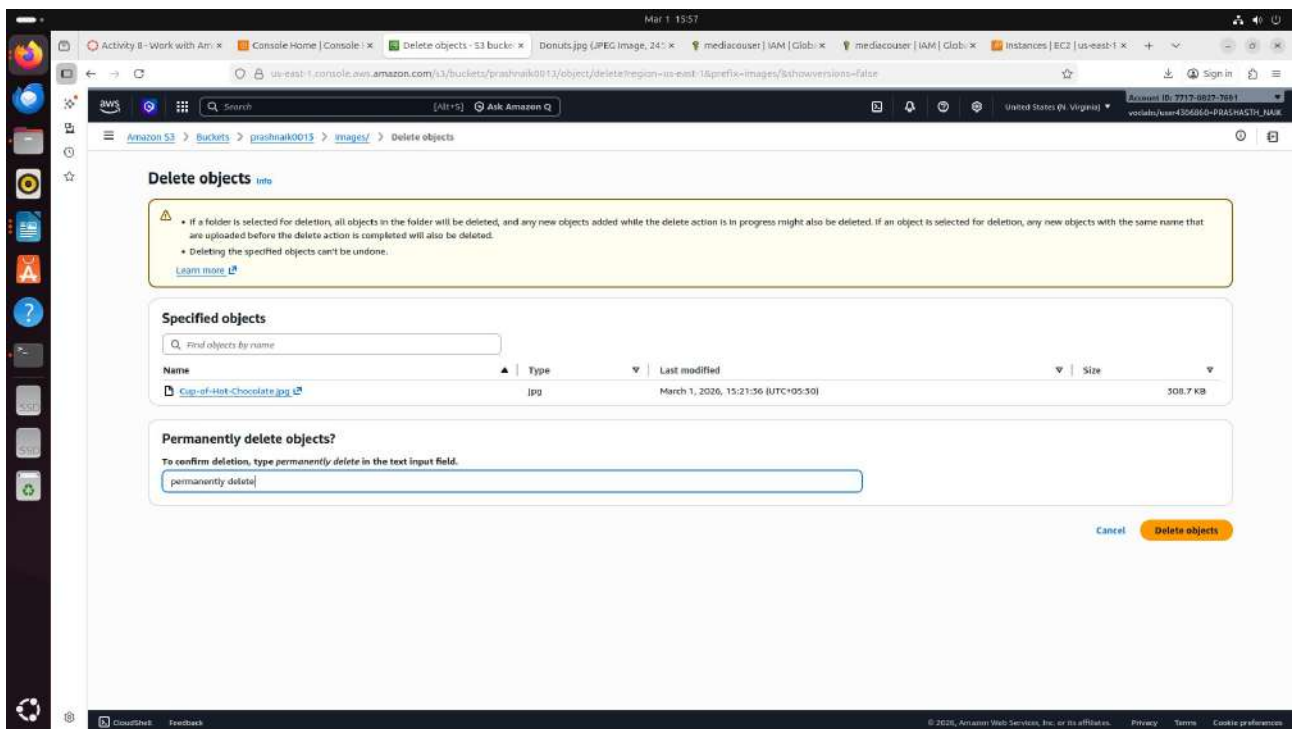
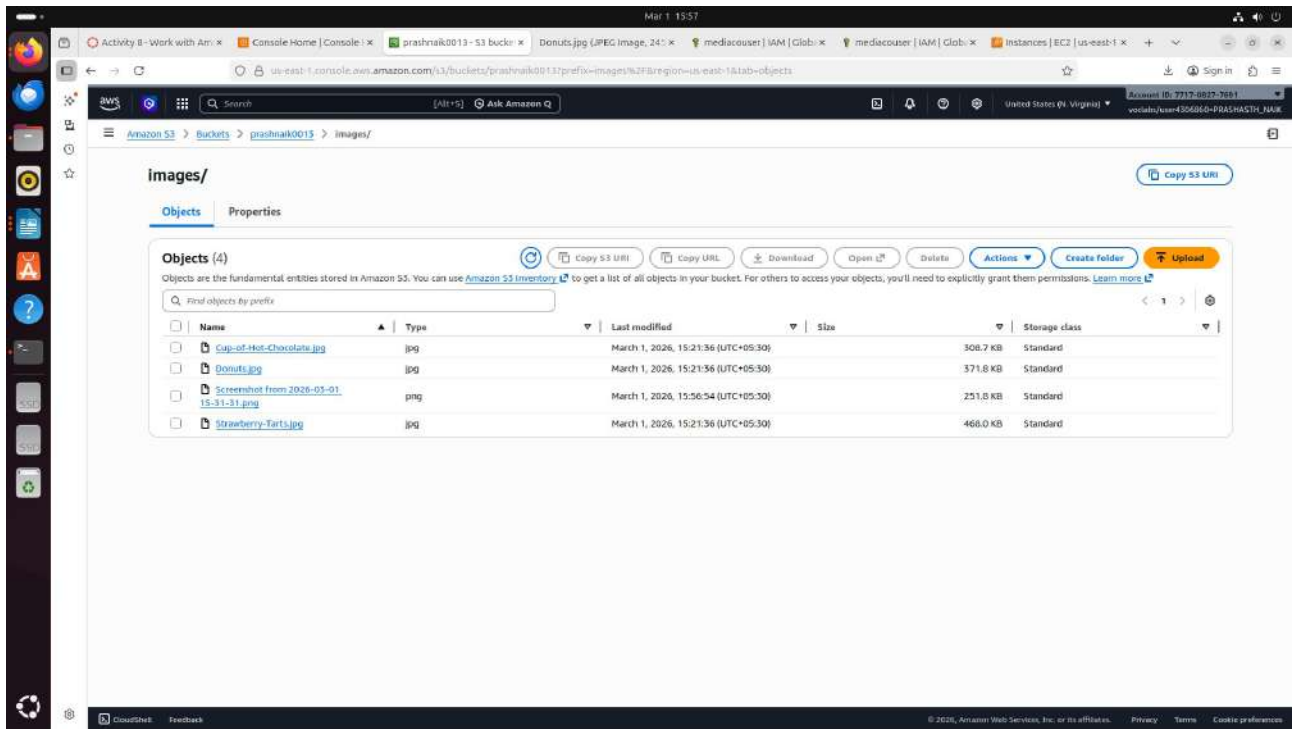


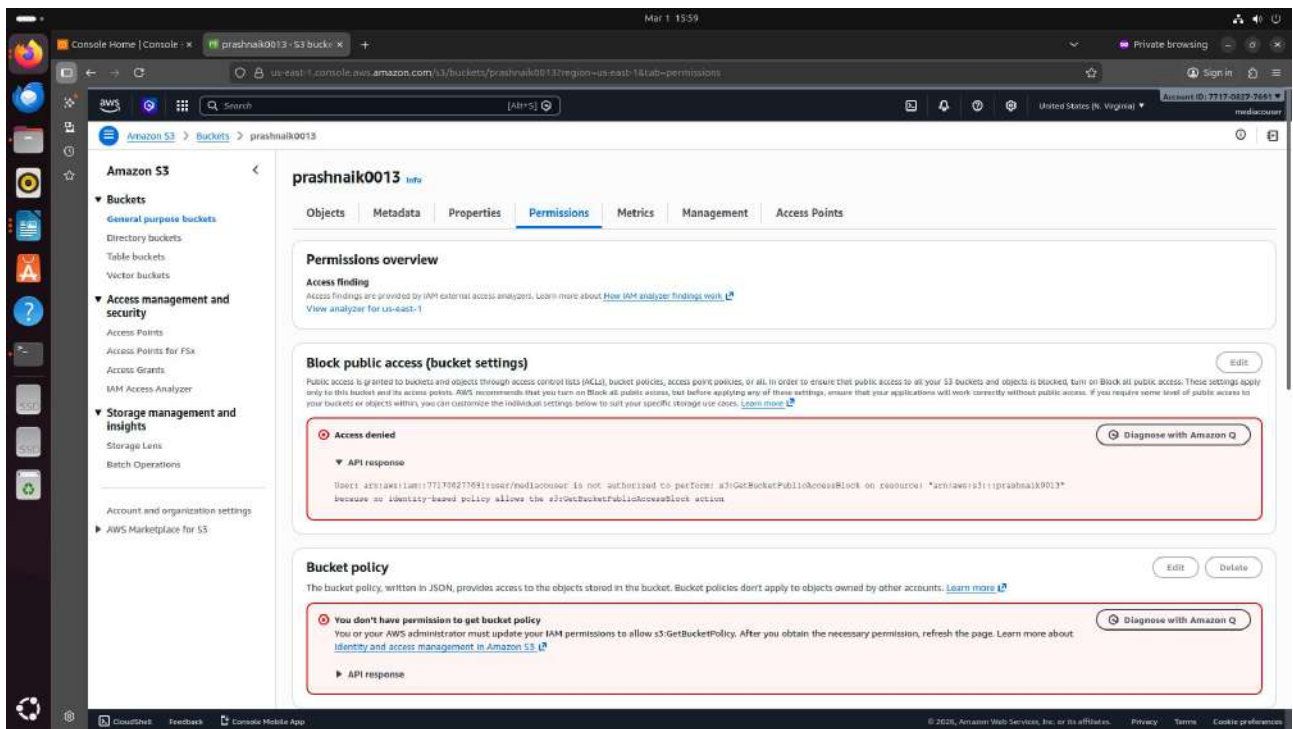
Step 4:

Open a new browser or an incognito tab and use the console link provided in IAM and login using valid username and password

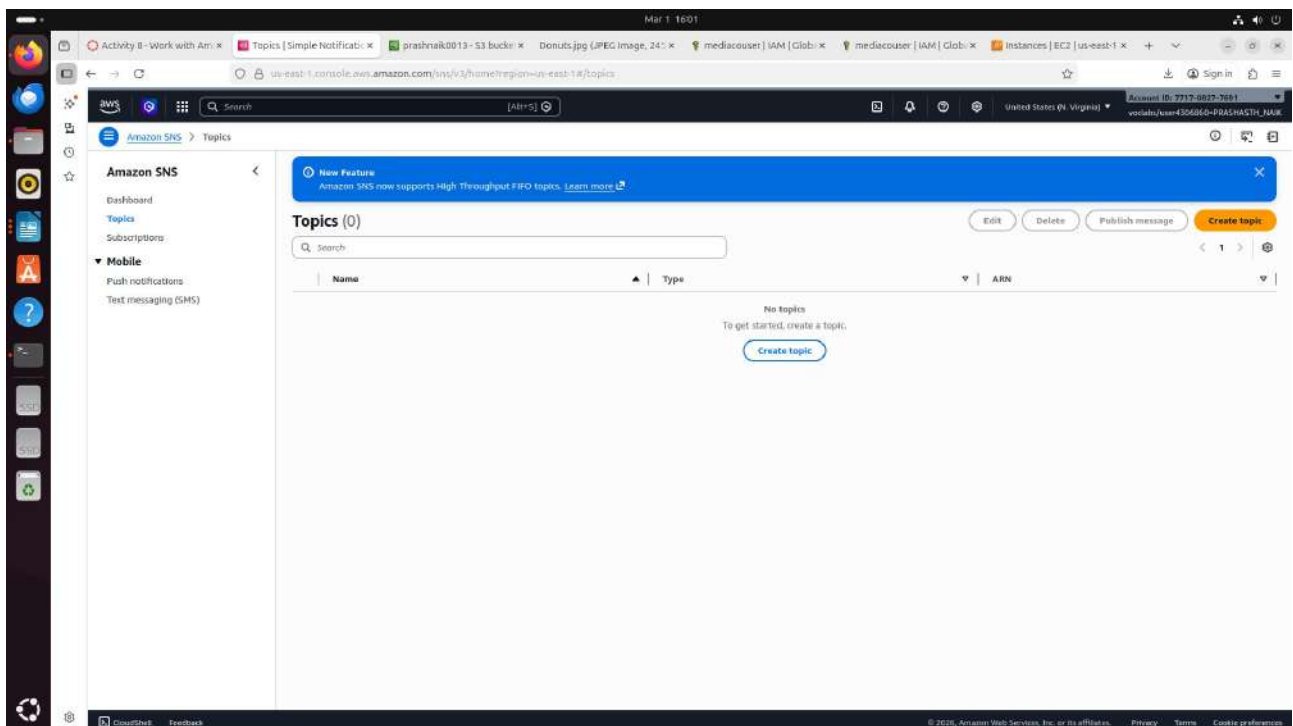
now observe the permissions of the mediacourse user in the console for the s3 bucket you created

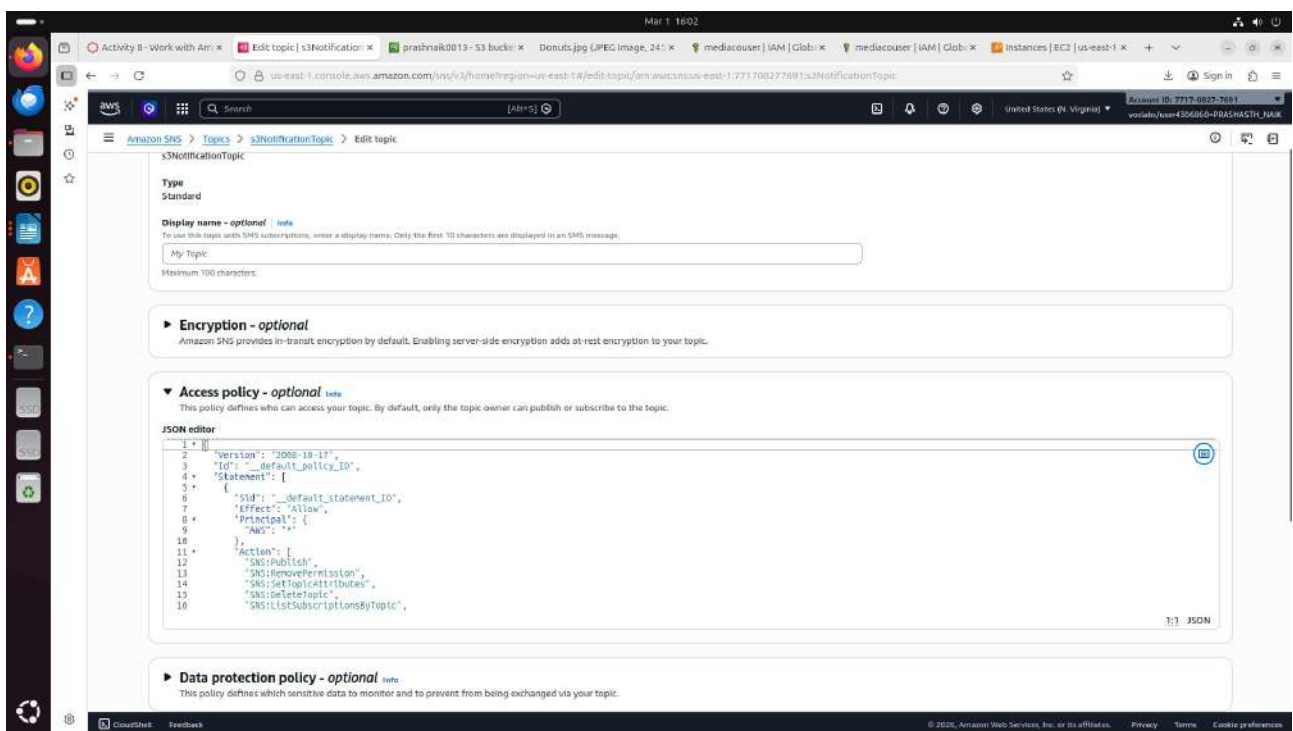
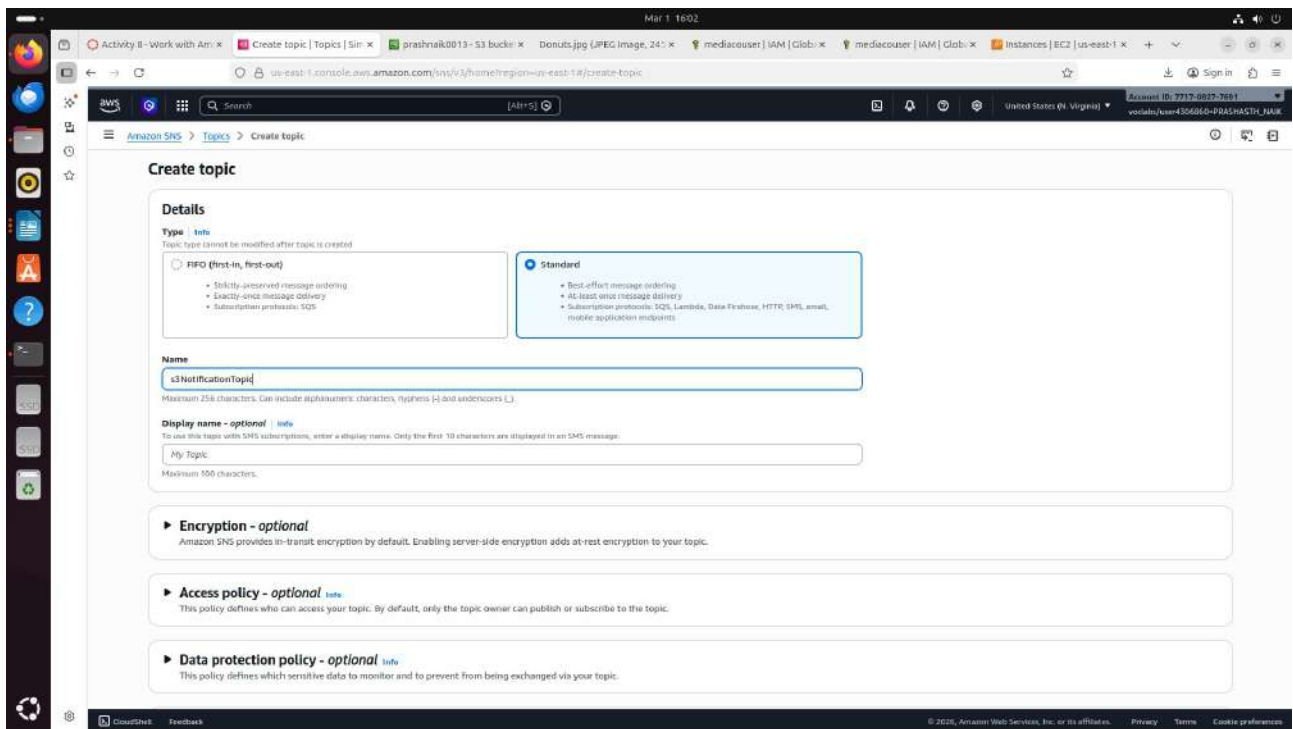






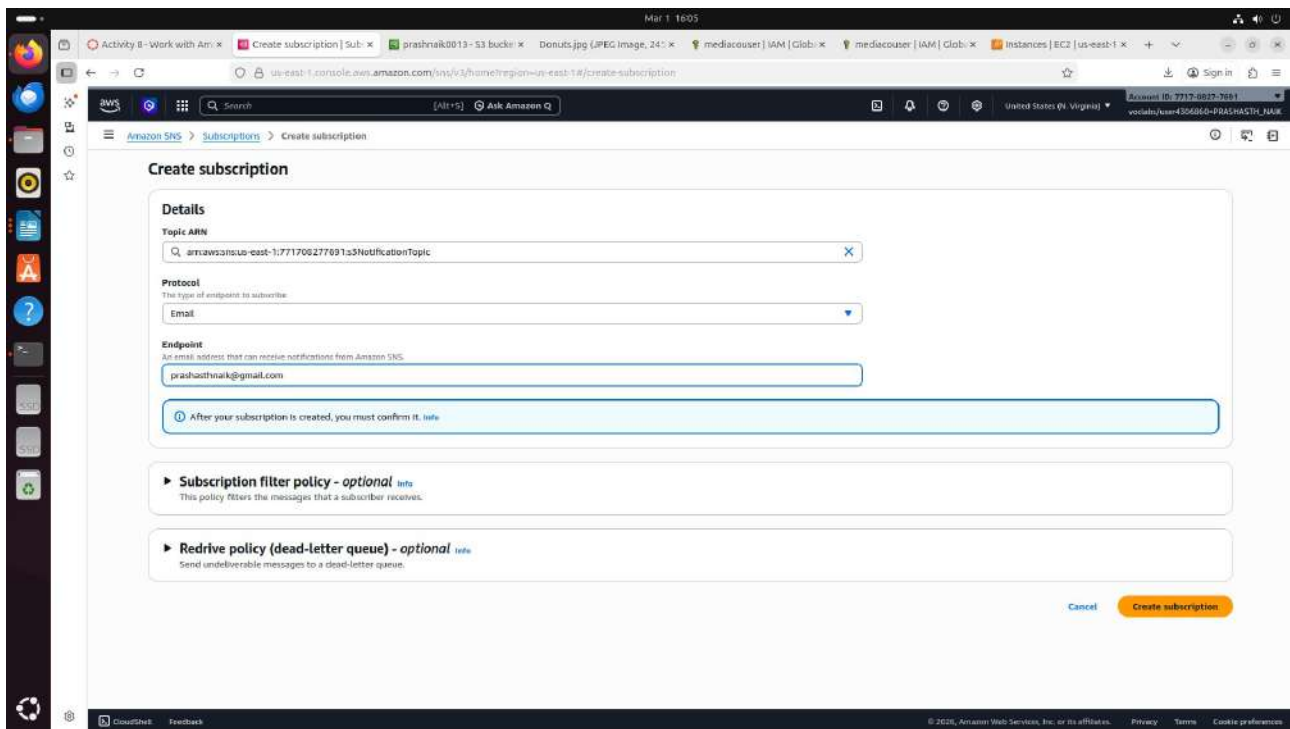
Step 5:
Open sns and create a topic of type standard . Edit the policy of the sns topic created and click save.





Step 6:

Create a subscription for your sns topic with protocol as email and enter your email id and click save once created go to your email and accept the subscription request



Step 7:

create a S3 bucket policy using cli

create a json file with the bucket policy and apply the policy once you apply the policy you should get a mail in your email

```
[ec2-user@ip-10-200-0-139 ~]$ nano s3EventNotification.json
[ec2-user@ip-10-200-0-139 ~]$ aws s3api put-bucket-notification-configuration --bucket prashnaik0013 --notification
[ec2-user@ip-10-200-0-139 ~]$
```

Now run the CLI command to update the S3 event notification you will receive an email through the SNS topic that you have created.



Step 8:

Now using aws configure login as the IAM user that you have created with the respective access key and secret access key

Step 9:

Now put an object into the S3 bucket using the put command you should receive an email through your SNS topic

```
[ec2-user@ip-10-200-0-47 ~]$ aws s3api put-object --bucket mompopcafe-pra013 --key images/Caramel-Delight.jpg --body ~/new-images/Caramel-Delight.jpg
{
  "ETag": "\"31ac30da619244b0ce786f106e4f3df7\"",
  "ServerSideEncryption": "AES256"
}
```



Now similarly try getting an object from S3


```
[ec2-user@ip-10-200-0-47 ~]$ aws s3api get-object --bucket mompopcafe-pra013 --key images/Donuts.jpg Donuts.jpg
{
  "AcceptRanges": "bytes",
  "ContentType": "image/jpeg",
  "LastModified": "Mon, 02 Mar 2026 11:40:13 GMT",
  "ContentLength": 380753,
  "ETag": "\"405b0bcc53cb5ab713c967dc1422b4f4\"",
  "ServerSideEncryption": "AES256",
  "Metadata": {}
}
```



AWS Notifications [info](#) [reply](#) [@sns.amazonaws.com](#)

5:23 PM (12 minutes ago)



[{"Records": [{"eventVersion": "2.1", "eventSource": "aws:s3", "awsRegion": "us-east-1", "eventTime": "2020-03-02T11:53:15.095Z", "eventName": "ObjectRemoved:Delete", "userIdentity": {"principalId": "AWS:AIDA3HLLPZC57ZBR5674X"}, "requestParameters": {"sourceIPAddress": "10.8.2.20.119"}, "responseElements": {"x-amz-request-id": "YC047BK7Z2V1R813", "x-amz-id-2": "pk8CVsCIEKKwPeT8r5Wz6HRm+eZCINWeMJC0XAzbsCRv1UFA0baOpR0XJQZT1n6kWta3Y+2qX2k0ga0ZVwGq+FW3PU0"}, "s3": {"s3SchemaVersion": "1.0", "configurationId": "YmJjJGJ0Th0Nzc0M0D0ZnEwLWYyMzA0NzF0QWZlZWZlNDJk", "bucket": {"name": "mompopcafe-pra013", "ownerIdentity": {"principalId": "A35N77HKC6Y4LUJ"}, "arn": "arn:aws:s3:::mompopcafe-pra013"}, "object": {"key": "images/Strawberry-Tarts.jpg", "sequencer": "0069A57A2B60FC668"}]]}]

...

Now attempt deleting an object

```
[ec2-user@ip-10-200-0-47 ~]$ aws s3api delete-object --bucket mompopcafe-pra013 --key images/Strawberry-Tarts.jpg
```

Finally try performing an unauthorized access of changing the permissions of the S3 bucket

```
[ec2-user@ip-10-200-0-47 ~]$ aws s3api put-object-acl --bucket mompopcafe-pra013 --key images/Donuts.jpg --acl public-read
```

An error occurred (AccessDenied) when calling the PutObjectAcl operation: User: arn:aws:iam::771708277691:user/mediacouser is not authorized to perform arn:aws:s3:::mompopcafe-pra013/images/Donuts.jpg" because public ACLs are prevented by the BlockPublicAcls setting in S3 Block Public Access.